7

Signal and System Analysis using the Laplace Transform

7.1 Introduction

In Chapter 6 we have studied that Fourier transform of a function $x(t)$ can be represented by continuous sum of exponential functions of the form $e^{j\Omega t}$. In addition the Fourier transform technique is applied for solving differential equation that relates input and output of a system. First, the Fourier transform is used to convert the differential equation into algebraic equation and then the Fourier transform of the output $Y(j\Omega)$ is obtained as a function of frequency response $H(j\Omega)$ and Fourier transform of input $X(j\Omega)$. The output is obtained as a function of time by taking inverse Fourier transform of $Y(j\Omega)$. But this technique is rarely used because Fourier transform does not exist for many functions. Moreover the Fourier transform cannot be used to analyse unstable systems.

An alternate approach is to redefine the transform and include an exponential convergence factor σ along with $j\Omega$. That is, the signal $x(t)$ is represented as a sum of complex exponentials of the form e^{st} where s is complex frequency denoted by $s = \sigma + j\Omega$.

The Fourier transform of a continuous time signal $x(t)$ can be represented as

$$X(j\Omega) = \int_{-\infty}^{\infty} x(t)e^{-j\Omega t}dt \tag{7.1}$$

where

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\Omega)e^{j\Omega t}d\Omega \tag{7.2}$$

Replacing $j\Omega$ by $\sigma + j\Omega$ in Eq. (7.1) yields

$$X(\sigma + j\Omega) = \int_{-\infty}^{\infty} x(t)e^{-(\sigma+j\Omega)t}dt \tag{7.3a}$$

and $x(t)$ can be expressed as

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\sigma + j\Omega)e^{(\sigma+j\Omega)t}d\Omega. \qquad (7.3b)$$

The term $\sigma + j\Omega$ is denoted by s. Then $d\Omega = \left(\frac{1}{j}\right)ds$.

Substituting these values in Eq. (7.3), we get

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt \qquad (7.4a)$$

and

$$x(t) = \frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty} X(s)e^{st}ds \qquad (7.4b)$$

Eq. (7.4a) and Eq. (7.4b) constitute complex Fourier transform pair or the bilateral Laplace transform pair.

To represent the Laplace and inverse Laplace transform we use following notations

$$x(t) \longleftrightarrow X(s)$$
$$L[x(t)] = X(s)$$

and

$$L^{-1}[X(s)] = x(t),$$

where $X(s)$ is Laplace transform of $x(t)$ and $x(t)$ is inverse Laplace transform of $X(s)$.

These equations are called Laplace transform pair;

7.2 Convergence of the Laplace Transfrom

The bilateral Laplace transform of a signal $x(t)$ exists if

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt \qquad (7.5)$$

is finite.

Substituting $s = \sigma + j\Omega$ in Eq. (7.5) yields

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-(\sigma+j\Omega)t}dt$$
$$= \int_{-\infty}^{\infty} x(t)e^{-\sigma t}e^{-j\Omega t}dt \qquad (7.6)$$

The above equation indicates that Laplace transform is the Fourier transform of $x(t)e^{-\sigma t}$. Therefore, a necessary condition for existence of the Laplace transfrom is absolute integrability of $x(t)e^{-\sigma t}$. That is, $X(s)$ exists if

$$\int_{-\infty}^{\infty} |x(t)e^{-\sigma t}|\,dt < \infty. \qquad (7.7)$$

The range of σ for which the Laplace transform converges is known as Region Of Convergence (ROC). Consider a signal $x(t) = e^{2t}u(t)$ as shown in Fig. 7.1. The Fourier transform for $x(t)$ does not exist because it is not absolutely integrable (Example 6.1 iv). Now multiply the signal $x(t)$ with $e^{-\sigma t}$. Then the resultant signal $x(t)e^{-\sigma t} = e^{(2-\sigma)t}u(t)$ is absolutely integrable for $\sigma > 2$. Therefore the Laplace transform of $x(t)$, which is Fourier transform of $x(t)e^{-\sigma t}$ exists for some values of σ. This concept is illustrated in Fig. 7.1.

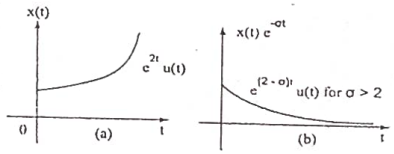

Fig. 7.1 (a) Exponential sequence $e^{2t}u(t)$ for which Fourier transform does not exist (b) The signal $x(t)e^{-\sigma t}$ which is absolutely integrable for $\sigma > 2$.

Thus we can conclude that the Laplace transform exist for signals for which Fourier transform does not exist. If $x(t)$ is absolutely integrable, then the Fourier transform of $x(t)$ can be obtained from its Laplace transform using the relation

$$X(j\Omega) = X(s)|_{s=j\Omega} \qquad (7.8)$$

Review Questions

1. What are the limitations of Fourier transform?
2. What is the relation between Fourier transform and Laplace transform?
3. Define Laplace transform. What is meant by ROC?

7.3 s - Plane

The Laplace variable s is a complex variable defined by $s = \sigma + j\Omega$, where σ is the real part of s denoted as Re(s) and the variable Ω is the imaginary part of s denoted as Im(s). The range of variation of s can be represented schematically as a s-plane. The horizontal axis of this plane is known as real axis, since it is defined by

$$s = \sigma + j0.$$

The vertical axis is known as imaginary axis and is represented by

$$s = 0 + j\Omega.$$

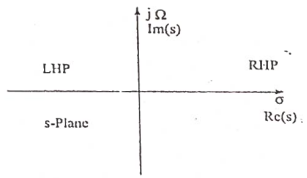

Fig. 7.1 s-plane.

The origin is represented by the complex zero $s = 0 + j0$. The $j\Omega$–axis (imaginary axis) divides the s-plane in two halves. The region to the left of the imaginary axis (for $-\infty < \sigma < 0$) is termed the left half plane (LHP) and the region to the right of imaginary axis (for $0 < \sigma < \infty$) is termed the right half plane (RHP). The whole complex plane is known as s-plane.

The most general form of the Laplace transform is a ratio of two polynomials

$$X(s) = \frac{N(s)}{D(s)} = \frac{b_M s^M + b_{M-1}s^{M-1} + \ldots + b_1 s + b_0}{s^N + a_{N-1}s^{N-1} + \ldots + a_1 s + a_0} \tag{7.9}$$

The numerator polynomial of $X(s)$ is $N(s) = b_M s^M + b_{M-1}s^{M-1} + \ldots + b_1 s + b_0$ with the degree M. The zeros of $X(s)$ are the values of s for which the transform is zero. They are found by setting the numerator polynomial to zero and solving the resulting equation. That is

$$N(s) = b_M s^M + b_{M-1}s^{M-1} + \ldots + b_1 s + b_0 = 0 \tag{7.10}$$

$$= b_M \left[s^M + \frac{b_{M-1}}{b_M} s^{M-1} + \ldots + \frac{b_1}{b_M} s + \frac{b_0}{b_M} \right] = 0$$

$$= b_M \prod_{k=1}^{M} (s - z_k) = 0. \tag{7.11}$$

The $\{z_k\}$ are the roots of the numerator polynomial and are zeros of $X(s)$. The zeros are represented in the s-plane with the 'o' symbol.

The denominator polynomial of $X(s)$ is

$$D(s) = s^N + a_{N-1}s^{N-1} + \ldots + a_1 s + a_0 \tag{7.12}$$

with degree N. Parameter N is also called the order of the transform $X(s)$.

The poles of the transform $X(s)$ are the values of s for which the transform is infinite. The poles are found by setting the denominator polynomial to zero and solving the resulting equation. That is

$$D(s) = s^N + a_{N-1}s^{N-1} + \ldots + a_1 s + a_0 = 0$$

$$= \prod_{k=1}^{N} (s - p_k) = 0. \tag{7.13}$$

The $\{p_k\}$ are the roots of the denominator polynomial and are poles of $X(s)$. The poles are represented in the s-plane with the '×' symbol.

Review Questions

1. What is s-plane?
2. What are poles and zeros? How they are represented in s-plane?

Solved Problem 7.1 Find the Laplace transform and ROC of the following signals

(i) $e^{-at}u(t)$

(ii) $-e^{-at}u(-t)$

Solution:

(i) Given

$$x(t) = e^{-at}u(t)$$

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt$$

$$= \int_{-\infty}^{\infty} e^{-at}u(t)e^{-st}dt \qquad \boxed{u(t)=1 \quad \text{for} \quad t \geq 0 \\ \quad\;\; =0 \quad \text{for} \quad t<0}$$

$$= \int_{0}^{\infty} e^{-(a+s)t}dt \qquad \boxed{Re(s+a)>0 \\ Re(s) \quad > -a}$$

We know that $s = \sigma + j\Omega$, therefore the above integral converges if $Re(s), \sigma > -a$. Now

$$X(s) = \frac{1}{-(a+s)}e^{-(a+s)t}\Big|_0^{\infty}$$

$$= \frac{1}{s+a}$$

$$\Rightarrow \quad L[e^{-at}u(t)] = \frac{1}{s+a} \quad \text{ROC}: \sigma > -a \qquad (7.14)$$

(ii) Given

$$x(t) = -e^{-at}u(-t)$$

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt$$

$$= \int_{-\infty}^{\infty} -e^{-at}u(-t)e^{-st}dt \qquad \boxed{u(-t)=0 \quad \text{for} \quad t>0 \\ \quad\;\; =1 \quad \text{for} \quad t \leq 0}$$

$$= -\int_{-\infty}^{0} e^{-(a+s)t}dt$$

$$= \int_{0}^{\infty} e^{(a+s)t}dt \qquad \boxed{Re(s+a)<0 \\ \Rightarrow Re(s)<-a}$$

Since $s = \sigma + j\Omega$, the integral converges if $Re(s)$, that is $\sigma < -a$ and

$$X(s) = \frac{1}{s+a} \qquad \text{ROC}: \sigma < -a \qquad (7.15)$$

From solved problem 7.1 we can note that the Laplace transforms of both signals are identical, but the ROCs are different. The ROC of $e^{-at}u(t)$ is $\sigma > -a$ and for $e^{-at}u(-t)$ is $\sigma < -a$. The ROCs of both signals are shown in Fig. 7.3.

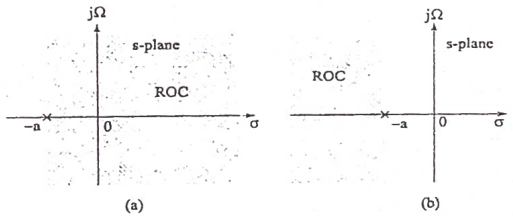

Fig. 7.3 **(a) ROC of $e^{-at}u(t)$** **(b) ROC of $-e^{-at}u(-t)$.**

The shaded region in Fig. 7.3a represents the set of points in the s- plane for which the Laplace transform of $e^{-at}u(t)$ converges. Similarly the shaded region in Fig. 7.3b indicates the ROC of $-e^{at}u(-t)$. The signal $e^{-at}u(t)$ is a causal signal and the signal $-e^{-at}u(-t)$ is a non-causal signal. In general, the ROC of a causal signal is Re $\{s\} > \alpha$, where α is some constant and the ROC of a noncausal signal is Re $\{s\} < \alpha$.

Solved Problem 7.2 Find the Laplace transform of the signal

May 2005, Nov 2005

$x(t) = e^{-3t}u(t) + e^{-2t}u(t)$ and find ROC.

What are the pole-zero locations?

Solution: Given $x(t) = e^{-3t}u(t) + e^{-2t}u(t)$

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt$$

$$= \int_{-\infty}^{\infty} e^{-3t}u(t)e^{-st}dt + \int_{-\infty}^{\infty} e^{-2t}u(t)e^{-st}dt$$

$$= \underbrace{\int_{0}^{\infty} e^{-(s+3)t}dt}_{\text{converges if } \mathrm{Re}(s) > -3} + \underbrace{\int_{0}^{\infty} e^{-(s+2)t}dt}_{\text{converges if } \mathrm{Re}(s) > -2}$$

converges if $Re(s) > -2$

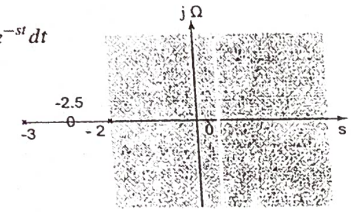

Fig.7.4 ROC of the signal $x(t)=e^{-3t}u(t)+e^{-2t}u(t)$

$$X(s) = \frac{-1}{s+3}e^{-(s+3)t}\Big|_0^\infty + \frac{1}{-(s+2)}e^{-(s+2)t}\Big|_0^\infty$$

$$= \frac{1}{s+3} + \frac{1}{s+2} \quad ROC: Re(s) > -2$$

$$= \frac{(s+2)+(s+3)}{(s+3)(s+2)}$$

$$X(s) = \frac{2s+5}{s^2+5s+6} \quad ROC: \sigma > -2$$

The poles can be obtained by equating the denominator polynomial to zero. That is $s^2+5s+6=0 \Rightarrow (s+2)(s+3)=0$. The poles are at $s=-2$ and $s=-3$. The zero is $s=-5/2$. The poles and zeros are located in s-plane, as shown in Fig.7.4

Solved Problem 7.3 Find the Laplace transform of the signal $x(t) = e^{-at}u(t) + e^{-bt}u(-t)$ and find ROC. What are pole locations?

Solution:

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt$$

$$= \int_{-\infty}^{\infty} e^{-at}u(t)e^{-st}dt + \int_{-\infty}^{\infty} e^{-bt}u(-t)e^{-st}dt$$

$$= \underbrace{\int_{0}^{\infty} e^{-(a+s)t}dt}_{\text{Converges if } \mathrm{Re}(s) > -a} + \underbrace{\int_{-\infty}^{0} e^{-(s+b)t}dt}_{\text{Converges if } \mathrm{Re}(s) < -b}$$

converges if $-a < \mathrm{Re}(s) < -b$

$$= \frac{-1}{s+b}e^{-(s+b)t}\Big|_{-\infty}^{0} + \frac{1}{-(s+a)}e^{-(s+a)t}\Big|_{0}^{\infty}$$

$$= \frac{1}{s+b} + \frac{1}{s+a}$$

$$\mathrm{ROC}: -a < Re(s) < -b \qquad (7.17)$$

The poles are at $s=-a$ and $s=-b$.

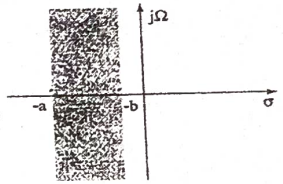

Fig.7.5 ROC of the signal $e^{-at}u(t) + e^{-bt}u(-t)$

Solved Problem 7.4 Find the Laplace transform of the signal

AU Nov 2003

$$x(t) = e^{-b|t|}$$

Solution:

Given $x(t) = e^{-b|t|}$

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt$$

$$= \int_{-\infty}^{\infty} e^{-b|t|}e^{-st}dt$$

$$= \int_{-\infty}^{0} e^{bt}e^{-st}dt + \int_{0}^{\infty} e^{-bt}e^{-st}dt$$

$$= \underbrace{\int_{-\infty}^{0} e^{-(s-b)t}dt}_{\text{Converges if } \mathrm{Re}(s) < b} + \underbrace{\int_{0}^{\infty} e^{-(s+b)t}dt}_{\text{Converges if } \mathrm{Re}(s) > -b}$$

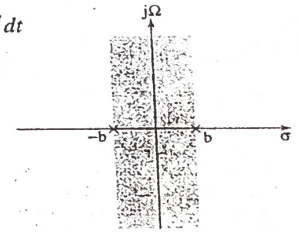

Fig.7.6 ROC of the signal $e^{-b|t|}$

$$X(s) = \frac{1}{-(s-b)}e^{-(s-b)t}\Big|_{-\infty}^{0} + \frac{1}{-(s+b)}e^{-(s+b)t}\Big|_{0}^{\infty}$$

$$= \frac{-1}{s-b} + \frac{1}{s+b} \quad \mathrm{ROC}: \quad -b < \mathrm{Re}(s) < b \qquad (7.18)$$

In this case the poles are at $s = \pm b$.

Properties of ROC

From solved problem (7.3) and (7.4) we can find that two different signals can have identical expressions for their Laplace transform and the signal are distinguishable only by ROC. That is the ROC of $X(s)$ depends on the nature of $x(t)$. In this section we will study the properties of ROC.

Property 1 The ROC of $X(s)$ consists of strips parallel to the $j\Omega-$ axis in the s-plane. (see Fig.7.5 and Fig.7.6)

Property 2 The ROC does not contain any poles.

Property 3 If $x(t)$ is of finite duration and is absolutely integrable, then the ROC is the entire s-plane.

Property 4 It $x(t)$ is a right-sided signal, that is $x(t) = 0$ for $t < t_0 < \infty$ then the ROC is of the form $Re(s) > \alpha_{max}$. Where α_{max} equals the maximum real part of any of the poles of $X(s)$.

Property 5 If $x(t)$ is a left-sided, that is $x(t) = 0$ for $t > t_1 > -\infty$, Then the ROC is of the form $Re(s) < \alpha_{min}$.
Where α_{min} equals the minimum real part of any of the poles of $X(s)$.

Property 6 If $x(t)$ is a two-sided signal, then the ROC is of the from $\alpha_1 < Re(s) < \alpha_2$.

Review Question

1. What is ROC of Laplace transform? State its properties.

AU Nov 2008

Solved Problem 7.5 Show that for a right-handed signal satisfying $x(t) = 0$ for $t < t_0, X(s)$ converges for some value of s, the ROC of $X(s)$ is $Re(s) > \alpha_{max}$ where α_{max} equals the maximum real part of any of the poles of $X(s)$.

Solution: Given a right-handed signal satisfying

$$x(t) = 0 \text{ for } t < t_0$$

Let us assume that $X(s)$ converges for $Re(s) = \sigma_0$. Then the magnitude of $X(s)$ satisfies

$$|X(s)| \leq \int_{-\infty}^{\infty} |x(t)e^{-st}|\, dt$$

$$= \int_{-\infty}^{\infty} |x(t)||e^{-\sigma_0 t}||e^{-j\Omega_0 t}\, dt$$

$$= \int_{-\infty}^{\infty} |x(t)|e^{-\sigma_0 t}\, dt$$

Since $x(t) = 0$ for $t < t_0$

$$|X(s)| = \int_{t_0}^{\infty} |x(t)|e^{-\sigma_0 t}\, dt < \infty$$

Let $Re(s) = \alpha > \sigma_0$. Then

$$\int_{t_0}^{\infty} |x(t)|e^{-\alpha t}\, dt = \int_{t_0}^{\infty} |x(t)|e^{-\sigma_0 t}\, e^{-(\alpha-\sigma_0)}dt$$

$$< e^{-(\alpha-\sigma_0)t_0} \int_{t_0}^{\infty} |x(t)|e^{-\sigma_0 t}\, dt < \infty$$

From above equation we find that $X(s)$ converges for $Re(s) = \alpha$. Since the ROC of $X(s)$ cannot include any poles of $X(s)$, then $Re(s) > \alpha_{max}$, where α_{max} equals the maximum real part of any of poles of $X(s)$.

Practice Problem 7.1

Find the Laplace transform $X(s)$ and sketch the pole-zero plot with the ROC for the following signals

(i) $x(t) = e^{-3t}u(t) + e^{-4t}u(t)$
(ii) $x(t) = e^{-4t}u(t) + e^{3t}u(-t)$
(iii) $x(t) = e^{-3t}u(t) + e^{-4t}u(-t)$

(i)**Ans:** Poles at $s = -3; -4$ $X(s) = \dfrac{2s+7}{(s+3)(s+4)}$ ROC : $Re(s) > -3$

(ii) **Ans:** Poles at $s = 3; -4$ $X(s) = \dfrac{-7}{(s+4)(s-3)}$ ROC $-4 < Re(s) < 3$.

(iii) **Ans:** Poles at $s = -3; 4$ $X(s)$ does not exist since ROC do not overlap.

7.4 The Unilateral Laplace Transform

The unilateral Laplace transform is applied to the signals that are causal; for the signals that is zero for time $t < 0$.

The unilateral Laplace transform of a signal $x(t)$ is defined by

$$X(s) = \int_0^\infty x(t)e^{-st}dt \tag{7.19}$$

$$= \int_0^\infty [x(t)e^{-\sigma t}]e^{-j\Omega t}dt \tag{7.20}$$

Because $|e^{j\Omega t}| = 1$, the integral converges if

$$\int_0^\infty |x(t)e^{-\sigma t}|dt < \infty. \tag{7.21}$$

Solved Problem 7.6 Find the unilateral Laplace transform of the following signals

(i) $x(t) = \delta(t)$ (ii) $x(t) = u(t)$

i) $X(s) = \int_0^\infty x(t)e^{-st}dt$

$$= \int_0^\infty \delta(t)e^{-st}dt = 1$$

$$\Rightarrow \boxed{L[\delta(t)] = 1} \tag{7.22}$$

ii) $x(t) = u(t)$

$$X(s) = \int_0^\infty x(t)e^{-st}dt = \int_0^\infty e^{-st}dt$$

$$= \frac{-1}{s}e^{-st}\Big|_0^\infty = \frac{1}{s}$$

$$\boxed{L[u(t)] = \frac{1}{s}} \tag{7.23}$$

Solved Problem 7.7 Find the unilateral Laplace transform of

(i) $x(t) = \cos\Omega_0 t$ (ii) $x(t) = \sin\Omega_0 t$

Solution:

(i) Given $x(t) = \cos\Omega_0 t = \left[\frac{e^{j\Omega_0 t} + e^{-j\Omega_0 t}}{2}\right]$

$$X(s) = \int_0^\infty x(t)e^{-st}dt = \int_0^\infty \left[\frac{e^{j\Omega_0 t} + e^{-j\Omega_0 t}}{2}\right]e^{-st}dt$$

$$= 0.5\left[\int_0^\infty e^{j\Omega_0 t}e^{-st}dt + \int_0^\infty e^{-j\Omega_0 t}e^{-st}dt\right]$$

$$= 0.5\left[\int_0^\infty e^{-(s-j\Omega_0)t}dt + \int_0^\infty e^{-(s+j\Omega_0)t}dt\right]$$

$$= 0.5\left[\frac{1}{s-j\Omega_0} + \frac{1}{s+j\Omega_0}\right]$$

$$= \frac{1}{2}\left[\frac{2s}{s^2+\Omega_0^2}\right] = \frac{s}{s^2+\Omega_0^2}$$

$$\boxed{L[\cos(\Omega_0 t)] = \frac{s}{s^2+\Omega_0^2}} \tag{7.24}$$

(ii) Given $x(t) = \sin\Omega_0 t$

$$= \frac{1}{2j}\left[e^{j\Omega_0 t} - e^{-j\Omega_0 t}\right]$$

$$L[x(t)] = L\left\{\frac{1}{2j}\left[e^{j\Omega_0 t} - e^{-j\Omega_0 t}\right]\right\}$$

$$= \frac{1}{2j}\left\{\int_0^\infty e^{j\Omega_0 t}e^{-st}dt - \int_0^\infty e^{-j\Omega_0 t}e^{-st}dt\right\}$$

$$= \frac{1}{2j}\left\{\frac{e^{-(s-j\Omega_0)t}}{-(s-j\Omega_0)}\Big|_0^\infty - \frac{e^{-(s+j\Omega_0)t}}{-(s+j\Omega_0)}\Big|_0^\infty\right\}$$

$$= \frac{1}{2j}\left[\frac{1}{s-j\Omega_0} - \frac{1}{s+j\Omega_0}\right] = \frac{1}{2j}\left[\frac{2j\Omega_0}{s^2+\Omega_0^2}\right] = \frac{\Omega_0}{s^2+\Omega_0^2}$$

$$\boxed{L[\sin\ \Omega_0 t] = \frac{\Omega_0}{s^2+\Omega_0^2}}$$

Practice Problem 7.2

Find the unilateral Laplace transform of the signal

$$x(t) = \sin^2 \Omega_0 t$$

Ans: $\frac{1}{2}\left[\frac{1}{s} - \frac{s}{s^2+4\Omega_0^2}\right]$

7.5 Properties of unilateral Laplace transform

7.5.1 Linearity

If $x_1(t)$ and $x_2(t)$ are two signals with Laplace transform $X_1(s)$ and $X_2(s)$, then

$$L[ax_1(t) + bx_2(t)] = aX_1(s) + bX_2(s) \tag{7.25}$$

where a and b are constants.

Proof

$$L[ax_1(t) + bx_2(t)] = \int_0^\infty [ax_1(t) + bx_2(t)]e^{-st}dt$$

$$= a\int_0^\infty x_1(t)e^{-st}dt + b\int_0^\infty x_2(t)e^{-st}dt$$

$$= aX_1(s) + bX_2(s)$$

Solved Problem 7.8 Find the Laplace transform of the signal

$$x(t) = 2e^{-2t}u(t) + 4e^{-4t}u(t)$$

Solution: Given $x(t) = 2e^{-2t}u(t) + 4e^{-4t}u(t)$

$$X(s) = L[x(t)] = L[2e^{-2t}u(t) + 4e^{-4t}u(t)]$$

Using linearity property

$$X(s) = 2L[e^{-2t}u(t)] + 4L[e^{-4t}u(t)]$$

$$= \frac{2}{s+2} + \frac{4}{s+4} \qquad \text{ROC}: Re(s) > -2$$

7.5.2 Transform of derivatives

$$\text{If} \quad L[x(t)] = X(s)$$

$$\text{then} \quad L\left[\frac{d}{dt}x(t)\right] = sX(s) - x(0^-) \tag{7.26}$$

Proof

From the definition of Laplace transform, we can write

$$L\left[\frac{d}{dt}x(t)\right] = \int_0^\infty \frac{d}{dt}x(t)e^{-st}dt \tag{7.27}$$

This equation may be integrated by parts by letting

$$u = e^{-st} \quad \text{and} \quad dv = dx(t)$$

$$\Rightarrow du = -se^{-st}dt \quad \text{and} \quad v = x(t)$$

$$\boxed{\int udv = vu - \int vdu}$$

$$L\left[\frac{d}{dt}x(t)\right] = e^{-st}x(t)\Big|_0^\infty - \int_0^\infty x(t)(-se^{-st})dt$$

$$= e^{-st}x(t)\Big|_0^\infty + s\int_0^\infty x(t)e^{-st}dt$$

$$= -x(0^-) + sX(s)$$

$$\Rightarrow \quad L\left[\frac{d}{dt}x(t)\right] = sX(s) - x(0^-) \tag{7.28}$$

Since $X(s)$ exists, $\lim_{t\to\infty} x(t)e^{-st} = 0$.

To find the transform of the second derivative, let us write

$$\frac{d^2x(t)}{dt^2} = \frac{d}{dt}\left[\frac{d}{dt}x(t)\right]$$

$$L\left[\frac{d^2x(t)}{dt^2}\right] = sL\left[\frac{d}{dt}x(t)\right] - \frac{dx}{dt}(0^-)$$
$$= s[sX(s) - x(0^-)] - \frac{dx}{dt}(0^-)$$
$$= s^2X(s) - sx(0^-) - \frac{dx}{dt}(0^-) \quad (7.29)$$

In the above expression the quantity $\frac{dx}{dt}(0^-)$ is the derivative of $x(t)$ evaluated at $t = 0^-$.

Similarly

$$L\left[\frac{d^nx(t)}{dt^n}\right] = s^nX(s) - s^{n-1}x(0^-) - s^{n-2}\frac{dx}{dt}(0^-) - \ldots - \frac{d^{n-1}x(0^-)}{dt^{n-1}} \quad (7.30)$$

7.5.3 Transform of the integrals

The Laplace transform of integral of $x(t)$ is given by

$$L\left[\int_{-\infty}^{t} x(\tau)d\tau\right] = \frac{X(s)}{s} + \frac{\int_{-\infty}^{0^-} x(\tau)d\tau}{s} \quad (7.31)$$

Proof.

$$\int_{-\infty}^{t} x(\tau)d\tau = \int_{-\infty}^{0^-} x(\tau)d\tau + \int_{0^-}^{t} x(\tau)d\tau \quad (7.32)$$

$$L\left[\int_{-\infty}^{t} x(\tau)d\tau\right] = L\left[\int_{-\infty}^{0^-} x(\tau)d\tau + \int_{0^-}^{t} x(\tau)d\tau\right] \quad (7.33)$$

$$L\left[\int_{0^-}^{t} x(\tau)d\tau\right] = \int_{0}^{\infty}\left[\int_{0^-}^{t} x(\tau)d\tau\right]e^{-st}dt$$

The integration may be carried out by parts $\boxed{\int udv = vu - \int vdu}$

where $u = \int_{0}^{t} x(\tau)d\tau \quad \Rightarrow du = x(t)$

and $dv = e^{-st}dt \Rightarrow v = \frac{-1}{s}e^{-st}$.

Hence

$$L\left[\int_{0^-}^{t} x(\tau)d\tau\right] = \frac{-1}{s}e^{-st}\int_{0^-}^{\tau} x(\tau)d\tau\Big|_{0}^{\infty} + \frac{1}{s}\underbrace{\int_{0^-}^{\infty} x(t)e^{-st}dt}_{X(s)} \quad (7.34)$$

The first term vanishes since e^{-st} approaches zero as t tends to infinity and at the lower limit

$$\int_{0^-}^{t} x(\tau)d\tau = 0$$

Then $L\left[\int_{0^-}^{t} x(\tau)d\tau\right] = \frac{X(s)}{s}$

When the first term in Eq. (7.32) is a constant then the Laplace transform of $\int_{-\infty}^{0^-} x(\tau)d\tau$ can be written as $\frac{\int_{-\infty}^{0^-} x(\tau)d\tau)}{s}$.

Therefore

$$L\left[\int_{-\infty}^{t} x(\tau)d\tau\right] = \frac{X(s)}{s} + \frac{\int_{-\infty}^{0^-} x(\tau)d\tau)}{s} \quad (7.35)$$

7.5.4 Scaling property

If $L[x(t)] = X(s)$

then $L[x(at)] = \frac{1}{|a|}X\left(\frac{s}{a}\right)$ (7.36)

Solved Problem 7.9 Given $x(t) = e^{-t}u(t)$, find inverse Laplace transform of $e^{-3s}X(2s)$

JNTU Nov 2004

Solution: We know $L[e^{-t}u(t)] = \frac{1}{s+1}$

Using time scaling property we have

$$L[x(at)] = \frac{1}{|a|}X\left(\frac{s}{a}\right)$$

$$\text{If } X(s) = \frac{1}{s+1}$$

$$X(2s) = \frac{1}{\frac{s}{2}+1} = \frac{2}{s+2} \Rightarrow 2X(2s) = \frac{4}{s+2}$$

$$L^{-1}[2X(2s)] = x\left(\frac{t}{2}\right) \qquad (\because a = \frac{1}{2})$$

$$L^{-1}[X(2s)] = \frac{1}{2}x\left(\frac{t}{2}\right)$$

Using time shifting property we have

$$L[x(t-t_0)] = e^{-st_0}X(s)$$

$$\Rightarrow L^{-1}[e^{-st_0}X(s)] = x(t-t_0)$$

$$\Rightarrow L^{-1}[e^{-3s}X(2s)] = \frac{1}{2}x\left(\frac{t-3}{2}\right) = e^{-\left(\frac{t-3}{2}\right)}u\left(\frac{t-3}{2}\right)$$

Practice Problem 7.3

Find Laplace transform of the signal $x(t) = e^{-t/2}\cos\left(\frac{\Omega_0 t}{2}\right)u(t)$

7.5.5 Time shift

$$\text{If} \quad L[x(t)] = X(s)$$

$$\text{then} \quad L[x(t-t_0)] = e^{-st_0}X(s) \tag{7.37}$$

A time shift of t_0 corresponds to multiplication by complex exponential e^{-st_0}.

Proof

$$L[x(t-t_0)] = \int_0^\infty x(t-t_0)e^{-st}dt$$

Let $t - t_0 = p$, then $dt = dp$ and

$$L[x(t-t_0)] = \int_{-t_0}^\infty x(p)e^{-(t_0+p)s}dp$$

$$= e^{-st_0}\int_0^\infty x(p)e^{-sp}dp$$

$$= e^{-st_0}X(s)$$

Solved Problem 7.10 Find the Laplace transform of $x(t) = e^{-5t}[u(t) - u(t-5)]$ and its ROC. JNTU Nov 2004

Solution:

Given

$$x(t) = e^{-5t}[u(t) - u(t-5)]$$

$$= e^{-5t}u(t) - e^{-5t}u(t-5)$$

$$= e^{-5t}u(t) - e^{-25}e^{-5(t-5)}u(t-5)$$

We know $L[e^{-at}u(t)] = \frac{1}{s+a}$ ROC : $Re(s) > -a$

$$L[e^{-5t}u(t)] = \frac{1}{s+5}, \text{ ROC : } Re(s) > -5$$

Using time shifting property we have

$$L[x(t-t_0)] = e^{-st_0}X(s)$$

$$\text{Hence } L[e^{-5(t-5)}u(t-5)] = e^{-5s}\frac{1}{s+5} \quad Re(s) > -5$$

$$\Rightarrow L[x(t)] = \frac{1}{s+5} - e^{-25}\frac{e^{-25}}{s+5}$$

$$= \frac{1-e^{-5(s+5)}}{s+5} \quad \text{ROC; } Re(s) > -5$$

Practice Problem 7.4

Find Laplace transform of the signal $x(t) = e^{-2t}u(t-2)$.

7.5.6 Frequency shift

$$\text{If} \quad L[x(t)] = X(s)$$

$$\text{then} \quad L\left[e^{-at}x(t)\right] = X(s+a) \tag{7.38}$$

Proof

$$L[e^{-at}x(t)] = \int_0^\infty e^{-at}x(t)e^{-st}dt$$

$$= \int_0^\infty x(t)e^{-(a+s)t}dt = X(s+a)$$

Similarly

$$L[e^{at}x(t)] = X(s-a) \tag{7.39}$$

Multiplication by a complex exponential in time domain introduces a shift in complex frequency s in frequency domain.

Solved Problem 7.11 Find Laplace transform of the signal $x(t) = e^{-at}\sin\Omega_0 t u(t)$ **AU May 2004**

Solution: We know that the Laplace transform of

$$\sin\Omega_0 t u(t) = \frac{\Omega_0}{s^2+\Omega_0^2}$$

Using frequency shifting property

$$L[e^{-at}\sin\Omega_0 t u(t)] = \frac{\Omega_0}{(s+a)^2+\Omega_0^2}$$

Solved Problem 7.12 Find the Laplace transform of $[4e^{-2t}\cos 5t - 3e^{-2t}\sin 5t]u(t)$ and its ROC. **JNTU 2005**

Solution:

$$\text{We have } L[\cos\Omega_0 t u(t)] = \frac{s}{s^2+\Omega_0^2}$$

$$\text{and} \quad L[\sin\Omega_0 t u(t)] = \frac{\Omega_0}{s^2+\Omega_0^2}$$

Using frequency shifting property

$$L\left[e^{-at}\cos\Omega_0 t u(t)\right] = \frac{s+a}{(s+a)^2+\Omega_0^2} \quad \text{ROC}: Re(s) > -a$$

$$\text{and} \quad L\left[e^{-at}\sin\Omega_0 t u(t)\right] = \frac{\Omega_0}{(s+a)^2+\Omega_0^2} \quad \text{ROC}: Re(s) > -a$$

we obtain $L[4e^{-2t}\cos 5t - 3e^{-2t}\sin 5t]u(t)$

$$= \frac{4(s+2)}{(s+2)^2+25} - \frac{15}{(s+2)^2+25}$$

Practice Problem 7.5

Find Laplace transform of the signal and its ROC. JNTU May 2004

$$x(t) = e^{-t}\sin(\Omega_0 t + \varphi)u(t)$$

7.5.7 Differentiation in the s-domain

$$\text{If} \quad L[x(t)] = X(s)$$

$$\text{then} \quad L[-tx(t)] = \frac{d}{ds}X(s) \tag{7.40}$$

Proof

$$X(s) = \int_0^\infty x(t)e^{-st}dt$$

Differentiate with respect to s on both sides

$$\frac{d}{ds}X(s) = \int_0^\infty x(t)\frac{d}{ds}(e^{-st})dt$$

$$= \int_0^\infty x(t)(-t)e^{-st}dt$$

$$= \int_0^\infty -tx(t)e^{-st}dt$$

$$= L[-tx(t)]$$

$$\Rightarrow \quad L[-tx(t)] = \frac{dX(s)}{ds}$$

In the same way

$$L[(-t)^n x(t)] = \frac{d^n X(s)}{ds^n} \tag{7.41}$$

Solved Problem 7.13 Find Laplace transform of the signal $x(t) = -te^{-2t}u(t)$ AU Nov 2006

Solution:

$$\text{We have } L[e^{-at}u(t)] = \frac{1}{s+a}$$

$$\Rightarrow L[e^{-2t}u(t)] = \frac{1}{s+2}$$

Using differentiation in s-domain property

$$L[-tx(t)] = \frac{dX(s)}{ds}$$

$$\Rightarrow L[-te^{-2t}u(t)] = \frac{d}{ds}\left[\frac{1}{s+2}\right]$$

$$= \frac{-1}{(s+2)^2}$$

Practice Problem 7.6

Find the Laplace transform of the signal $x(t) = t^2 \cos\Omega_0 t u(t)$

Solved Problem 7.14 Find Laplace transform of $x(t) = e^{-t}\cos(\Omega_0 t + \varphi)u(t)$ and its ROC. JNTU May 2004

Solution:

Given $x(t) = e^{-t}\cos(\Omega_0 t + \varphi)u(t)$

$$= e^{-t}[\cos\Omega_0 t \cos\varphi - \sin\Omega_0 t \sin\varphi]u(t)$$

$$L[x(t)] = L[(e^{-t}\cos\Omega_0 t\cos\varphi - e^{-t}\sin\Omega_0 t \sin\phi)u(t)]$$

$$= \cos\varphi\ L\ [e^{-t}\cos\Omega_0 t u(t)] - \sin\varphi L[e^{-t}\sin\Omega_0 t u(t)]$$

$$= \frac{\cos\varphi(s+1)}{(s+1)^2+\Omega_0^2} - \frac{\Omega_0\sin\varphi}{(s+1)^2+\Omega_0^2} \qquad \text{ROC}: Re(s) > -1$$

7.5.8 Time convolution

$$\text{If} \quad L[x_1(t)] = X_1(s)$$

and

$$L[x_2(t)] = X_2(s)$$

then

$$L[x_1(t) * x_2(t)] = X_1(s)X_2(s) \tag{7.42}$$

Proof

The signals $x_1(t)$ and $x_2(t)$ are causal.

$$\text{Therefore } x_1(t) * x_2(t) = \int_0^t x_1(\tau)x_2(t-\tau)d\tau$$

$$L[x_1(t) * x_2(t)] = \int_0^\infty \int_0^\infty x_1(\tau)x_2(t-\tau)u(t-\tau)e^{-st}d\tau dt$$

$$u(t-\tau) = 1 \text{ for } t \geq \tau$$
$$= 0 \text{ for } t < \tau$$

Note:

Here we used step function to change the limit from (0 to t) to(0 to ∞).

Let $t - \tau = p$, then $dt = dp$ and

$$L[x_1(t) * x_2(t)] = \int_0^\infty x_1(\tau)\int_0^\infty x_2(p)u(p)e^{-(p+\tau)s}d\tau\, dp$$

$$= \int_0^\infty x_1(\tau)\int_0^\infty x_2(p)u(p)e^{-sp}e^{-s\tau}dp\, d\tau$$

$$= \int_0^\infty x_1(\tau)X_2(s)e^{-s\tau}d\tau$$

$$= X_2(s)\int_0^\infty x_1(\tau)e^{-s\tau}d\tau$$

$$= X_2(s)X_1(s)$$

Thus the Laplace transform of the convolution of two signals is the product of their respective Laplace transforms.

Solved Problem 7.15 Given $x_1(t) = e^{-2t}u(t)$ and $x_2(t) = e^{-3t}u(t)$. Determine $Y(s)$ where $y(t) = x_1(t-2)^* x_2(-t+3)$ AU Dec 2006

Solution: Given

$$x_1(t) = e^{-2t}u(t) \text{ and } x_2(t) = e^{-3t}u(t)$$

$$X_1(s) = L[e^{-2t}u(t)] = \frac{1}{s+2} \text{ and } X_2(s) = L[e^{-3t}u(t)] = \frac{1}{s+3}$$

Let $x(t) = x_1(t-2)$ and $h(t) = x_2(-t+3)$, then

$$y(t) = x(t) * h(t)$$

Using time convolution property $Y(s) = X(s)H(s)$

$$X(s) = L[x_1(t-2)]$$

Using time shifting property

$$L[x_1(t-2)] = e^{-2s}X_1(s)$$
$$= e^{-2s}\left(\frac{1}{s+2}\right)$$

$H(s) = L[x_2(-t+3)]$

The signal $x_2(-t+3)$ can be obtained by time reversal of $x(t)$ and then shifting it right by 3 Units

Using time reversal property

$$L[x_2(-t)] = X_2(-s) = \frac{1}{-s+3}$$

Using time shifting property

$$H(s) = L[x_2(-t+3)] = e^{-3s}X_2(-s) = \frac{e^{-3s}}{3-s}$$
$$Y(s) = X(s)H(s) = \left(\frac{e^{-2s}}{s+2}\right)\left(\frac{e^{-3s}}{3-s}\right)$$

7.5.9 Frequency convolution

$$L[x_1(t)x_2(t)] = \frac{1}{2\pi j}\int_{c-j\infty}^{c+j\infty} X_1(u)X_2(s-u)du \qquad (7.43)$$

7.5.10 Initial value theorem

The initial value theorem allows us to calculate initial value of $x(t)$ i.e., $x(0)$ directly from the transform $X(s)$ without the need for finding the inverse transform of $X(s)$.

We know

$$L\left[\frac{d}{dt}x(t)\right] = \int_0^\infty \frac{d}{dt}x(t)e^{-st}dt$$
$$= sX(s) - x(0^-) \qquad (7.44)$$

If $x(t)$ is continuous at the origin (That is at $t = 0$), $\frac{dx}{dt}$ does not contain an impulse term at $t = 0$ and as $s \to \infty$ the integral goes to zero. Thus

$$\lim_{s\to\infty} sX(s) - x(0) = 0$$
$$\Rightarrow \lim_{s\to\infty} sX(s) = x(0)$$

If $x(t)$ is discontinuous at the origin, then $\frac{dx}{dt}$ contains an impulse term, namely $[x(0^+) - x(0^-)]\delta(t)$ and

$$\lim_{s\to\infty}\int_0^\infty \frac{dx}{dt}e^{-st}dt = x(0^+) - x(0^-)$$
$$\lim_{s\to\infty} sX(s) - x(0^-) = x(0^+) - x(0^-)$$

Therefore

$$\lim_{s\to\infty} sX(s) = x(0^+)$$
$$\Rightarrow \boxed{x(0^+) = \lim_{s\to\infty} sX(s)} \qquad (7.45)$$

7.5.11 Final value theorem

The value of $x(t)$ as t tends to infinity may also be found directly from the Laplace transform $X(s)$.

From Eq. (7.44) we obtain

$$\lim_{s\to 0}\int_0^\infty \frac{dx}{dt}e^{-st}dt = \lim_{s\to 0}[sX(s) - x(0^-)] \qquad (7.46)$$

If $sX(s)$ has all its singularities in the left half of the s-plane, then $\lim_{s\to 0} X(s)$ exists. Then Eq. (7.46) can be written as

$$\int_0^\infty \frac{dx}{dt}dt = \lim_{s\to 0}[sX(s) - x(0^-)]$$

Interchanging and applying limit on the left hand side, we have

$$\lim_{t\to\infty}[x(t) - x(0^-)] = \lim_{s\to 0}[sX(s) - x(0^-)]$$
$$\Rightarrow \lim_{t\to\infty} x(t) = \lim_{s\to 0} sX(s) \qquad (7.47)$$

A simple pole in $X(s)$ at the origin is permitted, and all other poles of $X(s)$ must be strictly in the left half of the s-plane. To apply final value theorem we must check whether any common factors are there between the numerator and denominator of $sX(s)$. If yes, we cancel the common factors and then check for the poles of the $sX(s)$. If any pole lies in the right half of s-plane then the final value theorem does not hold.

Solved Problem 7.16 Determine the initial value $x(0^+)$ for the following Laplace transforms

(i) $X(s) = \dfrac{3}{s^2+5s-1}$

(ii) $X(s) = \dfrac{2s+3}{s(s^2+5s+6)}$

Solution:

i) $x(0^+) = \lim\limits_{s\to\infty} sX(s)$

$$= \lim_{s\to\infty} s\left[\frac{3}{s^2+5s-1}\right] = \lim_{x\to 0}\frac{3x}{1+5x-x^2} = 0 \qquad \left(s=\frac{1}{x}\right)$$

ii) $x(0^+) = \lim\limits_{s\to\infty} sX(s)$

$$= \lim_{s\to\infty} s\left[\frac{2s+3}{s(s^2+5s+6)}\right]$$

$$= \lim_{s\to\infty}\frac{2s+3}{s^2+5s+6} = \lim_{x\to 0}\frac{2x+3x^2}{6x^2+5x+1} = 0 \qquad \left(s=\frac{1}{x}\right)$$

Solved Problem 7.17 Find $x(\infty)$ if $X(s)$ is given by

(i) $\dfrac{1}{(s-2)}$ (ii) $\dfrac{s-1}{s(s+1)}$ (iii) $\dfrac{1}{s^2+4}$

Solution:

i) The pole lies in right half of the s-plane, therefore $x(\infty) = \infty$

ii) $x(\infty) = \lim\limits_{s\to 0}\dfrac{s(s-1)}{s(s+1)} = -1$

iii) $X(s) = \dfrac{1}{s^2+4} = \dfrac{1}{(s+2j)(s-2j)}$

The poles are lying on the imaginary axis, therefore $x(\infty)$ is not defined.

Solved Problem 7.18 Find the initial and final values, for the following transforms

a) $\dfrac{s+5}{s^2+3s+2}$ (b) $\dfrac{s^2+5s+7}{s^2+3s+2}$

Solution:

a) Initialvalue

$$x(0) = \lim_{s\to\infty} sX(s)$$

$$= \lim_{s\to\infty}\frac{s(s+5)}{s^2+3s+2} = \lim_{s\to\infty}\frac{s^2+5s}{s^2+3s+2}$$

$$= \lim_{x\to 0}\frac{1+5x}{1+3x+2x^2} \qquad \left(s=\frac{1}{x}\right)$$

$$= 1$$

Final value

$$x(\infty) = \lim_{s\to 0} sX(s)$$

$$= \lim_{s\to 0}\frac{s(s+5)}{s^2+3s+2} = 0$$

b) $X(s) = \dfrac{s^2+5s+7}{s^2+3s+2}$

$$x(0) = \lim_{s\to\infty}\frac{s(s^2+5s+7)}{s^2+3s+2}$$

The initial value of the time function does not exist because $\lim\limits_{s\to\infty}\dfrac{s(s^2+5s+7)}{s^2+3s+2}$ does not exist.

The reason is that the inverse Laplace transform of $X(s)$ has the term $\delta(t)$.

The final value $x(\infty) = \lim\limits_{s\to 0}\dfrac{s(s^2+5s+7)}{s^2+3s+2} = 0$

Practice Problem 7.7

Find the initial and final values, for the following transforms

(i) $X(s) = \dfrac{s+3}{s^2+4s}$ (ii) $X(s) = \dfrac{5}{s^3+4s^2+6s+9}$

Solved Problem 7.19 The signal $x(t) = \sin 2t$ oscillates between +1 and -1 as $t \to \infty$. So it does not have a final value. Show that application of final value theorem gives an incorrect result for the signal. **(JNTU Nov 2003, 2005, 2006)**

Solution:

$$\text{Given } x(t) = \sin 2t$$

$$L[\sin 2t] = \frac{2}{s^2+4}$$

Find value theorem states that

$$\lim_{t\to\infty} x(t) = \lim_{s\to 0} sX(s)$$

$$x(\infty) = \lim_{s\to 0} s\left[\frac{2}{s^2+4}\right] = 0$$

The result shows that the final theorem gives an incorrect result.

Review Questions

1. State and prove convolution theorem of Laplace transform. AU 2008, JNTU 2005, AU May 2004
2. State and prove time differentiation and time integration properties of Laplace transform. March 2006, May 2005
3. State and prove time differentiation property of one-sided Laplace transform. March 2006
4. Explain time shifting property of Laplace transform. March 2005
5. State and prove the initial and final value theorems of Laplace transform. Nov 2004
6. State and prove the time scaling property of Laplace transform. AU Nov 2008

Solved Problem 7.20 Find the Laplace transform of the following

(i) $x(t) = u(t-2)$
(ii) $x(t) = t^2 e^{-2t} u(t)$ AU Nov 2009

Solution:

Given $x(t) = u(t-2)$

$$X(s) = \int_0^\infty x(t)e^{-st}dt$$

$$= \int_0^\infty u(t-2)e^{-st}dt$$

$$= \int_2^\infty e^{-st}dt$$

$$= \frac{1}{-s}e^{-st}\Big|_2^\infty = \frac{e^{-2s}}{s}$$

$u(t-2) = 0$ for $t < 2$
$= 1$ for $t \geq 2$

(or) we know that

$$L[u(t)] = \frac{1}{s}.$$

By using time shifting property, we can write

$$L[x(t-t_0)] = e^{-st_0}X(s)$$

Therefore

$$L[u(t-2)] = \frac{e^{-2s}}{s}$$

(ii) $x(t) = t^2e^{-2t}u(t)$

We know that

$$L[e^{-2t}u(t)] = \frac{1}{s+2}$$

Using differentiation in s-domain property, we have

$$L[(-t)^n x(t)] = \frac{d^nX(s)}{ds^n}.$$

For $n = 2$

$$L[t^2x(t)] = \frac{d^2X(s)}{ds^2}$$

$$\Rightarrow \quad L[t^2e^{-2t}u(t)] = \frac{d^2}{ds^2}\left[\frac{1}{s+2}\right]$$

$$= \frac{d}{ds}\left[\frac{-1}{(s+2)^2}\right]$$

$$= \frac{2}{(s+2)^3}$$

Solved Problem 7.21 Given the transform pair

$$L[x(t)] = \frac{2s}{s^2-2}$$

determine the Laplace transform of $x(2t)$.

Solution:

We know that if $L[x(t)] = X(s)$

then $L[x(at)] = \frac{1}{|a|}X\left(\frac{s}{a}\right)$.

Given

$$L[x(t)] = \frac{2s}{s^2-2}$$

By using the time scaling property, we can write

$$L[x(2t)] = \frac{1}{2}\frac{2\left(\frac{s}{2}\right)}{\left(\frac{s}{2}\right)^2-2} = \frac{1}{2}\frac{s}{\frac{s^2}{4}-2}$$
$$= \frac{2s}{s^2-8}$$

Solved Problem 7.22 Find the inverse Laplace transform of

(i) $X(s) = \frac{1}{s(s+2)}$

(ii) $X(s) = \frac{1+e^{-2s}}{3s^2+2s}$ JNTU 2003, 2005, 2006

Solution: Let

$$X_1(s) = \frac{1}{s+2}$$

Then

$$X(s) = \frac{X_1(s)}{s}.$$

$$x_1(t) = L^{-1}[X_1(s)] = L^{-1}\left[\frac{1}{s+2}\right] = e^{-2t}u(t).$$

(i) By using time integral property, we have

$$L\left[\int_{-\infty}^{t} x(\tau)d\tau\right] = \frac{X(s)}{s}$$

$$\Rightarrow L^{-1}\left[\frac{X(s)}{s}\right] = \int_{-\infty}^{t} x(\tau)d\tau$$

$\because \int_{-\infty}^{0^-} x(\tau)d\tau = 0$

$$L^{-1}\left[\frac{1}{s(s+2)}\right] = L^{-1}\left[\frac{X_1(s)}{s}\right] = \int_{-\infty}^{t} x_1(\tau)d\tau$$
$$= \int_{-\infty}^{t} e^{-2\tau}u(\tau)d\tau$$
$$= \int_{0}^{t} e^{-2\tau}d\tau$$
$$= \frac{-1}{2}e^{-2\tau}\Big|_0^t = \frac{1-e^{-2t}}{2}$$

(ii) Given

$$X(s) = \frac{1+e^{-2s}}{3s^2+2s} = \frac{1+e^{-2s}}{s(3s+2)} = \frac{X_1(s)}{s}$$

where $X_1(s) = \frac{1+e^{-2s}}{3s+2}$

$$= \frac{1}{3\left(s+\frac{2}{3}\right)} + \frac{1}{3}\frac{e^{-2s}}{s+\frac{2}{3}}$$

We have $L[e^{-at}u(t)] = \frac{1}{s+a} \Rightarrow L^{-1}\left[\frac{1}{s+\frac{2}{3}}\right] = e^{-\frac{2}{3}t}u(t)$

$$\Rightarrow x_1(t) = \frac{1}{3}e^{-\frac{2}{3}t}u(t) + \frac{1}{3}e^{-\frac{2}{3}(t-2)}u(t-2)$$

$$X(s) = \frac{X_1(s)}{s}$$

$$L^{-1}\left[\frac{X_1(s)}{s}\right] = L^{-1}[X(s)] = \int_{-\infty}^{t} x_1(\tau)d\tau$$

$$= \int_0^{\tau} \frac{1}{3} e^{-\frac{2}{3}t} dt + \int_2^{\infty} \frac{1}{3} e^{\frac{-2}{3}(t-2)} dt$$

$$= \underbrace{\frac{1}{2}[1 - e^{\frac{-2}{3}t}]}_{\text{for } t \geq 0} + \underbrace{\frac{1}{2}[1 - e^{\frac{-2}{3}(t-2)}]}_{\text{for } t > 2}$$

$$x(t) = \frac{1}{2}[1 - e^{-\frac{2}{3}t}]u(t) + \frac{1}{2}[1 - e^{\frac{-2}{3}(t-2)}]u(t-2)$$

Solved Problem 7.23 Find Laplace transform of the signal

(i) $x(t) = \delta(t - t_0)$ (ii) $x(t) = u(t - t_0)$ (iii) $x(t) = \delta(at + b)$

Solution:

(i)

Given $x(t) = \delta(t - t_0)$

$$X(s) = \int_{-\infty}^{\infty} \delta(t - t_0) e^{-st} dt$$

$$= e^{-st_0}$$

(ii)

Given $X(t) = u(t - t_0)$

$$X(s) = \int_{-\infty}^{\infty} u(t - t_0) e^{-st} dt$$

$$= \int_{t_0}^{\infty} e^{-st} dt = \frac{-1}{s} e^{-st} \Big|_{t_0}^{\infty}$$

$$\frac{1}{s}[-e^{-st_0}] = \frac{e^{-st_0}}{s}$$

(iii) We have $L[\delta(t)] = 1$

Using time scaling property we have

$$L[x(at)] = \frac{1}{|a|} X(s/a)$$

Hence $L[\delta(at)] = \frac{1}{|a|}$

We can write $\delta(at + b) = \delta[a(t + b/a)] \Rightarrow L[\delta(at + b)] = \frac{1}{|a|} e^{sb/a}$

Solved Problem 7.24 Determine the Laplace transform of the signal AU Nov 2009

$$x(t) = \sin(\pi t); 0 < t < 1$$
$$= 0 \quad \text{otherwise}$$

Solution:

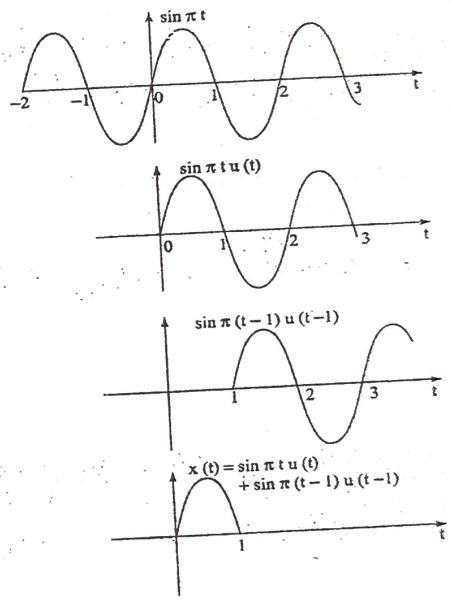

Fig. 7.7

The time period of the signal $\sin \pi t$ is

$$T = \frac{2\pi}{\pi} = 2.$$

From Fig. 7.7, we find that

$$x(t) = \sin \pi t u(t) + \sin \pi (t-1) u(t-1)$$

We know that $L[\sin \pi t u(t)] = \dfrac{\pi}{s^2+\pi^2}$

By using time shifting property, we can write

$$L[\sin \pi (t-1) u(t-1)] = e^{-s} \frac{\pi}{s^2+\pi^2}$$

$$L[\sin \pi t u(t) + \sin \pi (t-1) u(t-1)] = \frac{\pi}{s^2+\pi^2} + e^{-s} \frac{\pi}{s^2+\pi^2}$$

$$= \frac{\pi}{s^2+\pi^2}(1+e^{-s})$$

Solved Problem 7.25 Find the Laplace transform of

$$x(t) = e^{-2t} \sin 2t u(t)$$

Solution: We know that

$$L[\sin 2t u(t)] = \frac{2}{s^2+4}$$

$$\boxed{L[\sin \Omega_0 t u(t)] = \frac{\Omega_0}{s^2+\Omega_0^2}}$$

By using frequency shifting property, we have

$$L[e^{-at} x(t)] = X(s+a)$$

Similarly

$$L[e^{-2t} \sin 2t u(t)] = \frac{2}{(s+2)^2+4}$$

$$= \frac{2}{s^2+4s+8}$$

Solved Problem 7.26 For the following waveforms find the Laplace transform

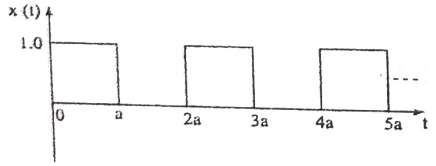

Fig. 7.8

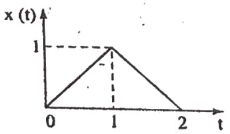

Fig. 7.9

Solution:

From Fig. 7.10 the given waveform can be expressed mathematically as follows

$$x(t) = u(t) - u(t-a) + u(t-2a) - u(t-3a) + u(t-4a) - u(t-5a) + u(t-6a) - u(t-7a) \ldots\ldots$$

$$L[u(t)] = \frac{1}{s}$$

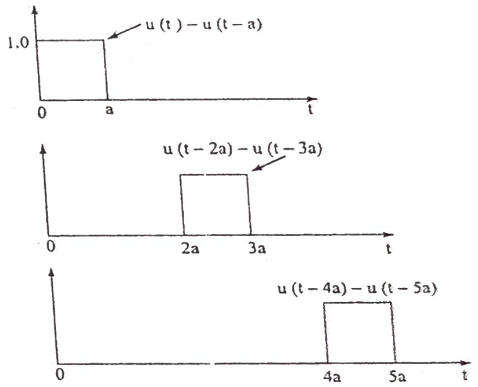

Fig. 7.10

By using time shifting property, we have

$$L[u(t-t_o)] = \frac{e^{-st_o}}{s}.$$

Therefore

$$X(s) = \left[\frac{1}{s} - \frac{e^{-as}}{s} + \frac{e^{-2as}}{s} - \frac{e^{-3as}}{s} + \frac{e^{-4as}}{s} - \frac{e^{-5as}}{s} \ldots\right]$$

$$= \frac{1}{s}\left[(1 + e^{-2as} + e^{-4as} + \ldots) - (e^{-as} + e^{-3as} + \ldots)\right]$$

$$= \frac{1}{s}\left[\frac{1}{1-e^{-2as}} - \frac{e^{-as}}{1-e^{-2as}}\right]$$

$$= \frac{1}{s}\left[\frac{1-e^{-as}}{1-e^{-2as}}\right] = \frac{1}{s}\left[\frac{1}{1+e^{-as}}\right]$$

(ii) Given

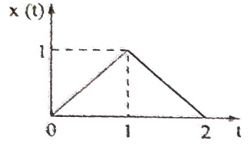

Fig. 7.11

$$x(t) = r(t) - 2r(t-1) + r(t-2)$$

$$r(t) = t \quad \text{for} \quad t \geq 0$$
$$= 0 \quad \text{for} \quad t < 0$$

$$L[r(t)] = \frac{1}{s^2}$$

By using time shifting property

$$L[r(t-1)] = \frac{e^{-s}}{s^2}.$$

Similarly

$$L[r(t-2)] = \frac{e^{-2s}}{s^2}$$

$$\Rightarrow L[x(t)] = L[r(t) - 2r(t-1) + r(t-2)]$$
$$= L[r(t)] - 2L[r(t-1)] + L[r(t-2)]$$
$$= \frac{1}{s^2} - \frac{2e^{-s}}{s^2} + \frac{e^{-2s}}{s^2}$$
$$= \frac{1}{s^2}[1 - 2e^{-s} + e^{-2s}]$$
$$= \frac{1}{s^2}(1 - e^{-s})^2$$
$$= \left[\frac{1 - e^{-s}}{s}\right]^2$$

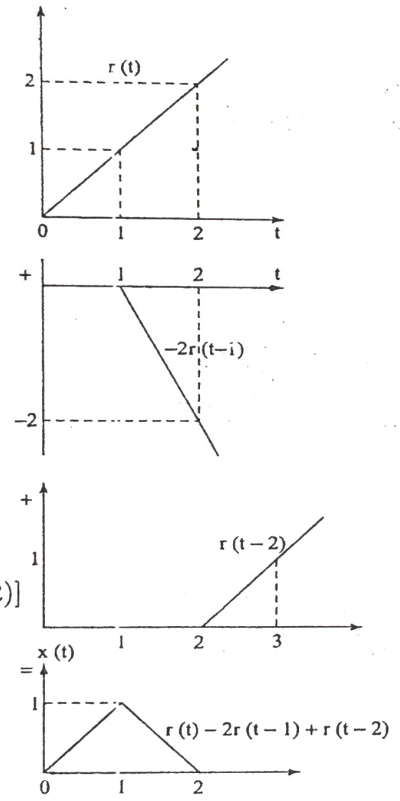

Fig. 7.12

Table 7.1. Properties of Laplace transform

Property	$x(t)$	$X(s)$
	$x_1(t)$	$X_1(s)$
	$x_2(t)$	$X_2(s)$
Linearity	$a_1x_1(t) + a_2x_2(t)$	$a_1X_1(s) + a_2X_2(s)$
Time shifting	$x(t - t_0)$	$e^{-st_0}X(s)$
Frequency shifting	$x(t)e^{j\Omega_0 t}$	$X(s - \Omega_0)$
Time differentiation	$\frac{dx(t)}{dt}$	$sX(s) - x(0^-)$
	$\frac{d^2x(t)}{dt^2}$	$s^2X(s) - sx(0^-) - \frac{dx(0^-)}{dt}$
	$\frac{d^nx(t)}{dt^n}$	$s^nX(s) - s^{n-1}x(0^-) - \ldots\ldots - \frac{d^{n-1}}{dt^{n-1}}x(0^-)$
Time integration	$\int_{0^-}^{t} x(\tau)d\tau$	$\frac{X(s)}{s}$
	$\int_{-\infty}^{t} x(\tau)d\tau$	$\frac{X(s)}{s} + \frac{1}{s}\int_{-\infty}^{0^-} x(\tau)d\tau$
Frequency differentiation	$-tx(t)$	$\frac{dX(s)}{ds}$
Frequency integration	$\frac{x(t)}{t}$	$\int_{s}^{\infty} X(s)ds$
Scaling	$x(at)$	$\frac{1}{\lvert a\rvert}X\left(\frac{s}{a}\right)$
Time convolution	$x_1(t) * x_2(t)$	$X_1(s)X_2(s)$
Frequency convolution	$x_1(t)x_2(t)$	$X_1(s) * X_2(s)$
Initial value theorem	$x(0^-)$	$\lim_{s\to\infty} sX(s)$
Final value theorem	$x(\infty)$	$\lim_{s\to 0} sX(s)$

Table 7.2 Laplace Transform of some standard signals

$x(t)$	$X(s)$
$\delta(t)$	1
1	$\frac{1}{s}$
t	$\frac{1}{s^2}$
t^n	$\frac{n!}{s^{n+1}}$
e^{-at}	$\frac{1}{s+a}$
te^{-at}	$\frac{1}{(s+a)^2}$
$t^n e^{-at}$	$\frac{n!}{(s+a)^{n+1}}$
$\sin \Omega_0 t$	$\frac{\Omega_0}{s^2+\Omega_0^2}$
$\cos \Omega_0 t$	$\frac{s}{s^2+\Omega_0^2}$
$e^{-at} \sin \Omega_0 t$	$\frac{\Omega_0}{(s+a)^2+\Omega_0^2}$
$e^{-at} \cos \Omega_0 t$	$\frac{s+a}{(s+a)^2+\Omega_0^2}$

Solved Problem 7.27 A signal has Laplace transform

$$X(s) = \frac{s+2}{s^2+4s+5}$$

Find Laplace transforms $Y(s)$, of the following signals

(a) $y_1(t) = tx(t)$ (b) $y_2(t) = e^{-t}x(t)$ (c) $y_3(t) = x(t) * x(t)$

Solution:

a) Given $X(s) = \frac{s+2}{s^2+4s+5}$

By using differentiation in s-domain property, we have

$$Y_1(s) = L[y_1(t)] = L[tx(t)] = \frac{-d}{ds}X(s)$$

$$= \frac{-d}{ds}\left[\frac{s+2}{s^2+4s+5}\right]$$

$$= -\left[\frac{(s^2+4s+5)-(s+2)(2s+4)}{(s^2+4s+5)^2}\right]$$

$$= -\left[\frac{(s^2+4s+5-2s^2-8s-8)}{(s^2+4s+5)^2}\right] = \frac{s^2+4s+3}{(s^2+4s+5)^2}$$

b) $y_2(t) = e^{-t}x(t)$

We have frequency shifting property, we know that

$$L[e^{-at}x(t)] = X(s+a)$$

$$L[e^{-t}x(t)] = X(s+1)$$

$$Y_2(s) = \frac{(s+1)+2}{(s+1)^2+4(s+1)+5} = \frac{s+3}{s^2+6s+10}$$

c) By using convolution property, we have

$$L[x(t) * x(t)] = X(s) \cdot X(s)$$

$$= [X(s)]^2$$

$$= \left[\frac{s+2}{s^2+4s+5}\right]^2$$

Solved Problem 7.28 If $x(t)$ is even function, prove that $X(s) = X(-s)$ and if $x(t)$ is odd, prove that $X(s) = -X(-s)$

Solution:

$$X(s) = F[x(t)] = \int_{-\infty}^{\infty} x(t)e^{-st}\,dt$$

$$X(-s) = \int_{-\infty}^{\infty} x(t)e^{st}\,dt$$

Let $t = -p$ then $dt = -dp$

$$X(-s) = \int_{-\infty}^{\infty} x(-p)e^{-sp}\,(dp)$$

$$= \int_{-\infty}^{\infty} x(-p)e^{-sp}\,dp$$

$$= \int_{-\infty}^{\infty} x(-t)e^{-st}\,dt$$

Since $x(t) = x(-t)$ for even

$$X(-s) = \int_{-\infty}^{\infty} x(t)e^{-st}\,dt$$
$$= X(s)$$

We have

$$X(-s) = \int_{-\infty}^{\infty} x(-t)e^{-st}\,dt$$

For odd signal $x(t) = -x(-t)$. Therefore

$$X(-s) = -\int_{-\infty}^{\infty} x(t)e^{-st}\,dt$$
$$= -X(s)$$

Practice Problem 7.8

A signal has Laplace transform

$$X(s) = \frac{2}{s^2+4s+3}$$

Find Laplace transforms of the following signals

(a) $y(t) = x(2t)$ (b) $y_2(t) = t^2x(t)$

7.6 Inversion of Unilateral Laplace Transform

The direct method of finding Laplace transform using Equation

$$x(t) = \frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty} X(s)e^{st}\,ds \qquad (7.48)$$

requires an understanding of complex variable. There are other methods to find inverse Laplace transform that uses one-to-one relationship between a signal and its unilateral Laplace transform. One such method is using partial fraction expansion.

Let

$$X(s) = \frac{N(s)}{D(s)} = \frac{b_M s^M + b_{M-1}s^{M-1} + \ldots + b_1 s + b_0}{s^N + a_{N-1}s^{N-1} + \ldots + a_1 s + a_0} \qquad (7.49)$$

where $M < N$.

To find partial fraction expansion, first we find the roots of the denominator polynomial, say $p_1, p_2 \cdots p_k$. These roots are the poles of $X(s)$. Now we can express $X(s)$ as

$$X(s) = \frac{N(s)}{D(s)} = \frac{b_M s^M + b_{M-1}s^{M-1} + \ldots + b_1 s + b_0}{\prod_{k=1}^{N}(s-p_k)} \qquad (7.50)$$

7.6.1 Distinct poles

If all the poles p_k are distinct, then we may write $X(s)$ as a sum of single pole terms given by

$$X(s) = \frac{C_1}{s-p_1} + \frac{C_2}{s-p_2} + \ldots + \frac{C_i}{s-p_i} + \ldots + \frac{C_k}{s-p_k} \qquad (7.51)$$

Multiply both sides of Eq. (7.51) by $(s-p_i)$ and substitute $s = p_i$

$$(s-p_i)X(s)\Big|_{s=p_i} = \Big|(s-p_i)\frac{C_1}{s-p_1} + (s-p_i)\frac{C_2}{s-p_2} + \ldots + (s-p_i)\frac{C_i}{(s-p_i)}$$
$$+ \ldots + (s-p_i)\frac{C_k}{(s-p_k)}\Big|_{s=p_i}$$
$$= C_i$$

That is, the coefficients C_i in Eq. (7.51) can be found by using

$$C_i = (s-p_i)X(s)\Big|_{s=p_i} \qquad (7.52)$$

For example if

$$X(s) = \frac{2}{(s-1)(s-2)}$$

then we can write

$$X(s) = \frac{C_1}{s-1} + \frac{C_2}{s-2}$$

where

$$C_1 = (s-1)X(s)\Big|_{s=1} = (s-1)\frac{2}{(s-1)(s-2)}\Big|_{s=1} = -2$$

$$C_2 = (s-2)X(s)\Big|_{s=2} = (s-2)\frac{2}{(s-1)(s-2)}\Big|_{s=2} = 2$$

Therefore

$$X(s) = \frac{-2}{s-1} + \frac{2}{s-2}$$

7.6.2 Multiple poles

In the case of multiple roots say p repeats ℓ times, then the expansion of Eq. (7.50) must include terms

$$X(s) = \frac{C_1'}{(s-p)^{\ell}} + \frac{C_2'}{(s-p)^{\ell-1}} + \frac{C_3'}{(s-p)^{\ell-2}} + \cdots \frac{C_{\ell}'}{s-p} \tag{7.53}$$

where $C_1' = (s-p)^{\ell} \times (s)\big|_{s=p}$; $C_2' = \frac{1}{1!}\frac{d}{ds}(s-p)^{\ell} \times (s)\Big|_{s=p}$

$$C_3' = \frac{1}{2!}\frac{d^2}{ds^2}(s-p)^{\ell} \times (s)\Big|_{s=p}$$

Let $X(s) = \dfrac{2s+1}{(s+2)^3}$

$$= \frac{C_1'}{(s+2)^3} + \frac{C_2'}{(s+2)^2} + \frac{C_3'}{(s+2)} \tag{7.54}$$

By using Eq. (7.54), we have

$$C_1' = (s+2)^3 X(s)\Big|_{s=-2}$$

$$= \cancel{(s+2^3)} \quad \frac{(2s+1)}{\cancel{(s+2^3)}}\Big|_{s=-2} = 2(-2)+1 = -3$$

$$C_2' = \frac{d}{ds}\left[(s+2)^3 X(s)\right]\Big|_{s=-2}$$

$$= \frac{d}{ds}\left[\quad \cancel{(s+2^3)} \quad \frac{2s+1}{\cancel{(s+2^3)}} \right]\Big|_{s=-2} = 2$$

$$C_3' = \frac{1}{(3-1)!}\frac{d^2}{ds^2}\left[\quad \cancel{(s+2^3)} \quad \frac{(2s+1)}{\cancel{(s+2^3)}} \right]\Big|_{s=-2}$$

$$= \frac{1}{2!}\left[\frac{d^2}{ds^2}(2s+1)\right]\Big|_{s=-2} = 0$$

$$\Rightarrow X(s) = \frac{2}{(s+2)^2} - \frac{3}{(s+2)^3}$$

7.6.3 Complex roots

If $X(s)$ has complex poles then partial fraction of the $X(s)$ can be expressed as

$$X(s) = \frac{C_1}{s-p_1} + \frac{C_1^*}{s-p_1^*} \tag{7.55}$$

where C_1^* is complex conjugate of C_1 and p_1^* is complex conjugate of p_1. In otherwords, complex conjugate poles result in complex conjugate coefficients.

Let $X(s) = \dfrac{s}{s^2+2s+2}$

$$= \frac{C_1}{s+1+j1} + \frac{C_2}{s+1-j1}$$

Roots of s^2+2s+2

$s_{1,2} = s^2+2s+2 = 0 \; \frac{-2\pm\sqrt{-4}}{2} = -1 \pm j1$

$$C_1 = (s+1+j1)X(s)\Big|_{s=-1-j1}$$

$$= (s+1+j1)\frac{s}{(s+1+j1)(s+1-j1)}\Big|_{s=-1-j1} = \frac{-1-j1}{-j2}$$

$$= \frac{1+j1}{j2} = 0.5 - j0.5$$

$$C_2 = (s+1-j1)\frac{s}{(s+1+j1)(s+1-j1)}\Big|_{s=-1+j1} = \frac{-1+j1}{j2} = 0.5 + j0.5$$

$$\Rightarrow \quad C_2 = C_1^*$$

Therefore complex conjugate poles results in complex conjugate coefficients.

Alternative method

The above procedure of finding partial fraction involves manipulation of complex numbers. When the denominator has complex roots, the complex pair can be expressed as a single quadratic factor instead of having first - order partial fractions.

That is, if $s=-a\pm jb$, we can retain the quadratic factor

$$\begin{aligned}D(s)&=(s+a+jb)(s+a-jb)\\&=(s+a)^2+b^2\end{aligned}$$

Let us consider

$$X(s)=\frac{N(s)}{D(s)}=\frac{2s+3}{(s+2)(s^2+4s+8)}$$

The roots of the denominator $D(s)$ are $-2;-2\pm j2$.

Instead of expressing $X(s)$ with first - order partial fractions as

$$X(s)=\frac{A}{s+2}+\frac{B}{s+2+j2}+\frac{B^*}{s+2-j2}$$

we can retain the quadratic factor s^2+4s+8 and express $X(s)$ as

$$\begin{aligned}X(s)&=\frac{2s+3}{(s+2)(s^2+4s+8)}=\frac{A}{s+2}+\frac{Bs+C}{s^2+4s+8}\\&=\frac{A(s^2+4s+8)+(s+2)(Bs+C)}{(s+2)(s^2+4s+8)}\end{aligned}$$

Equating numerator we get

$$\begin{aligned}2s+3&=A(s^2+4s+8)+(s+2)(Bs+C)\\&=(A+B)s^2+(4A+2B+C)s+8A+2C\end{aligned}$$

Now equating the coefficients of s^2, s and constant yields

$$\begin{aligned}A+B&=0\\4A+2B+C&=2\\8A+2C&=3\end{aligned}$$

$$\Rightarrow A=\frac{-1}{4};B=\frac{1}{4};C=\frac{5}{2}$$

$$X(s)=\frac{-1}{4(s+2)}+\frac{\frac{1}{4}s+\frac{5}{2}}{s^2+4s+8}$$

$$\begin{array}{l}4A+2B+C=2\times 2\\8A+2C=3\\\hline 8A+4B+2C=4\\8A+2C=3\\\hline 4B=1\\B=\frac{1}{4}\\A=\frac{-1}{4}\\C=5/2\end{array}$$

Solved Problem 7.29 Find the inverse Laplace transform of the following

(i) $X(s)=\dfrac{s}{s^2+5s+6}$

(ii) $X(s)=\dfrac{3s^2+8s+6}{(s+2)(s^2+2s+1)}$

(iii) $X(s)=\dfrac{2s+1}{(s+1)(s^2+2s+2)}$

Solution:

(i) $X(s)=\dfrac{s}{s^2+5s+6}$

$$\begin{aligned}&=\frac{s}{(s+2)(s+3)}\\&=\frac{A}{s+2}+\frac{B}{s+3}\end{aligned}$$

$$A=\left.\frac{\cancel{(s+2)}\ s}{\cancel{(s+2)}\ (s+3)}\right|_{s=-2}$$

$$=\frac{(-2)}{(-2+3)}=-2$$

$$B=\left.\frac{\cancel{(s+3)}\ s}{(s+2)\ \cancel{(s+3)}}\right|_{s=-3}$$

$$=\frac{-3}{(-3+2)}=3$$

$$X(s)=\frac{-2}{(s+2)}+\frac{3}{s+3}$$

Taking inverse Laplace transform we get

$$x(t)=-2e^{-2t}u(t)+3e^{-3t}u(t)$$

(ii) Given

$$X(s)=\frac{3s^2+8s+6}{(s+2)(s^2+2s+1)}=\frac{3s^2+8s+6}{(s+2)(s+1)^2}$$

$$=\frac{A}{s+2}+\frac{B}{s+1}+\frac{C}{(s+1)^2}$$

$$A=\cancel{(s+2)}\left.\frac{3s^2+8s+6}{\cancel{(s+2)}(s+1)^2}\right|_{s=-2}$$

$$=\frac{3(-2)^2+8(-2)+6}{(-2+1)^2}$$

$$=\frac{3(4)-16+6}{1}=2$$

$$B=\frac{1}{1!}\frac{d}{ds}\left[\cancel{(s+1)^2}\frac{3s^2+8s+6}{(s+2)\cancel{(s+1)^2}}\right]$$

$$=\left.\frac{(s+2)(6s+8)-(3s^2+8s+6)}{(s+2)^2}\right|_{s=-1}$$

$$=\frac{(-1+2)(-6+8)-(3-8+6)}{(-1+2)^2}$$

$$=1$$

$$C=\cancel{(s+1)^2}\left.\frac{3s^2+8s+6}{\cancel{(s+1)^2}(s+2)}\right|_{s=-1}$$

$$=\frac{3(-1)^2+8(-1)+6}{(-1+2)}=1$$

$$X(s)=\frac{2}{s+2}+\frac{1}{s+1}+\frac{1}{(s+1)^2}$$

$$\boxed{\begin{aligned}L^{-1}\left[\tfrac{1}{s+2}\right]&=e^{-2t}u(t)\\ L^{-1}\left[\tfrac{1}{s+1}\right]&=e^{-t}u(t)\\ L^{-1}\left[\tfrac{1}{(s+2)^2}\right]&=te^{-t}u(t)\end{aligned}}$$

Taking inverse Laplace transform we get

$$x(t)=2e^{-2t}u(t)+e^{-t}u(t)+te^{-t}u(t)$$

(iii) $X(s)=\dfrac{2s+1}{(s+1)(s^2+2s+2)}$

$$=\frac{A}{s+1}+\frac{B}{s-(-1+j1)}+\frac{B^*}{s-(-1-j1)}$$

$\dfrac{-2\pm\sqrt{4-8}}{2}=\dfrac{-2\pm j2}{2}=-1\pm j1$

$$A=(s+1)\left.\frac{(2s+1)}{(s+1)(s^2+2s+2)}\right|_{s=-1}$$

$$=\frac{2(-1)+1}{(-1)^2+2(-1)+2}=\frac{-1}{1}=-1$$

$$B=(s+1-j)\left.\frac{2s+1}{(s+1)(s+1+j)(s+1-j)}\right|_{s=-1+j}$$

$$=\frac{2(-1+j)+1}{(-1+j+1)(-1+j+1+j)}$$

$$=\frac{-1+2j}{j(2j)}=\frac{-1}{2}(-1+2j)=0.5-j$$

$$\Rightarrow X(s)=\frac{-1}{s+1}+\frac{0.5-j}{s-(-1+j1)}+\frac{0.5+j}{s-(-1-j1)}$$

Taking inverse Laplace-transform we get

$$x(t)=-e^{-t}u(t)+(0.5-j)e^{(-1+j1)t}u(t)+(0.5+j)e^{(-1-j1)t}u(t)$$

$$=-e^{-t}u(t)+(0.5-j)e^{-t}e^{jt}u(t)+(0.5+j)e^{-t}e^{-jt}u(t)$$

$$=-e^{-t}u(t)+0.5e^{-t}e^{jt}u(t)-je^{-t}e^{jt}u(t)+0.5e^{-t}e^{-jt}u(t)$$
$$+je^{-t}e^{-jt}u(t)$$

$$=-e^{-t}u(t)+0.5e^{-t}(e^{jt}+e^{-jt})u(t)-je^{-t}(e^{jt}-e^{-jt})u(t)$$

$$=-e^{-t}u(t)+e^{-t}\cos tu(t)+2e^{-t}\sin tu(t)$$

$$=-e^{-t}u(t)+e^{-t}(\cos t+2\sin t)u(t)$$

$$\boxed{\begin{aligned}\frac{e^{jt}+e^{-jt}}{2}&=\cos t\\ e^{jt}-e^{-jt}&=2j\sin t\end{aligned}}$$

(or)

$$X(s)=\frac{2s+1}{(s+1)(s^2+2s+2)}=\frac{A}{s+1}+\frac{Bs+C}{s^2+2s+2}$$

$$\frac{2s+1}{(s+1)(s^2+2s+2)}=\frac{A}{s+1}+\frac{Bs+C}{s^2+2s+2}$$

$$=\frac{A(s^2+2s+2)+(Bs+C)(s+1)}{(s+1)(s^2+2s+2)}$$

$$=\frac{(A+B)s^2+(2A+B+C)s+(2A+C)}{(s+1)(s^2+2s+2)}$$

Comparing the numerators of LHS and RHS we get

$$A+B=0$$
$$2A+B+C=2$$
$$2A+C=1$$
$$\Rightarrow \quad B=1$$
$$A=-1$$
$$C=3$$

$$L^{-1}\left[\frac{s+a}{(s+a)^2+b^2}\right]=e^{-at}\cos bt$$
$$L^{-1}\left[\frac{b}{(s+a)^2+b^2}\right]=e^{-at}\sin bt$$

$$X(s)=\frac{-1}{s+1}+\frac{s+3}{s^2+2s+2}$$

$$=\frac{-1}{s+1}+\frac{s+3}{(s+1)^2+1}$$
$$=\frac{-1}{s+1}+\frac{s+1}{(s+1)^2+1}+\frac{2}{(s+1)^2+1^2}$$

Taking inverse Laplace transform

$$x(t)=-e^{-t}u(t)+e^{-t}\cos tu(t)+2e^{-t}\sin tu(t)$$

Practice Problem 7.9

Find the inverse Laplace transform of the following functions.

(i) $X(s)=\dfrac{s+2}{s(s+1)^2}$

(ii) $X(s)=\dfrac{s+1}{s(s+2)^2(s^2+4s+5)}$

(iii) $X(s)=\dfrac{4s^2+8s+10}{(s+2)(s^2+2s+5)}$

(iv) $X(s)=\dfrac{3s^2+22s+27}{(s^2+3s+2)(s^2+2s+5)}$

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Solved Problem 7.30 Use the convolution theorem of Laplace transform to find $y(t)=x_1(t)*x_2(t)$ where $x_1(t)$ and $x_2(t)$ are given below.

(a) $x_1(t)=e^{-3t}u(t)$ and $x_2(t)=u(t-2)$

(b) $x_1(t)=\cos(4t)u(t)$ and $x_2(t)=\sin(2t)u(t)$

Solution:

From convolution property we have, $L[x_1(t)*x_2(t)]=X_1(s)X_2(s)$

(a) $x_1(t)=e^{-3t}u(t)$

$$X_1(s)=L[e^{-3t}u(t)]=\frac{1}{s+3}$$

$$x_2(t)=u(t-2)$$

$$X_2(s)=L[u(t-2)]=\frac{e^{-2s}}{s}$$

$$L[x_1(t)*x_2(t)]=X_1(s)X_2(s)=\frac{e^{-2s}}{s(s+3)}$$

$$L[u(t)]=\frac{1}{s}$$

Using time shifting property

$$L[u(t-2)]=\frac{e^{-2s}}{s}$$

Let

$$Y(s)=\frac{e^{-2s}}{s(s+3)}$$
$$=e^{-2s}Y_1(s)$$

where

$$Y_1(s)=\frac{1}{s(s+3)}$$
$$Y_1(s)=\frac{A}{s}+\frac{A}{s+3}$$
$$A=s\left.\frac{1}{s(s+3)}\right|_{s=0}=\frac{1}{3}$$
$$B=(s+3)\left.\frac{1}{s(s+3)}\right|_{s=-3}=\frac{-1}{3}$$
$$Y_1(s)=\frac{1}{3s}-\frac{1}{3(s+3)}$$
$$y_1(t)=\left[\frac{1}{3}-\frac{1}{3}e^{-3t}\right]u(t)$$

By using time shifting property we can write

$$y(t)=L^{-1}[e^{-2s}Y_1(s)]=y_1(t-2)$$
$$\Rightarrow \quad y(t)=\frac{1}{3}u(t-2)-\frac{1}{3}e^{-3(t-2)}u(t-2)$$

(b) $x_1(t)=\cos(4t)u(t)$; $X_1(s)=\frac{s}{s^2+16}$

$$x_2(t)=\sin(2t)u(t); X_2(s)=\frac{2}{s^2+4}$$

$$L[\cos(\Omega_0 t)u(t)]=\frac{s}{s^2+\Omega_0^2}$$
$$L[\sin(\Omega_0 t)u(t)]=\frac{\Omega_0}{s^2+\Omega_0^2}$$

$$L[x_1(t)*x_2(t)] = X_1(s)X_2(s)$$
$$= \frac{s}{s^2+16}\cdot\frac{2}{s^2+4}$$
$$= \frac{2s}{(s^2+16)(s^2+4)}$$

Let $Y(s) = \dfrac{2s}{(s^2+16)(s^2+4)}$

$$= \frac{As+B}{s^2+16} + \frac{Cs+D}{s^2+4}$$
$$2s = (As+B)(s^2+4)+(Cs+D)(s^2+16)$$
$$\Rightarrow \quad = As^3+Bs^2+4As+4B+Cs^3+Ds^2+16Cs+16D$$
$$= (A+C)s^3+(B+D)s^2+(4A+16C)s+4B+16D$$

Comparing coefficients of s^3, s^2, s and constant terms we get

$$A+C=0$$
$$B+D=0$$
$$4A+16C=2$$
$$4B+16D=0$$

Solving for A, B, C and D we get $A=\dfrac{-1}{6}$ $B=0$ $C=\dfrac{1}{6}$ and $D=0$

$$\Rightarrow \quad X(s) = \frac{-1}{6}\left[\frac{s}{s^2+16} - \frac{s}{s^2+4}\right]$$
$$x(t) = \frac{-1}{6}[\cos 4t - \cos 2t]u(t)$$

Practice problem 7.10

Use the convolution theorem of Laplace transform to find $y(t) = x_1(t) * x_2(t)$ where $x_1(t)$ and $x_2(t)$ are given below

(i) $x_1(t) = e^{-3t}u(t)$ and $x_2(t) = e^{-2t}u(t)$.

(ii) $x_1(t) = tu(t)$ and $x_2(t) = \delta(t-1) - \delta(t-4)$

7.7 Inversion of the Bilateral Laplace Transform

So far we discussed the inversion of unilateral Laplace transform, which can be used only for causal signals. If $x(t)$ is a non-causal signal then to find $x(t)$ from $X(s)$, we must know the region of convergence of $X(s)$. The location of pole along with ROC determines, whether a given pole corresponds to a positive or negative time portion of $x(t)$. From solved problem (7.1) we find that if the ROC is Re $(s) > a$, then the signal $x(t)$ is a causal signal. On the other hand if the ROC is Re $(s) < a$, then the signal $x(t)$ is a non-causal signal. In other words, if a pole of $X(s)$ lies to the left of the ROC, then this pole gives rise to a causal signal and if a pole of $X(s)$ lies to the right of the ROC, then this poles gives a non-causal signal. For example let us consider a two-sided signal $x(t)$ whole Laplace transform is given by

$$X(s) = \frac{2}{(s+3)(s+2)} \quad \text{ROC: } -3 < \text{Re}(s) < -2$$

The ROC is shown in Fig. 7.13.

Here, the pole $s=-3$ lies to the left of ROC, hence the pole gives rise to a causal signal. The poles $s=-2$ lies to the right of ROC, hence the pole gives rise to a non causal signal. We now expand $X(s)$ in partial fraction to obtain.

$$X(s) = \frac{2}{s+2} - \frac{2}{s+3}$$
$$\text{ROC: } -3 < \text{Re}(s) < -2$$

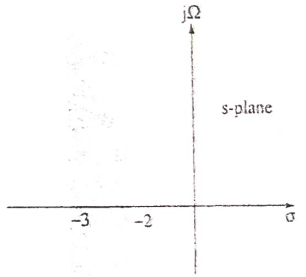

Fig. 7.13

Now

$$L^{-1}\left[\frac{2}{s+2}\right] = -2e^{-2t}u(-t)$$

and

$$L^{-1}\left[\frac{2}{s+3}\right] = 2e^{-3t}u(t)$$

Therefore

$$x(t) = -2e^{-2t}u(-t) - 2e^{-3t}u(t)$$

Solved Problem 7.31 Find the inverse Laplace transform of

$$X(s) = \frac{2}{(s+4)(s-1)}$$ if the region of convergence is

(a) $-4 < \mathrm{Re}(s) < 1$

(b) $\mathrm{Re}(s) > 1$

(c) $\mathrm{Re}(s) < -4$

Solution:

Given

$$X(s) = \frac{2}{(s+4)(s-1)} = \frac{A}{s+4} + \frac{B}{s-1}$$

$$A = \cancel{(s+4)}\left.\frac{2}{\cancel{(s+4)}(s-1)}\right|_{s=-4} = \frac{-2}{5}$$

$$B = \cancel{(s-1)}\left.\frac{2}{(s+4)\cancel{(s-1)}}\right|_{s=1} = \frac{2}{5}$$

$$\Rightarrow X(s) = \frac{-2}{5}\frac{1}{s+4} + \frac{2}{5(s-1)}$$

(a) The $X(s)$ has poles at -4 and 1. The strip of ROC is $-4 < \mathrm{Re}(s) < 1$ as shown in Fig. 7.14. The pole at -4, which is at the left of the strip of ROC, corresponds to the causal signal and the pole at 1 to the right of the strip of ROC corresponds to anticausal signal. Therefore

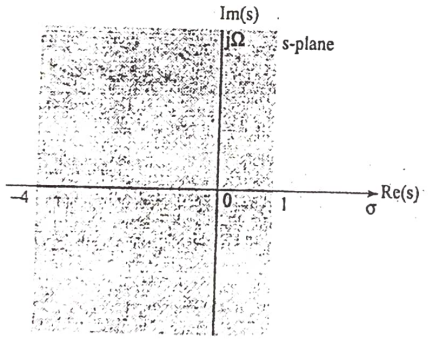

Fig. 7.14

$$x(t) = \frac{-2}{5}e^{-4t}u(t) - \frac{2}{5}e^{t}u(-t)$$

(b) The ROC is $\mathrm{Re}(s) > 1$.

Both poles lies to the left of the ROC, so both poles correspond to causal signal. Therefore

$$x(t) = \frac{-2}{5}e^{-4t}u(t) + \frac{2}{5}e^{t}u(t)$$

(c) The ROC is $\mathrm{Re}(s) < -4$

Both poles lie to the right of the ROC. So both poles correspond to anti-causal signals. Therefore

$$x(t) = \frac{2}{5}e^{-4t}u(-t) - \frac{-2}{5}e^{t}u(-t)$$

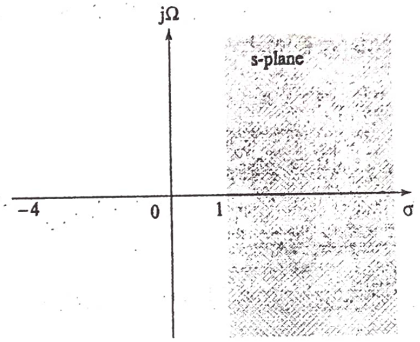

Fig. 7.15

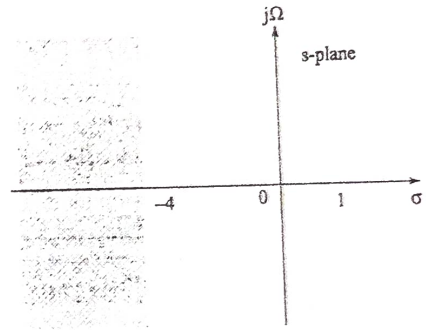

Fig. 7.16

Solved Problem 7.32 Find the signal whose bilateral transform is

$$X(s) = \frac{1}{(s+5)(s+1)} \qquad -5 < \mathrm{Re}(s) < -1$$

Solution:

$$X(s) = \frac{1}{(s+5)(s+1)} = \frac{A}{s+5} + \frac{B}{s+1}$$

$$A = (s+5)\left.\frac{1}{(s+5)(s+1)}\right|_{s=-5} = \frac{-1}{4}$$

$$B = (s+1)\left.\frac{1}{(s+5)(s+1)}\right|_{s=-1} = \frac{1}{4}$$

$$X(s) = \frac{-1}{4}\cdot\frac{1}{s+5} + \frac{1}{4}\cdot\frac{1}{s+1}$$

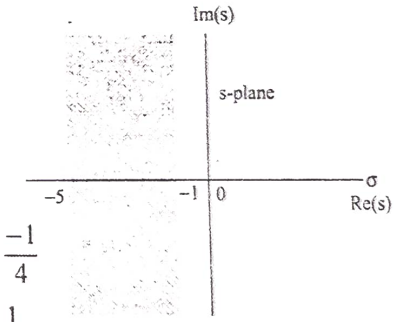

Fig. 7.17

ROC is $-5 < \mathrm{Re}(s) < -1$ shown in Fig. 7.17

The pole -5 is left to the region of convergence so this correspond to causal signal and the pole -1 is right of the ROC so this corresponds to anticausal signal.

$$x(t) = \frac{-1}{4}e^{-5t}u(t) - \frac{1}{4}e^{-t}u(-t)$$

Review Questions

Write in detail about partial fraction expansion method of finding inverse Laplace transform.

Practice Problem 7.11

Find the signal $x(t)$ whose bilateral transform are as given below

(i) $X(s) = \frac{2}{(s^2+1)(s+1)}$ $-1 < \text{Re}\,(s) < 0$

(ii) $X(s) = \frac{2s}{(s+3)(s+1)}$ for (a) $-1 < \text{Re}(s) < -3$ (b) $\text{Re}\,(s) < -3$ (c) $\text{Re}(s) > -1$

7.8 Solution of Differential Equations Using Laplace Transform

In section 4.2 we studied the classical method of solving differential equations. In this section we will find the solution of differential equations using the Laplace transform. We already know that the total response of a system can be expressed as the sum of two components. That is, natural response and force response. The natural response of the system is due to initial conditions of the system where as the forced response of the system is due to input alone. The total response of the system is equal to the forced response when initial conditions are zero.

Let us consider an LTI system with input $x(t)$ and output $y(t)$ described by a differential equation of the form

$$a_N\frac{d^N y(t)}{dt^N}+a_{N-1}\frac{d^{N-1}y(t)}{dt^{N-1}}+\ldots+a_0 y(t)=b_M\frac{d^M x(t)}{dt^M}+b_{M-1}\frac{d^{M-1}x(t)}{dt^{M-1}}+\ldots+b_0 x(t) \tag{7.56}$$

The above differential equation can be solved using time differential property of the Laplace transform which replaces $\frac{d^N y(t)}{dt^N}$ by $s^N Y(s)$, $\frac{d^{N-1}y(t)}{dt^{N-1}}$ by $s^{N-1}Y(s)$ etc. by assuming zero intial conditions on the variables $x(t)$ and $y(t)$.

That is

$$y(0^-)=\frac{dy(0^-)}{dt}=\ \ldots\ =\frac{d^{n-1}y(0^-)}{dt^{n-1}}=0 \tag{7.57}$$

and

$$x(0^-)=\frac{dx(0^-)}{dt}=\ \ldots\ =\frac{d^{n-1}x(0^-)}{dt^{n-1}}=0 \tag{7.58}$$

Now applying Laplace transform on both sides to Eq. (7.56) and assuming initial conditions to zero, we get

$$a_N s^N Y(s)+a_{N-1}s^{N-1}Y(s)+\ldots+a_0 Y(s)=b_M s^M X(s)+b_{M-1}s^{M-1}X(s)+\ldots+b_0 X(s) \tag{7.59}$$

$$Y(s)[a_N s^N+a_{N-1}s^{N-1}+\ldots+a_0]=X(s)[b_M s^M+b_{M-1}s^{M-1}+\ldots+b_0]$$

The above equation can be written as

$$\frac{Y(s)}{X(s)}=\frac{b_M s^N+b_{M-1}s^{M-1}+\ldots+b_0}{a_N s^N+a_{N-1}s^{N-1}+\ldots+a_0} \tag{7.60}$$

The ratio of two polynomials in Eq. (7.60) is known as transfer function of the system denoted by $H(s)$. That is the transfer function $H(s)$ of the system is the ratio of the Laplace transform of the output signal to the Laplace transform of the input signal with all initial conditions to zero.

Thus

$$H(s)=\frac{Y(s)}{X(s)}=\frac{L[\text{Zero state responce}]}{L[\text{input}]} \tag{7.61}$$

From Eq. (7.61) we find the Laplace transform of zero state response (forced response) $Y(s)$ is the product of $X(s)$ and $H(s)$, where $X(s)$ is the Laplace transform of input and $H(s)$ is the transfer function. The function $H(s)$ depends upon the coefficients of differential equation. It does not depend on the input signal or the initial energy storage.

From Eq. (7.61) we have

$$Y(s)=X(s)H(s) \tag{7.62}$$

The inverse Laplace Transform of Eq. (7.62)

$$y(t)=L^{-1}[X(s)H(s)] \tag{7.63}$$

is known as zero state response.

If the input $x(t)=\delta(t)$, then $X(s)=1$ which results

$$Y(s)=H(s)$$

Then

$$y(t)=L^{-1}[H(s)]=h(t) \tag{7.64}$$

is the impulse response of the system.

Let us consider a second order system

$$\frac{d^2y(t)}{dt^2}+\frac{3dy(t)}{dt}+2y(t)=x(t) \tag{7.65}$$

Applying Laplace transform on both sides with all zero initial conditions yields

$$s^2Y(s)+3sY(s)+2Y(s)=X(s) \tag{7.66}$$

$$\Rightarrow \quad \frac{Y(s)}{X(s)}=\frac{1}{s^2+3s+2} \tag{7.67}$$

If the input is an impulse then

$$\begin{aligned} y(t)=h(t)&=L^{-1}\left[\frac{1}{s^2+3s+2}\right]\\ &=L^{-1}\left[\frac{1}{(s+1)(s+2)}\right]\\ &=L^{-1}\left[\frac{1}{s+1}-\frac{1}{s+2}\right]\\ &=e^{-t}-e^{-2t} \end{aligned} \tag{7.68}$$

The impulse response

$$h(t)=e^{-t}-e^{-2t} \tag{7.69}$$

Let the initial conditions are not zero and

$$y(0^-)=\frac{d}{dt}y(0^-)=1$$

then the Laplace transform of Eq. (7.65) is

$$\left[s^2Y(s)-sy(0^-)-\frac{dy(0^-)}{dt}\right]+3[sY(s)-y(0^-)]+2Y(s)=X(s)$$

$$(s^2+3s+2)Y(s)=(s+3)y(0^-)+\frac{dy(0^-)}{dt}+X(s)$$

$$Y(s)=\underbrace{\frac{(s+3)y(0^-)+\frac{dy}{dt}(0^-)}{s^2+3s+2}}_{\text{natural response}}+\underbrace{\frac{X(s)}{s^2+3s+2}}_{\text{Forced response}} \tag{7.70}$$

The first term on the right-hand side of the response depends on initial conditions, hence it is natural response. The second term is the response due to input and is the forced response. Substituting the initial condition $y(0^-)=\frac{dy(0^-)}{dt}=1$ and $X(s)=1$ in Eq. (7.70) we get

$$\begin{aligned} Y(s)&=\frac{s+4}{s^2+3s+2}+\frac{1}{s^2+3s+2}\\ &=\frac{s+4}{(s+1)(s+2)}+\left(\frac{1}{s+1}-\frac{1}{s+2}\right) \end{aligned}$$

$$\Rightarrow \quad y(t)=\underbrace{3e^{-t}-2e^{2t}}_{\text{natural response}}+\underbrace{e^{-t}-e^{-2t}}_{\text{forced response}} \tag{7.71}$$

$$=4e^{-t}-3e^{-2t} \tag{7.72}$$

So far we have seen two types of representations of a system: The differential equation and the transfer function. The differential equation representation of a system is function of time variable 't'. So this representation is known as time domain representation. Similarly the transfer function representation is function of complex variable s. So we can say this representation of the system is in the frequency domain representation.

The transfer function of the system is the ratio of two polynomials. The order of the transfer function is the degree of the denominator, which is N. If the order of the numerator is less than the order of the denominator polynomial $M \leq N$, then the transfer function is proper, on the other hand if $M > N$ then the transfer function is not proper. We already studied that the roots of the numerator polynomial are called zeros of the transfer function and the roots of the denominator polynomial are called poles of the transfer function. If the poles and zeros are graphically represented as location in the s-plane, then we call this diagram a pole-zero diagram.

Review Questions

1. Define natural response and forced response of a system.
2. Define transfer function.

Solved Problem 7.33 Plot the pole-zero diagram of the following transfer functions

(i) $\frac{s+2}{s^2+2s+2}$ (ii) $\frac{s+3}{s(s^2+4)(s+2)(s+1)}$

Solution:

(i) Given the transfer function

$$H(s)=\frac{s+2}{s^2+2s+2}$$

The zeros can be obtained by equating the numerator polynomial to zero

$$s+2=0$$
$$s=-2$$

The poles can be obtained by equating the denominator polynomial to zero

$$s^2+2s+2=0$$
$$s_{1,2}=\frac{-2\pm\sqrt{4-8}}{2}=-1\pm j$$

The poles are at $-1\pm j$ and the zero is at -2. The pole-zero diagram is shown in Fig. 7.18.

(ii) Given

$$H(s)=\frac{s+3}{s(s^2+4)(s+2)(s+1)}$$

The zero is at $s=-3$ and the poles are at $s=0; s=\pm j2; s=-2; s=-1$

The pole-zero diagram is shown in Fig.7.19.

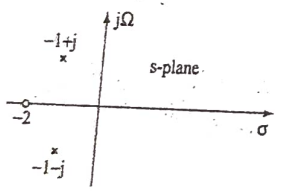

Fig. 7.18 Pole-zero diagram of Example 7.21(i)

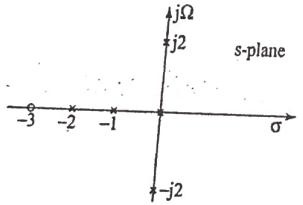

Fig. 7.19 Pole-zero diagram of Example 7.21(ii)

Solved Problem 7.34 Obtain the transfer function of the system if $y(t)=e^{-t}-2e^{-2t}+e^{-3t}$ and $x(t)=e^{-0.5t}$

Solution:

Given

$$x(t)=e^{-0.5t}\Rightarrow X(s)=\frac{1}{s+0.5}$$
$$y(t)=e^{-t}-2e^{-2t}+e^{-3t}$$
$$Y(s)=\frac{1}{s+1}-\frac{2}{s+2}+\frac{1}{s+3}$$
$$=\frac{(s^2+5s+6)-2(s^2+4s+3)+(s^2+3s+2)}{(s+1)(s+2)(s+3)}$$
$$=\frac{2}{(s+1)(s+2)(s+3)}$$

The transfer function

$$H(s)=\frac{Y(s)}{X(s)}=\frac{\dfrac{2}{(s+1)(s+2)(s+3)}}{\dfrac{1}{(s+0.5)}}$$
$$=\frac{2(s+0.5)}{(s+1)(s+2)(s+3)}$$

Solved Problem 7.35 Find the forced response of the system with transfer function $H(s)=\dfrac{1}{(s+0.5)(s+1)}$ for a unit step input.

Solution:

Given $H(s)=\dfrac{1}{(s+1)(s+0.5)}$

$$x(t)=u(t)$$
$$\Rightarrow\quad X(s)=\frac{1}{s}$$
$$H(s)=\frac{Y(s)}{X(s)}=\frac{1}{(s+0.5)(s+1)}$$
$$Y(s)=\frac{1}{s(s+0.5)(s+1)}$$
$$=\frac{C_1}{s}+\frac{C_2}{s+1}+\frac{C_3}{s+0.5}$$

Solving for C_1, C_2 and C_3 we get

$$Y(s) = \underbrace{\frac{2}{s}}_{\text{Due to pole of input}} + \underbrace{\frac{2}{s+1} - \frac{4}{s+0.5}}_{\text{Due to poles of transfer function}}$$

Taking inverse Laplace transform on both sides we get

$$y(t) = \underbrace{2}_{\text{Due to input pole}} + \underbrace{2e^{-t} - 4e^{-0.5t}}_{\text{Due to poles of transfer function}}$$

$\uparrow$ Steady state response

$\uparrow$ Transient response

From solved problem 7.35 we can find that the forced response consists of two parts. One part is due to the poles of the system and the other is due to input. The response due to the poles of the system vanishes after a long time. It is often called the transient response, and the response due to the pole of $X(s)$ remain and is often referred to as the steady state response.

Solved Problem 7.36 Find the transfer function of the following.

(i) An ideal differentiator.
(ii) An ideal integrator.
(iii) An ideal delay of T seconds.

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Solution:

(i) For an ideal differentiator the relationship between the $x(t)$ input and output $y(t)$ is given by

$$y(t) = \frac{dx(t)}{dt}$$

The Laplace transform of above equation is given by

$$Y(s) = sX(s) - x(0^-)$$

$$= sX(s) \quad \boxed{\text{Since } x(0^-) = 0}$$

$$\Rightarrow \quad H(s) = \frac{Y(s)}{X(s)} = s$$

(ii) For an ideal integrator the output is given by

$$y(t) = \int_{-\infty}^{t} x(\tau)d\tau.$$

If the initial state is zero then

$$y(t) = \int_{0}^{t} x(\tau)d\tau$$

Applying Laplace transform yields

$$Y(s) = \frac{X(s)}{s}$$

$$\Rightarrow \quad H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s}$$

(iii) For an ideal delay of T seconds the output

$$y(t) = x(t - T)$$

Applying Laplace transform on both sides we get

$$Y(s) = e^{-sT}X(s)$$

$$\Rightarrow \quad H(s) = \frac{Y(s)}{X(s)} = e^{-sT}$$

Solved Problem 7.37 For each of the systems described by the following differential equations, find the system transfer function

(i) $\frac{d^2y(t)}{dt^2} + 11\frac{dy(t)}{dt} + 24y(t) = 5\frac{dx(t)}{dt} + 3x(t)$

(ii) $\frac{d^3y(t)}{dt^3} + 6\frac{d^2y(t)}{dt^2} + 11\frac{dy(t)}{dt} + 6y(t) = 3\frac{d^2x(t)}{dt} + 7\frac{dx(t)}{dt} + 5x(t)$

Solution:

(i) $\frac{d^2y(t)}{dt^2} + 11\frac{dy(t)}{dt} + 24y(t) = 5\frac{dx(t)}{dt} + 3x(t)$

Applying Laplace transform on both sides and assuming initial zero conditions we get

$$s^2Y(s)+11sY(s)+24Y(s)=5sX(s)+3X(s)$$
$$Y(s)[s^2+11s+24]=X(s)[5s+3]$$

with zero initial conditions
$$L\left[\frac{d^ny(t)}{dt^n}\right]=s^nY(s)$$

The transfer function

$$H(s)=\frac{Y(s)}{X(s)}=\frac{5s+3}{s^2+11s+24}$$

(ii) Given

$$\frac{d^3y(t)}{dt^3}+6\frac{d^2y(t)}{dt^2}+11\frac{dy(t)}{dt}+6y(t)=3\frac{d^2x(t)}{dt}+7\frac{dx(t)}{dt}+5x(t)$$

Applying Laplace transform on both sides and assuming zero initial conditions, we get

$$s^3Y(s)+6s^2Y(s)+11sY(s)+6Y(s)=3s^2X(s)+7sX(s)+5X(s)$$
$$Y(s)[s^3+6s^2+11s+6]=X(s)[3s^2+7s+5]$$

The transfer function is

$$H(s)=\frac{Y(s)}{X(s)}=\frac{3s^2+7s+5}{s^3+6s^2+11s+6}$$

Solved Problem 7.38 Using Laplace transform, solve the following differential equations

(i) $\frac{d^2y(t)}{dt^2}+3\frac{dy(t)}{dt}+2y(t)=\frac{d}{dt}x(t)$ if $y(0^-)=2;\frac{dy(0^-)}{dt}=1$ and $x(t)=e^{-t}u(t)$

(ii) $\frac{d^3y(t)}{dt^3}+\frac{7d^2y(t)}{dt}+\frac{16dy(t)}{dt}+12y(t)=x(t)$ if $\frac{dy(0^-)}{dt}=0;\frac{d^2y(0^-)}{dt^2}=0$ $y(0^-)=0$ and $x(t)=\delta(t)$

(i) Given

$$\frac{d^2y(t)}{dt^2}+3\frac{dy(t)}{dt}+2y(t)=\frac{d}{dt}x(t)$$

Taking Laplace transform on both sides we get

$$\left[s^2Y(s)-sy(0^-)-\frac{dy}{dt}(0^-)\right]+3[sY(s)-y(0^-)]+2Y(s)=sX(s)-x(0^-)$$

$$y(0^-)=2;\frac{dy(0^-)}{dt}=1$$

$$(s^2Y(s)-2s-1)+3[sY(s)-2]+2Y(s)=sX(s)$$

$$Y(s)[s^2+3s+2]=2s+7+sX(s)$$

Given $x(t)=e^{-t}u(t)$

$$X(s)=\frac{1}{s+1}$$

$$\Rightarrow\quad Y(s)[s^2+3s+2]=2s+7+\frac{s}{s+1}$$

$$Y(s)=\frac{2s+7}{s^2+3s+2}+\frac{s}{(s+1)(s^2+3s+2)}$$

$$=\frac{(2s+7)(s+1)+s}{(s+1)(s^2+3s+2)}$$

$$=\frac{2s^2+10s+7}{(s+1)(s^2+3s+2)}$$

$$=\frac{2s^2+10s+7}{(s+1)^2(s+2)}$$

$$=\frac{A}{s+1}+\frac{B}{(s+1)^2}+\frac{C}{s+2}$$

$$A=\frac{1}{1!}\frac{d}{ds}\left[(s+1)^2\frac{2s^2+10s+7}{(s+1)^2(s+2)}\right]\Bigg|_{s=-1}$$

$$=\frac{(s+2)(4s+10)-(2s^2+10s+7)}{(s+2)^2}\Bigg|_{s=-1}$$

$$=7$$

$$B=(s+1)^2\frac{2s^2+10s+7}{(s+1)^2(s+2)}\Bigg|_{s=-1}=-1$$

$$C=(s+2)\frac{2s^2+10s+7}{(s+1)^2(s+2)}\Bigg|_{s=-2}=-5$$

$$Y(s)=\frac{7}{s+1}-\frac{1}{(s+1)^2}-\frac{5}{s+2}$$

Taking inverse Laplace transform on both sides, we get

$$y(t) = 7e^{-t}u(t) - te^{-t}u(t) - 5e^{-2t}u(t)$$

(ii) $\dfrac{d^3y(t)}{dt^3} + 7\dfrac{d^2y(t)}{dt^2} + 16\dfrac{dy(t)}{dt} + 12y(t) = x(t)$

Applying Laplace transform on both sides we get

$$[s^3Y(s) - s^2y(0^-) - s\frac{dy(0^-)}{dt^2} - \frac{d^2y(0^-)}{dt^2}] + 7[s^2Y(s) - sy(0^-) - \frac{dy(0^-)}{dt}] + 16[sY(s) - y(0^-)] + 12Y(s) = X(s) \quad (7.73)$$

Given

$$y(0^-) = 0$$

$$\frac{dy(0^-)}{dt} = 0 \quad \text{and} \quad \frac{d^2y(0^-)}{dt^2} = 0$$

Substituting these values in Eq. (7.73), we get

$$s^3Y(s) + 7s^2Y(s) + 16sY(s) + 12Y(s) = X(s)$$

$$Y(s)[s^3 + 7s^2 + 16s + 12] = X(s)$$

$$\frac{Y(s)}{X(s)} = \frac{1}{s^3 + 7s^2 + 16s + 12}$$

For $x(t) = \delta(t); X(s) = 1$. Therefore

$$Y(s) = \frac{1}{s^3 + 7s^2 + 16s + 12}$$

$$= \frac{1}{(s+3)(s+2)^2}$$

$$= \frac{A}{s+2} + \frac{B}{(s+2)^2} + \frac{C}{s+3}$$

$$A = \frac{1}{1!}\frac{d}{ds}\left[(s+2)^2\frac{1}{(s+3)(s+2)^2}\right]\Big|_{s=-2}$$

-3	1	7	16	12
	0	-3	-12	-12
	1	4	4	0

$$\Rightarrow s^3 + 7s^2 + 6s + 12 = (s+3)(s^2 + 4s + 4) = (s+3)(s+2)^2$$

$$= \frac{d}{ds}\left[\frac{1}{s+3}\right]\Big|_{s=-2}$$

$$= \frac{-1}{(s+3)^2}\Big|_{s=-2} = -1$$

$$B = (s+2)^2\frac{1}{(s+3)(s+2)^2}\Big|_{s=-2} = 1$$

$$C = (s+3)\frac{1}{(s+3)(s+2)^2}\Big|_{s=-3} = 1$$

$$Y(s) = \frac{-1}{s+2} + \frac{1}{(s+2)^2} + \frac{1}{s+3}$$

$$\Rightarrow \quad y(t) = -e^{-2t}u(t) + te^{-2t}u(t) + e^{-3t}u(t)$$

Practice Problem 7.12

Using the Laplace transform, solve the following differential equations

(i) $\dfrac{d^3y(t)}{dt^3} - 2\dfrac{d^2y(t)}{dt^2} + \dfrac{dy(t)}{dt} - 2y(t) = x(t)$

$\dfrac{d^2y(0^-)}{dt^2} = \dfrac{dy(0^-)}{dt} = y(0^-) = 1$ and $x(t) = e^{-t}u(t)$

(ii) $\dfrac{d^2y(t)}{dt^2} + y(t) = \cos t\ ;\ y(0^-) = \dfrac{dy(0^-)}{dt} = 0$

Solved Problem 7.39 A system is described by the following differential equation

$$\frac{d^2y(t)}{dt^2} + 7\frac{dy(t)}{dt} + 12y(t) = x(t)$$

Determine the response of the system to a unit step applied at $t = 0$. The initial conditions are $y(0^-) = -2$; $\dfrac{dy}{dt}(0^-) = 0$

Solution:

$$\frac{d^2y(t)}{dt^2} + 7\frac{dy(t)}{dt} + 12y(t) = x(t)$$

Applying Laplace-transform on both sides, we have

$$\left[s^2Y(s) - sy(0^-) - \frac{dy}{dt}(0^-)\right] + 7[sY(s) - y(0^-)] + 12Y(s) = X(s)$$

Substituting $y(0^-) = -2$ and $\frac{dy}{dt}(0^-) = 0$, we obtain

$$s^2Y(s) + 2s + 7sY(s) + 14 + 12Y(s) = X(s)$$

$$(s^2 + 7s + 12)Y(s) + 2s + 14 = X(s)$$

a) For a unit step input

$x(t) = u(t)$

$X(s) = \frac{1}{s}$

$$\Rightarrow Y(s) = \frac{1}{s(s^2+7s+12)} - \frac{(2s+14)}{s^2+7s+12}$$

$$= \frac{1-2s^2-14s}{s(s+3)(s+4)}$$

$$= \frac{A}{s} + \frac{B}{s+3} + \frac{C}{s+4}$$

$$= \frac{1}{12s} - \frac{25}{3(s+3)} + \frac{25}{4(s+4)}$$

$$A = s\frac{1-2s^2-14s}{s(s+3)(s+4)}\Big|_{s=0}$$
$$= \frac{1}{12}$$

$$B = (s+3)\frac{(1-2s^2-14s)}{s(s+3)(s+4)}\Big|_{s=-3}$$
$$= \frac{1-18+42}{(-3)(1)} = -\frac{25}{3}$$

$$C = \frac{(s+4)(1-2s^2-14s)}{s(s+3)(s+4)}\Big|_{s=-4}$$
$$= \frac{1-32+56}{(-4)(-1)} = \frac{25}{4}$$

Taking inverse Laplace transform

$$y(t) = \frac{1}{12} - \frac{25}{3}e^{-3t}u(t) + \frac{25}{4}e^{-4t}u(t)$$

Solved Problem 7.40 For a system with transfer function

$$H(s) = \frac{s+5}{s^2+5s+6}$$

find the zero-state response if the input $x(t)$ is $e^{-3t}u(t)$

Solution: Given

$$H(s) = \frac{s+5}{s^2+5s+6}$$

$$\frac{Y(s)}{X(s)} = \frac{s+5}{(s+2)(s+3)}$$

For

$$x(t) = e^{-3t}u(t) \quad X(s) = \frac{1}{s+3}$$

$$\Rightarrow Y(s) = \frac{s+5}{(s+2)(s+3)^2}$$

$$= \frac{A}{s+2} + \frac{B}{s+3} + \frac{C}{(s+3)^2}$$

$$A = (s+2)\frac{s+5}{(s+2)(s+3)^2}\Big|_{s=-2} = \frac{-2+5}{(-2+3)^2} = 3$$

$$B = \frac{1}{1!}\frac{d}{ds}\left[(s+3)^2\frac{s+5}{(s+2)(s+3)^2}\right]\Big|_{s=-3}$$

$$= \frac{(s+2)-(s+5)}{(s+2)^2}\Big|_{s=-3}$$

$$= \frac{-3}{(-3+2)^2} = -3$$

$$C = (s+3)^2\frac{s+5}{(s+2)(s+3)^2}\Big|_{s=-3} = -2$$

$$Y(s) = \frac{3}{s+2} - \frac{3}{s+3} - \frac{2}{(s+3)^2}$$

Taking inverse Laplace transform, we have

$$y(t) = 3e^{-2t}u(t) - 3e^{-3t}u(t) - 2te^{-3t}u(t)$$

Solved Problem 7.41 Find the impulse and the step response of the following systems.

(i) $H(s) = \frac{10}{s^2+6s+10}$ (ii) $H(s) = \frac{s+2}{s^2+5s+4}$

(i)

$$H(s) = \frac{10}{s^2+6s+10}$$

$$\frac{Y(s)}{X(s)} = \frac{10}{s^2+6s+10}$$

For an impulse $x(t) = \delta(t); X(s) = 1$

$$\Rightarrow Y(s) = \frac{10}{s^2+6s+10}$$

$$= \frac{10}{(s+3)^2+1^2}$$

Taking inverse Laplace transform we get

$$y(t) = 10e^{-3t}\sin tu(t)$$

(ii) For a unit step input

$$x(t) = u(t)$$

$$X(s) = \frac{1}{s}$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{10}{s^2+6s+10}$$

$$Y(s) = \frac{10}{s(s^2+6s+10)}$$

$$= \frac{A}{s} + \frac{Bs+C}{s^2+6s+10}$$

$$\Rightarrow \quad \frac{10}{s(s^2+6s+10)} = \frac{A(s^2+6s+10)+s(Bs+C)}{s(s^2+6s+10)}$$

$$(A+B)s^2 + (6A+C)s + 10A = 10$$

Comparing the coefficient of s^2, s and constant we get

$$A+B=0\ ;\ 6A+C=0$$

$$10A = 10$$

$$A = 1$$

$$B = -1$$

$$C = -6$$

$$Y(s) = \frac{1}{s} + \frac{-s-6}{s^2+6s+10}$$

$$= \frac{1}{s} - \frac{(s+6)}{s^2+6s+10}$$

$$= \frac{1}{s} - \frac{s+6}{(s+3)^2+1}$$

$$= \frac{1}{s} - \left\{\frac{s+3}{(s+3)^2+1} + \frac{3}{(s+3)^2+1}\right\}$$

Taking inverse Laplace transform we get

$$y(t) = u(t) - \{e^{-3t}\cos tu(t) + 3e^{-3t}\sin tu(t)\}$$

$$= [1 - e^{-3t}\{\cos t + 3\sin t\}]\,u(t)$$

(iii) $H(s) = \dfrac{s+2}{s^2+5s+4}$

$$\frac{Y(s)}{X(s)} = \frac{s+2}{(s+1)(s+4)}$$

For an impulse $x(t) = \delta(t)$
$X(s) = 1$

$$\Rightarrow \quad Y(s) = \frac{s+2}{(s+1)(s+4)}$$

$$= \frac{A}{s+1} + \frac{B}{s+4}$$

$$A = (s+1)\left.\frac{s+2}{(s+1)(s+4)}\right|_{s=-1} = \frac{1}{3}$$

$$B = (s+4)\left.\frac{(s+2)}{(s+1)(s+4)}\right|_{s=-4} = \frac{2}{3}$$

$$Y(s) = \frac{1}{3(s+1)} + \frac{2}{3(s+4)}$$

Taking inverse Laplace transform, we have

$$y(t) = \frac{1}{3}e^{-t}u(t) + \frac{2}{3}e^{-4t}u(t)$$

For a unit step input

$$x(t) = u(t)$$

$$X(s) = \frac{1}{s}$$

Then

$$Y(s) = \frac{s+2}{s(s+1)(s+4)}$$

$$= \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+4}$$

$$A = s\left.\frac{s+2}{s(s+1)(s+4)}\right|_{s=0} = \frac{1}{2}$$

$$B = (s+1)\left.\frac{s+2}{s(s+1)(s+4)}\right|_{s=-1} = \frac{-1}{3}$$

$$C=(s+4)\frac{s+2}{s(s+1)(s+4)}\bigg|_{s=-4}=\frac{-1}{6}$$

$$Y(s)=\frac{1}{2s}-\frac{1}{3(s+1)}-\frac{1}{6(s+4)}$$

Taking inverse Laplace transform

$$y(t)=\frac{1}{2}u(t)-\frac{1}{3}e^{-t}u(t)-\frac{1}{6}e^{-4t}u(t)$$

Solved Problem 7.42 For the transfer function

$$H(s)=\frac{s+10}{s^2+3s+2}$$

find the response due to input $x(t)=\sin(2t)u(t)$

Solution:

Given

$$x(t)=\sin(2t)u(t)$$

$$x(t)=\frac{2}{s^2+4}$$

$$H(s)=\frac{Y(s)}{X(s)}=\frac{s+10}{(s^2+3s+2)}$$

$$\Rightarrow\quad Y(s)=\frac{s+10}{(s+1)(s+2)}\cdot\frac{2}{s^2+4}$$

$$=\frac{2(s+10)}{(s+1)(s+2)(s^2+4)}$$

$$=\frac{A}{s+1}+\frac{B}{s+2}+\frac{Cs+D}{s^2+4}$$

$$\Rightarrow\quad 2(s+10)=A(s+2)(s^2+4)+B(s+1)(s^2+4)+(Cs+D)(s+1)(s+2)$$

$$=A(s^3+2s^2+4s+8)+B(s^3+s^2+4s+4)+C(s^3+3s^2+2s)+D(s^2+3s+2)$$

$$=(A+B+C)s^3+(2A+B+3C+D)s^2+(4A+4B+2C+3D)s+8A+4B+2D$$

$$A+B+C=0;\ 2A+B+3C+D=0;\ 4A+4B+2C+3D=2;\ 8A+4B+2D=20$$

Solving for A,B,C and D we get

$$A=\frac{18}{5};B=-2;C=\frac{-8}{5};D=\frac{-2}{5}$$

$$Y(s)=\frac{18}{5(s+1)}-\frac{2}{s+2}-\left(\frac{\frac{8}{5}s+\frac{2}{5}}{s^2+4}\right)$$

$$=\frac{18}{5(s+1)}-\frac{2}{s+2}-\frac{8}{5}\frac{s}{s^2+4}-\frac{2}{5}\frac{1}{s^2+4}$$

$$=\frac{18}{5(s+1)}-\frac{2}{s+2}-\frac{8}{5}\left(\frac{s}{s^2+4}\right)-\frac{1}{5}\left(\frac{2}{s^2+4}\right)$$

$$y(t)=\frac{18}{5}e^{-t}-2e^{-t}-\frac{8}{5}\cos 2t-\frac{1}{5}\sin 2t$$

Solved Problem 7.43 Using the Laplace transform find the impulse response of an LTI system described by differential equation

$$\frac{d^2y(t)}{dt^2}-\frac{dy(t)}{dt}-2y(t)=x(t)$$

Solution:

Apply Laplace transform for differential assuming initial conditions to zero

$$s^2Y(s)-sY(s)-2Y(s)=X(s)$$

$$Y(s)[s^2-s-2]=X(s)$$

$$H(s)=\frac{Y(s)}{X(s)}=\frac{1}{s^2-s-2}=\frac{1}{(s-2)(s+1)}$$

$$=\frac{A}{s-2}+\frac{B}{s+1}$$

$$=\frac{1}{3}\frac{1}{s-2}-\frac{1}{3}\frac{1}{s+1}$$

$$h(t)=\frac{1}{3}e^{2t}u(t)-\frac{1}{3}e^{-t}u(t)$$

Solved Problem 7.44 Find $y(t)$ for the differential equation

$$\frac{dy(t)}{dt} + 5y(t) = x(t)$$

where $x(t) = 3e^{-2t}u(t)$ and $y(0^+) = -2$

Solution:

Applying Laplace transform on both sides we get

$$sY(s) - y(0^+) + 5Y(s) = X(s)$$

$$y(0^+) = -2 \text{ and } X(s) = \frac{3}{s+2}$$

$$\Rightarrow Y(s)[s+5] = \frac{3}{s+2} - 2$$

$$Y(s) = \frac{3}{(s+2)(s+5)} - \frac{2}{s+5}$$

$$= \frac{1}{s+2} - \frac{1}{s+5} - \frac{2}{s+5}$$

$$\Rightarrow y(t) = e^{-2t}u(t) - 3e^{-5t}u(t)$$

Solved Problem 7.45 The output response $y(t)$ of a continuous time LTI system is $2e^{-3t}u(t)$ when the input $x(t)$ is $u(t)$. Find the system function.

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Solution:

Given

$$y(t) = 2e^{-3t}u(t) \quad \text{and} \quad x(t) = u(t)$$

Applying Laplace transform we get

$$Y(s) = \frac{2}{s+3}$$

$$X(s) = \frac{1}{s}$$

The system function $H(s) = \frac{Y(s)}{X(s)} = \frac{2}{s+3} / \frac{1}{s} = \frac{2s}{s+3}$

Solved Problem 7.46 Determine the response of the system with impulse response $h(t) = u(t)$ for the input $x(t) = e^{-2t}u(t)$ AU May 2004

Solution:

$$h(t) = u(t); x(t) = e^{-2t}u(t)$$

$$H(s) = \frac{1}{s} \text{ and } X(s) = \frac{1}{s+2}$$

$$Y(s) = H(s)X(s) = \frac{1}{s(s+2)} = \frac{1}{2}\left(\frac{1}{s} - \frac{1}{s+2}\right)$$

$$y(t) = \frac{1}{2}\left[1 - e^{-2t}\right]u(t)$$

Solved Problem 7.47 Find the output response of an LTI system with impulse response $h(t) = \delta(t-3)$ for input $x(t) = \cos 4t + \cos 7t$

Solution:

$$h(t) = \delta(t-3); x(t) = \cos 4t + \cos 7t$$

$$H(s) = e^{-3s}\ ; X(s) = \frac{s}{s^2+16} + \frac{s}{s^2+49}$$

$$Y(s) = X(s)H(s)$$

$$= e^{-3s}\frac{s}{s^2+16} + e^{-3s}\frac{s}{s^2+49}$$

Taking inverse Laplace transform

$$y(t) = \cos 4(t-3) + \cos 7(t-3)$$

7.9 Analysis of Electrical Networks Using Laplace Transform

An electrical network is an LTI system if all the elements are modelled as linear and time-invariant. A network consists of elements like resistors, capacitors and inductors. A network can be modelled using fundamental laws. The model of the network can be obtained by writing integro-differential equations. Then the network can be analysed by solving these equations using Laplace transform.

There is another way of analysing an electrical network without writing integro differential equations. In this method each element in the network is represented in frequency domain there by transforming the time domain electrical network to frequency domain. This transformed network is then solved by writing algebraic equations.

7.9.1 Initial conditions

The frequency domain representation of each element can be found by writing voltage-current relationship of the element in time domain and then applying Laplace transform with initial conditions. These initial conditions must be considered at time $t = 0$, when some switiching operation occurs. The initial conditions are the initial voltage across the capacitor (or charge on capacitor) and the initial current through the inductor. If these initial voltages and currents are zero then the system is initially relaxed.

1. If the initial voltage across the capacitor is not zero then the voltage across the capacitor will be same before and after the instant of switching. If the switching operation is represented from $t = 0^-$ to $t = 0^+$, then

$$V_C(0^+) = V_C(0^-) \tag{7.74}$$

2. If the initial current through an inductor is not zero then the current through the inductor will be same before and after the instant of switching. That is

$$i_L(0^+) = i_L(0^-). \tag{7.75}$$

7.9.2 Transformed form of elements

1. **Resistor:** If a current $i(t)$ is flowing through a resistor R ohms, then the voltage $v_R(t)$ across the resistor is given by

$$v_R(t) = Ri(t) \tag{7.76}$$

The transformed form of the resistor is shown in Fig. 7.21.

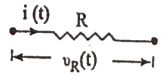


Fig. 7.20 A resistor.

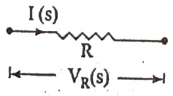


Fig. 7.21 Resistor in s-domain.

Capacitance

For a capacitance of C farad, the voltage current relationship is given by

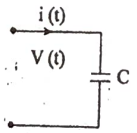


Fig. 7.22 A capacitor.

$$v(t) = \frac{1}{C}\int_{-\infty}^{t} i(t)dt = \frac{1}{C}\int_{0}^{t} i(t)dt + V_C(0^-) \tag{7.77}$$

Applying Laplace transform

$$V(s) = \frac{1}{Cs}I(s) + \frac{V_C(0^-)}{s} \tag{7.78}$$

or

$$I(s) = CsV(s) - CV_C(0^-) \tag{7.79}$$

Now the transformed network is

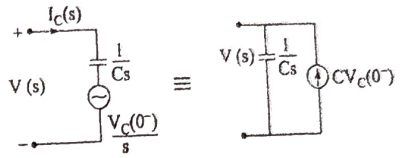


Fig. 7.23 Capacitor in s-domain.

If the initial condition is zero, then from Eq. (7.78) we have

$$V(s) = \frac{1}{Cs}I(s) \tag{7.80}$$

and transformed form of capacitor with zero initial condition is

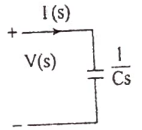


Fig. 7.24 Capacitor in s-domain with zero initial conditions.

Inductance

For an inductance of L Henry, the voltage-current relationship is given by

$$v_L(t) = L\frac{d}{dt}i_L(t) \tag{7.81}$$

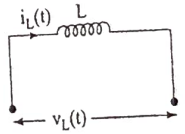


Fig. 7.25 An inductor.

Using differential property of Laplace transform we get

$$V_L(s) = L[sI_L(s) - i_L(0^-)] \tag{7.82}$$

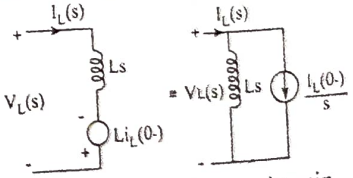


Fig. 7.26 Inductor in s-domain.

or

$$I_L(s) = \frac{V(s)}{Ls} + \frac{i_L(0^-)}{s} \tag{7.83}$$

with initial condition to zero

$$V_L(s) = sLI_L(s) \qquad (7.84)$$

and the transformed form of element is

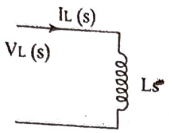
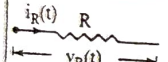
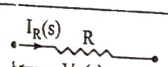
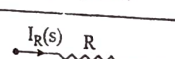
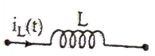
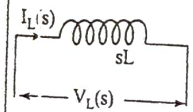
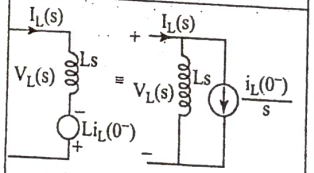
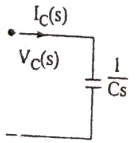
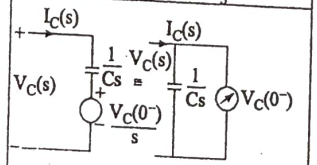


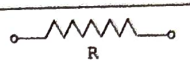
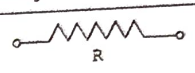
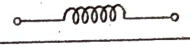
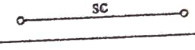
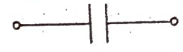
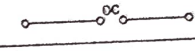
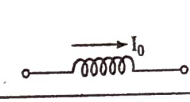
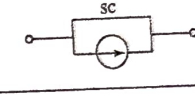
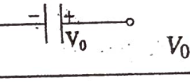
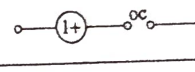
Fig. 7.27 Inductor in s-domain with zero initial conditions.

The fundamental electrical elements and their s- domain representation and condition of the elements at steady state are given in Table 7.2.

Table 7.2 Representation of elements in s-domain

Element	Time domain representation	Transformed Network	
		With zero initial conditions	With non-zero initial conditions
Resistor	 $v_R(t) = Ri_R(t)$	 $V_R(s) = RI_R(s)$	 $V_R(s) = RI_R(s)$
Inductor	 $v_L(t) = \frac{Ldi(t)}{dt}$	 $V_L(s) = LsI_L(s)$	 $V_L(s) = LsI_L(s) - Li_L(0^-)$ or $I_L(s) = \frac{V_L(s)}{Ls} + \frac{i_L(0^-)}{s}$
Capacitor	$v_C(t) = \frac{1}{C}\int i_C(t)dt$	 $V_C(s) = \frac{I_C(s)}{Cs}$	 $V_C(s) = \frac{I_C(s)}{Cs} + \frac{V_C(0^-)}{S}$ $I_C(s) = V_C(s)Cs - CV_C(0^-)$

Steady state conditions: The steady state conditions of the elements are given below.

Element	Steady state equivalent
	
	
	
	
 $V_0 = \frac{q_0}{C}$	

Solved Problem 7.48 Find the step response of the circuit shown in Fig. 7.28.

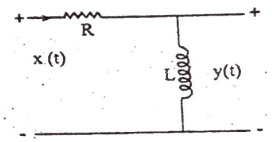


Fig. 7.28 An RL circuit.

Solution:

Given

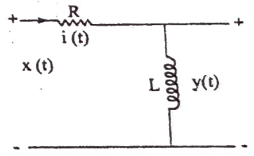


Fig. 7.29

The mathematical representation of the circuit is given by

$$x(t) = Ri(t) + L\frac{di(t)}{dt} \qquad (7.85)$$

The output

$$y(t) = L\frac{di(t)}{dt} \qquad (7.86)$$

Taking Laplace transform on both sides for Eq. (7.85) yields

$$X(s) = RI(s) + LsI(s)$$
$$\Rightarrow \quad X(s) = I(s)(R + Ls)$$

Taking Laplace transform for Eq. (7.86) we get

$$Y(s) = sLI(s)$$
$$= sL\frac{X(s)}{R+Ls}$$
$$\frac{Y(s)}{X(s)} = \frac{Ls}{R+Ls} = \frac{s}{s+R/L}$$

For unit step input

$$X(s) = \frac{1}{s}.$$

Therefore

$$Y(s) = \frac{1}{s} \cdot \frac{s}{s+R/L} = \frac{1}{s+R/L} \tag{7.87}$$

Taking inverse Laplace transformation on both sides

$$y(t) = e^{-(R/L)t} \tag{7.88}$$

Solved Problem 7.49 Find the impulse response of the following circuits

(i)

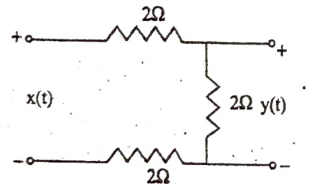


Fig. 7.30

(ii)

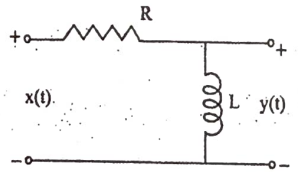


Fig. 7.31

(iii)

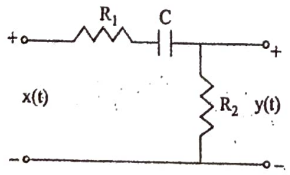


Fig. 7.32

Solution:

(i) Given the circuit

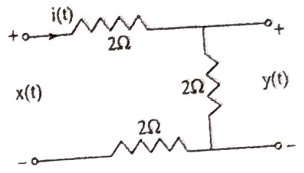


Fig. 7.33

The input voltage $x(t) = 6i(t)$

$$y(t) = 2i(t)$$
$$= \frac{2x(t)}{6}$$
$$= \frac{1}{3}x(t)$$

If the input $x(t) = \delta(t)$ then

$$h(t) = \frac{1}{3}\delta(t) \tag{7.89}$$

(ii)

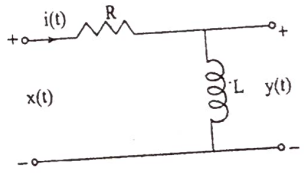


Fig. 7.34

The differential equation governing the response of the circuit is

$$Ri(t) + L\frac{di(t)}{dt} = x(t)$$

Applying Laplace transform on both sides with zero initial condition yields

$$RI(s) + LsI(s) = X(s) \tag{7.90}$$
$$I(s)[R + Ls] = X(s)$$
$$I(s) = \frac{X(s)}{R+LS} \tag{7.91}$$

we know $y(t) = L\frac{di(t)}{dt}$ (7.92)

$$\Rightarrow \quad Y(s) = LsI(s)$$

$$= Ls\frac{X(s)}{R+Ls}$$

$$\Rightarrow \quad \frac{Y(s)}{X(s)} = H(s) = \frac{Ls}{R+Ls} \tag{7.93}$$

The impulse response

$$h(t) = L^{-1}[H(s)]$$

$$= L^{-1}\left[\frac{sL}{R+Ls}\right]$$

$$= L^{-1}\left[\frac{s}{s+R/L}\right]$$

$$= L^{-1}\left[1 - \frac{R/L}{s+R/L}\right]$$

$$= \delta(t) - \frac{R}{L}e^{-(R/L)t} \tag{7.94}$$

(ii) Given

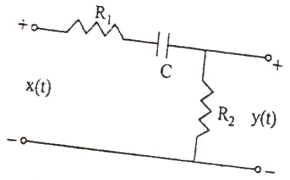


Fig. 7.35

The transformed network with zero initial condition is shown in Fig. 7.36.

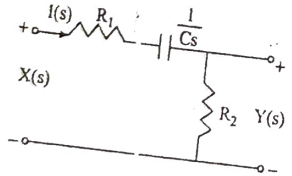


Fig. 7.36

Applying KVL, we have

$$X(s) = R_1I(s) + \frac{1}{Cs}I(s) + R_2I(s)$$

$$= I(s)\left[R_1 + \frac{1}{Cs} + R_2\right] \tag{7.95}$$

$$= R_2\left[\frac{X(s)}{R_1+R_2+\frac{1}{Cs}}\right]$$

$$= \frac{R_2CsX(s)}{(R_1+R_2)Cs+1} \tag{7.97}$$

The transfer function is

$$H(s) = \frac{Y(s)}{X(s)} = \frac{R_2Cs}{(R_1+R_2)C\left[s+\frac{1}{(R_1+R_2)C}\right]}$$

$$= \frac{R_2s}{(R_1+R_2)\left[s+\frac{1}{(R_1+R_2)C}\right]}$$

The impulse response is

$$h(t) = L^{-1}[H(s)]$$

$$= L^{-1}\left\{\frac{R_2}{R_1+R_2}\left[1 - \frac{\frac{1}{(R_1+R_2)C}}{s+\frac{1}{(R_1+R_2)C}}\right]\right\}$$

$$= \frac{R_2}{R_1+R_2}\left[\delta(t) - \frac{1}{(R_1+R_2)C}e^{-\frac{1}{(R_1+R_2)C}}\right] \tag{7.98}$$

Solved Problem 7.50

Obtain the voltage across the resistor as a function of time for $t > 0$. Assume that $i(0) = v_c(0) = 0$.

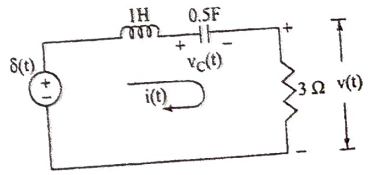


Fig. 7.37

Solution:

The transformed circuit with zero initial condition is given by

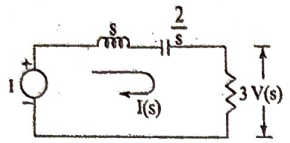


Fig. 7.38

From the circuit we have

$$I(s)\left[s+\frac{2}{s}+3\right]=1 \tag{7.99}$$

$$I(s)=\frac{1}{\frac{s^2+3s+2}{s}}=\frac{s}{s^2+3s+2}$$

$$=\frac{s}{(s+1)(s+2)}$$

$$=-\frac{1}{s+1}+\frac{2}{s+2}$$

$$i(t)=-e^{-t}+2e^{-2t} \tag{7.100}$$

The voltage across resistor $v(t)=3i(t)=3\left[-e^{-t}+2e^{-2t}\right]$ (7.101)

Solved Problem 7.51

Solve for the current in the circuit shown if the switch changes from position 1 to position 2 at $t=0$. Assume that the switch has been in position 1 since $t=-\infty$.

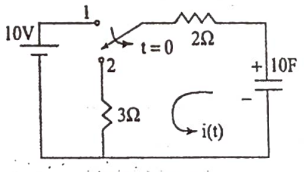


Fig. 7.39

Solution:

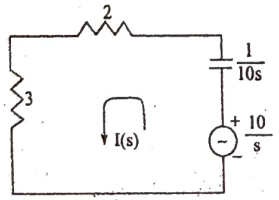


Fig. 7.40

When the switch is at position 1 for a long time the capacitor is charged to 10 V as the polarity shown in Fig.7.39. At position 2, the transformed network is shown in Fig.7.40.

From the transformed circuit we have

$$\frac{10}{s}=\left(5+\frac{1}{10s}\right)I(s) \tag{7.102}$$

$$I(s)=\frac{10}{s\left(5+\frac{1}{10s}\right)}=\frac{10}{s\frac{(50s+1)}{10s}}$$

$$=\frac{100s}{50s\left(s+\frac{1}{50}\right)}=\frac{100}{50\left(s+\frac{1}{50}\right)}$$

$$\Rightarrow \quad i(t)=2e^{-t/50} \tag{7.103}$$

Solved Problem 7.52

In the circuit shown the switch is at position 1 for a very long time and at $t=0$, the switch moved to position 2. Solve for current $i(t)$, for $t\geq 0$ using the Laplace transform.

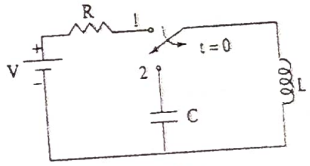


Fig. 7.41

Solution:

The switch is at position 1 for a long time, therefore the current

$$i(0^-)=\frac{V}{R}. \tag{7.104}$$

When the switch is moved to position 2, the transformed circuit is given by

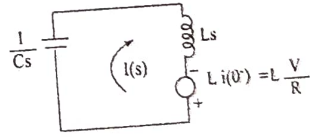


Fig. 7.42

By using Fig. 7.42 we can write

$$\frac{1}{Cs}I(s)+LsI(s)=\frac{LV}{R} \tag{7.105}$$

$$I(s)\left[\frac{1+LCs^2}{Cs}\right]=\frac{LV}{R}$$

$$I(s)=\frac{LCVs}{R(1+LCs^2)}$$

$$=\frac{Vs}{R(s^2+1/LC)}$$

$$=\frac{V}{R}\frac{s}{s^2+\left(\frac{1}{\sqrt{LC}}\right)^2}$$

$$\Rightarrow i(t)=\frac{V}{R}\cos\left(\frac{1}{\sqrt{LC}}\right)t \text{ for } t\geq 0$$

$$=\frac{V}{R}\cos\left(\frac{1}{\sqrt{LC}}\right)t\ \ u(t) \tag{7.106}$$

Solved Problem 7.53

Find the response of the circuit to the input

$$x(t)=r(t)-2r(t-1)+r(t-2)$$

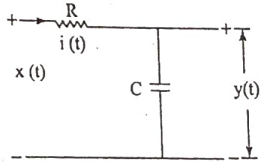


Fig. 7.43

Solution:

From Fig. 7.43 we have

$$x(t)=Ri(t)+\frac{1}{C}\int_{-\infty}^{t}i(t)dt \tag{7.107}$$

and $$y(t)=\frac{1}{C}\int_{-\infty}^{t}i(t)dt \tag{7.108}$$

Taking Laplace transform, we get

$$X(s)=RI(s)+\frac{1}{Cs}I(s)\quad \text{(Let initial voltage across capacitor is zero)}$$

and $Y(s)=\frac{1}{Cs}X(s)$

$$I(s)\left[R+\frac{1}{Cs}\right]=X(s)$$

$$Y(s)=\frac{1}{Cs}\frac{X(s)}{\left(R+\frac{1}{Cs}\right)}$$

$$\frac{Y(s)}{X(s)}=\frac{1}{RCs+1}=\frac{1}{RC\left(s+\frac{1}{RC}\right)}$$

For input $x(t)=r(t)$

$$X(s)=\frac{1}{s^2}$$

Therefore

$$Y(s)=\frac{1}{RCs^2\left(s+\frac{1}{RC}\right)}$$

$$=-\frac{RC}{s}+\frac{1}{s^2}+\frac{RC}{s+\frac{1}{RC}}$$

$$y(t)=-RC+t+RC(e^{-t/RC})$$

$$=t-RC\left[1-e^{-t/RC}\right] \tag{7.109}$$

That is, for input $r(t)$

$$y(t)=t-RC(1-e^{-t/RC})$$

By using time shifting property the output $y(t)$ due to input $r(t-1)$ and $r(t-2)$ can be obtained as follows

$$(t-1)-RC\left[1-e^{-(t-1)/RC}\right] \tag{7.110}$$

and

$$(t-2)-RC\left[1-e^{-(t-2)/RC}\right] \tag{7.111}$$

respectively.

Now the output of the circuit due to input $r(t)-2r(t-1)+r(t-2)$ can be obtained using superposition property.

That is

$$y(t)=\left[t-RC(1-e^{-t/RC})\right]-2\left\{(t-1)-RC\left[1-e^{-(t-1)/RC}\right]\right\}$$
$$+\left\{(t-2)-RC\left[1-e^{-(t-2)/RC}\right]\right\}$$
$$=RCe^{-t/RC}-2RCe^{-(t-1)/RC}+RCe^{-(t-2)/RC} \tag{7.112}$$

Solved Problem 7.54

Find whether the system given in Fig. 7.44 is BIBO stable or not

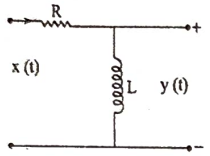


Fig. 7.44

Solution:

The system is said to be BIBO stable if it satisfies the condition

$$\int_{-\infty}^{\infty} |h(t)|dt < \infty \qquad (7.113)$$

The impulse response of the given circuit is

$$h(t) = \delta(t) - \frac{R}{L} e^{-(R/L)t} u(t) \quad \text{(see solved probelm 7.49ii)} \qquad (7.114)$$

$$\int_{-\infty}^{\infty} |h(t)|\, dt = \int_{0}^{\infty} \left[\delta(t) - \frac{R}{L} e^{-(R/L)t}\right] dt$$

$$= \int_{0}^{\infty} \delta(t)\, dt - \frac{R}{L} \int_{0}^{\infty} e^{-(R/L)t} dt$$

$$= 1 - \frac{R}{L} \frac{e^{-(R/L)t}}{-R/L}\Big|_0^{\infty}$$

$$= 1 + e^{-(R/L)t}\Big|_0^{\infty}$$

$$= 1 + (0 - 1)$$

$$= < \infty$$

Therefore the system is stable.

Practice Problem 7.13

Find the transfer function and impulse response of each system given below

i) $\frac{dy(t)}{dt} + 10y(t) = 3x(t)$

ii) $\frac{d^2y(t)}{dt^2} + 11\frac{dy(t)}{dt} + 30y(t) = x(t) + \frac{d}{dt}x(t)$

iii) $\frac{d^2y(t)}{dt^2} + 7\frac{dy(t)}{dt} + 6y(t) = 7x(t) + 11\frac{d}{dt}x(t)$

Practice Problem 7.14

For the circuit shown in Fig. E7.1 find $y(t)$

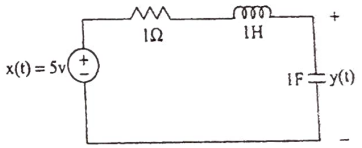


Fig. E.7.1

Solved Problem 7.55

i) Find the impulse response of the system shown in Fig. 7.45.

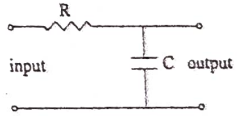


Fig. 7.45

ii) If $x(t) \leftrightarrow X(s)$, derive the Fourier transform of $x(t-1)$ and $e^{-at}\, u(t)$ in terms of $X(s)$

Anna University Nov. 2002, AU Nov 2009

Solution:

Given

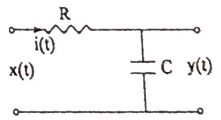


Fig. 7.46

The differential equation governing the response of the circuit is

$$Ri(t) + \frac{1}{C}\int_{-\infty}^{t} i(t)\,dt = x(t) \qquad (7.115)$$

Applying Laplace transform on both sides with zero initial conditions yields

$$RI(s) + \frac{1}{Cs}I(s) = X(s)$$

$$I(s) = \frac{X(s)}{R + \frac{1}{Cs}}$$

$$= \frac{Cs}{RCs+1}X(s) \qquad (7.116)$$

We know that

$$y(t) = \frac{1}{C}\int_{-\infty}^{t} i(t)dt \qquad (7.117)$$

$$\Rightarrow \quad Y(s) = \frac{1}{Cs}I(s) \quad \text{(initial voltage across capacitor is zero)}$$

$$= \frac{1}{Cs}\cdot\frac{Cs}{RCs+1}X(s)$$

$$\Rightarrow \quad H(s) = \frac{Y(s)}{X(s)} = \frac{1}{RCs+1} \qquad (7.118)$$

The impulse response

$$h(t) = L^{-1}[H(s)]$$

$$= L^{-1}\left[\frac{1}{RCs+1}\right]$$

$$= \frac{1}{RC}L^{-1}\left[\frac{1}{s+\frac{1}{RC}}\right]$$

$$= \frac{1}{RC}e^{-t/RC} \quad \text{for } t \geq 0$$

or

$$h(t) = \frac{1}{RC}e^{-t/RC}u(t) \qquad (7.119)$$

(ii) Refer to time shift & frequency shift properties of Laplace transform in section (7.5.5) and (7.5.6) respectively

Practice Problem 7.15

Find the response of the RC low-pass system for the step input

Hint : RC Low – pass system is shown in Fig.7.45

$$H(s) = \frac{1}{RCs+1} \quad ; \text{for step input } X(s) = \frac{1}{s}$$

$$Y(s) = \frac{1}{s(RCs+1)} \quad ; y(t) = L^{-1}\left[\frac{1/RC}{s(s+1/RC)}\right]$$

$$y(t) = \frac{1}{RC}\left[1 - e^{-t/RC}\right]$$

Solved Problem 7.56

Find the output of the system shown in Fig.7.47 for the input $e^{-2t}u(t)$ using Laplace transform (Anna University APRIL 2003)

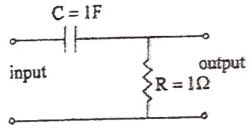


Fig. 7.47

Solution:

Given

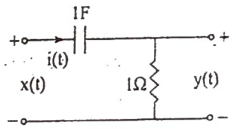


Fig. 7.48

The differential equation representation of the circuit is

$$x(t) = \frac{1}{C}\int_{-\infty}^{t} i(t)dt + Ri(t) \qquad (7.120)$$

$$= \int_{-\infty}^{t} i(t)dt + i(t) \qquad (7.121)$$

Taking Laplace transform on both sides with zero initial conditions, we get

$$X(s) = \frac{I(s)}{s} + I(s)$$

$$= I(s)\left[\frac{1+s}{s}\right] \qquad (7.122)$$

The output

$$y(t) = Ri(t)$$
$$= i(t) \quad (7.123)$$
$$\Rightarrow \quad Y(s) = I(s)$$
$$= \frac{s}{s+1}X(s) \quad (7.124)$$

For input $x(t) = e^{-2t}u(t), X(s)$ is

$$x(t) = e^{-2t}u(t)$$
$$X(s) = \frac{1}{s+2} \quad (7.125)$$

By substituting Eq. (7.125) in Eq. (7.124) we get

$$Y(s) = \frac{s}{s+1} \cdot \frac{1}{s+2} \quad (7.126)$$
$$= \frac{A}{s+1} + \frac{B}{s+2}$$
$$A = (s+1)\left.\frac{s}{(s+1)(s+2)}\right|_{s=-1} = -1$$
$$B = (s+2)\left.\frac{s}{(s+1)(s+2)}\right|_{s=-2} = 2$$
$$Y(s) = \frac{-1}{s+1} + \frac{2}{s+2} \quad (7.127)$$

Taking inverse Laplace transform on both sides, we get

$$y(t) = -e^{-t} + 2e^{-2t} \quad \text{for } t \geq 0$$
$$= -e^{-t}u(t) + 2e^{-2t}u(t) \quad (7.128)$$

Solved Problem 7.57 Find the inverse Laplace transform of

(Anna University April 2003)

$$X(s) = \frac{2s^2+9s-47}{(s+1)(s^2+6s+25)}$$

Solution:

Given

$$X(s) = \frac{2s^2+9s-47}{(s+1)(s^2+6s+25)}$$

$$\frac{2s^2+9s-47}{(s+1)(s^2+6s+25)} = \frac{A}{s+1} + \frac{Bs+C}{s^2+6s+25}$$
$$= \frac{A(s^2+6s+25)+(Bs+C)(s+1)}{(s+1)(s^2+6s+25)}$$
$$= \frac{As^2+6As+25A+Bs^2+Cs+Bs+C}{(s+1)(s^2+6s+25)}$$
$$= \frac{(A+B)s^2+(6A+B+C)s+25A+C}{(s+1)(s^2+6s+25)}$$

Comparing L.H.S and R.H.S. we get

$$A+B=2$$
$$6A+B+C=9$$
$$25A+C=-47$$

Solving for A, B and C we get

$$A = \frac{-27}{10}; \quad B = \frac{47}{10}; \quad C = \frac{41}{2} \quad (7.129)$$

$$X(s) = \frac{-27}{10(s+1)} + \frac{\frac{47}{10}s + \frac{41}{2}}{s^2+6s+25} \quad (7.130)$$
$$= \frac{-27}{10}\frac{1}{s+1} + \frac{47}{10}\left[\frac{s+\frac{205}{47}}{(s+3)^2+4^2}\right]$$
$$= \frac{-27}{10} \cdot \frac{1}{s+1} + \frac{47}{10}\left[\frac{s+3}{(s+3)^2+4^2} + \frac{16}{47} \cdot \frac{4}{(s+3)^2+4^2}\right]$$

Taking inverse Laplace transform on both sides we get

$$x(t) = \frac{-27}{10}e^{-t} + \frac{47}{10}e^{-3t}\cos 4t + \frac{8}{5}e^{-3t}\sin 4t$$
$$= \frac{-27}{10}e^{-t} + e^{-3t}\left[\frac{47}{10}\cos 4t + \frac{8}{5}\sin 4t\right] \quad (7.131)$$

7.10 Stability

The stability concept in time domain is discussed in detail in section (4.8). Recall that, the necessary condition for a system to be stable is its impulse response should be absolutely integrable.We now turn our attention to study the stability concept in s-domain. We know that the transfer function of a system $H(s)$ is the Laplace transform of impulse response $h(t)$. In time domain, the stability concept is studied using impulse response $h(t)$ of the system. In the same way, the transfer function $H(s)$ can be used to study the stability in s- domain.

Let us consider the transfer function of a causal time-invariant system

$$H(s) = \frac{b_M s^M + b_{M-1} s^{M-1} + \cdots + b_0}{s^N + a_{N-1} + \cdots + a_0} \tag{7.132}$$

The denominator of the transfer function is known as characteristic equation of the system and the roots of characteristic equation are poles. The denominator polynomial of $H(s)$ can be factored and we can express $H(s)$ in partial fraction form as

$$H(s) = \frac{A_1}{s - s_1} + \frac{A_2}{s - s_2} + \cdots + \frac{A_N}{s - s_N} \tag{7.133}$$

The poles of the transfer function $H(s)$ can be either real, imaginary or complex. The poles may lie in left half of s-plane or right half of s-plane. Depending on the location and type of poles the system has different types of impulse responses $(h(t) = L^{-1}[H(s)])$.

Real poles

If the pole is a real number $s_k = a$ and is located in the left half of the s-plane as shown in Fig. 7.49a the corresponding impulse response is $A_k e^{-at} u(t)$. This indicates that after a long time the impulse response lies away to zero. If a is positive, then the pole is in the right half of the s-plane and the corresponding impulse response is $A_k e^{at} u(t)$. That is, the impulse response increases without any bound as t increases. Thus a stable system must have real poles of $H(s)$ in the left half of the s-plane.

If the pole is at origin, then the response is a constant value for $t \geq 0$.

Complex poles

If the poles are complex conjugate of the form $s = -a \pm jb$, then the transfer function $H(s)$ in partial fraction form is

$$\frac{A_1}{s + a + jb} + \frac{A_1^*}{s + a - jb} \tag{7.134}$$

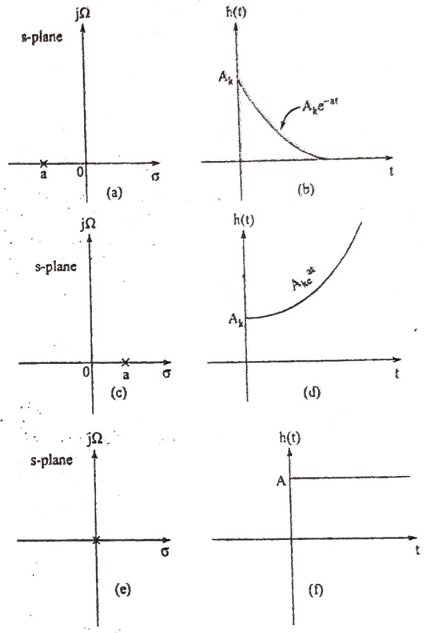


Fig. 7.49

If real part a is negative, then the poles are on the left half of the s-plane as shown in Fig. 7.50a. The corresponding impulse response is of the form $Ae^{-at}\cos(\Omega t + \varphi)$. If a is positive, then the poles are in the right half of the s-plane and the corresponding impulse response is $Ae^{at}\cos(\Omega t + \varphi)$. Note that the left plane poles have impulse response that is decaying sinusoidally and approaching zero as t increases where as the right plane poles corresponds to time function that is increasing sinusoidally without bound. If the poles are on $j\Omega$-axis, then the impulse response is of the form $A\cos(\Omega t + \varphi)$. In this case, there is no damping and the response continuously oscillates between two limits as shown in Fig. 7.50f.

Multiple order poles

For multiple real poles in the left half of plane, the response is of the form

$$(A_1 + A_2 t)e^{at} \tag{7.135}$$

If a is negative, the exponential time function decreases, faster than the linearly increasing time factor. The response ultimately lies out. If a is positive then the response increases exponentially with out bound.

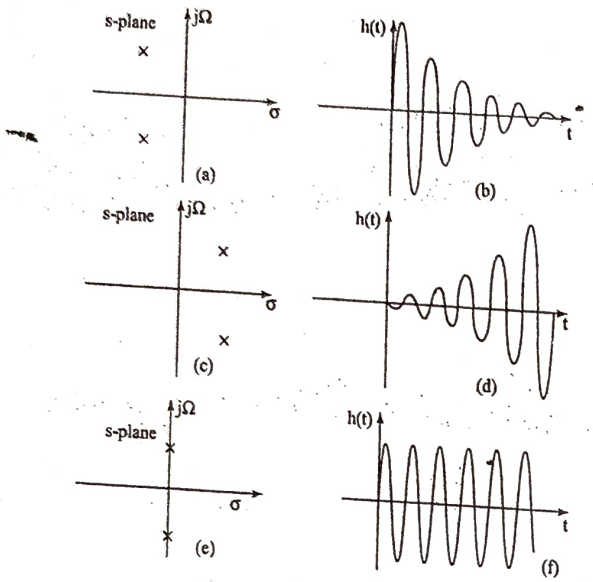


Fig. 7.50

For multiple poles on the imaginary axis, the response function is of the form

$$h(t) = A_1 \cos(\Omega t + \varphi_1) + A_2 t(\cos \Omega t + \varphi_2) \tag{7.136}$$

The first term on the right hand side is a sustained oscillation function. The second term is a time-modulated oscillatory function that increases with time. Therefore the response is unbounded.

Review Questions

1. What is the condition on poles location for system stability?
2. Draw the impulse response of a second order system for different pole location in s-plane.

Relationship between Laplace transform and Fourier transform

In this section we will study the relationship between the Laplace transform and Fourier transform.

We know that the Laplace transform of a signal $x(t)$ is given by

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt \tag{7.137}$$

and the Fourier transform of $x(t)$ is given by

$$X(j\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt \tag{7.138}$$

The Eq. (7.137) and Eq. (7.138) are identical when $s = j\Omega$. In the s-plane this corresponds to the points on the imaginary axis. Hence $X(j\Omega)$ is equal to $X(s)$ when $s = j\Omega$. For $X(j\Omega)$ to exist, the signal $x(t)$ must be absolutely integrable. Therefore, if $x(t)$ is absolutely integrable, we may obtain the Fourier transform from the Laplace transform by setting $\sigma = 0$. That is

$$X(j\Omega) = X(s)\big|_{s=j\Omega} \tag{7.139}$$

7.11 Block Diagram Representation

A system consists of number of components or elements. Each element is represented by a block. All the blocks are interconnected by lines with arrows indicating the flow of signals from one block to another. By inspecting a block diagram of a system gives an overall idea of the system structure and interrelationships that exists among the various components.

Definition: A block diagram is interconection of subsystems representing certain basic mathematical operations in such a way that the overall diagram obeys the system's mathematical model.

The basic elements of the block diagram is the single block shown in Fig. 7.51a. The input to the block is $x(t)$ and the output is $y(t)$. Here the input and output are represented as a function of time. However it is advisable to use frequency domain representation of signals as shown in Fig. 7.51b, where $X(s)$ is the Laplace transform of the input, $Y(s)$ is the Laplace transform of the output and $H(s)$ is the transfer function of the element.

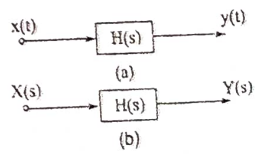


Fig. 7.51 A single block

The output of the system is given by

$$Y(s) = H(s)\, X(s) \tag{7.140}$$

Other elements generally used in block diagram are summer, gain and multiplier.

7.11.1 Summer

The addition and subtraction of signals are represented by a summer, or summing junction. Pictorially they are represented by a small circle that has at least two input terminals and one output terminal. Next to each input terminal is a plus or minus symbol indicating the sign associated with the variable. The output variable is the algebraic sum of all the input variables.

For example, in Fig. 7.52 the inputs to the summer are $X_1(s)$, $X_2(s)$ and $X_3(s)$ where $X_1(s)$ and $X_2(s)$ are assigned + sign and $X_3(s)$ is assigned − sign. Therefore the output of the summer is $X_1(s) + X_2(s) - X_3(s)$.

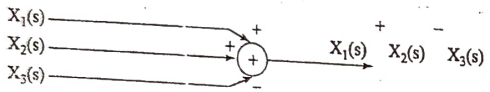


Fig. 7.52 A Summer

7.11.2 Gain

The multiplication of a variable by a constant is represented by an arrow. The gain can be positive or negative. Its magnitude can be larger or smaller than 1.

Fig. 7.53

Transfer function

From section (7.8) we recall that the Laplace transform of the output is product of transfer function and Laplace transform of input which we conveniently represent as

$$Y(s) = X(s)\, H(s) \tag{7.141}$$

A block containing the transfer function is indicated as shown in Fig. 7.54.

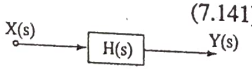


Fig. 7.54

7.11.3 Feed back

Fig. 7.55 shows a single loop feedback system. The open loop transfer function of the system is in $G(s)$ and feedback path transfer function is $H(s)$. The system input and output are $X(s)$ and $Y(s)$ respectively. Here the output is fedback to the input. The overall transfer function can be obtained by simplifying the algebraic equation corresponding to the block diagram.

First note that $$C(s) = G(s)\, E(s) \tag{7.142}$$

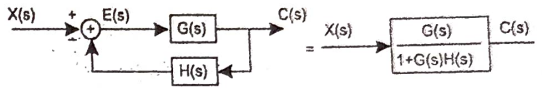


Fig. 7.55 Feed back

and $$E(s) = X(s) - H(s)\, G(s) \tag{7.143}$$

Substituting Eq. (7.143) into Eq. (7.142) yields

$$E(s) = X(s) - H(s)\, G(s) E(s)$$
$$\Rightarrow \quad E(s)\,[1 + G(s)\, H(s)] = X(s)$$
$$E(s) = \frac{X(s)}{1 + G(s) H(s)} \tag{7.144}$$

Substituting Eq. (7.144) in Eq. (7.142)

$$C(s) = \frac{G(s)\, X(s)}{1 + G(s)\, H(s)} \tag{7.145}$$

The overall transfer function

$$T(s) = \frac{C(s)}{X(s)} = \frac{G(s)}{1 + G(s)\, H(s)} \tag{7.146}$$

7.11.4 Integrator

The integrator block is used in realization of systems. If the input and output of the integrator block are $x(t)$ and $y(t)$, then the output obeys the relationship

$$y(t) = y(0^+) + \int_0^t x(t)\, d\tau \tag{7.147}$$

Setting $y(0^+)$ to zero and taking Laplace transform gives

$$Y(s) = \frac{X(s)}{s}, \tag{7.148}$$

which can be represented by block diagram as shown in Fig. 7.56

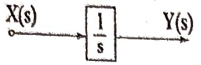


Fig. 7.56

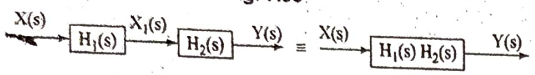


Fig. 7.57

7.11.5 Cascade connection of blocks

Two blocks are said to be in cascade when the output of one goes as input to the other. Fig. 7.57 shows the cascade connection of two blocks.

From Fig. 7.57, we find that

$$\frac{Y(s)}{X_1(s)} = H_2(s) \text{ and } \frac{X_1(s)}{X(s)} = H_1(s) \tag{7.149}$$

The overall transfer function of the system is

$$\frac{Y(s)}{X(s)} = \frac{Y(s)}{X_1(s)} \cdot \frac{X_1(s)}{X(s)} = H_1(s)\, H_2(s) \tag{7.150}$$

That is the transfer function of blocks in cascade is the product of the transfer function of individual blocks.

7.11.6 Parallel connection of blocks

Two systems are said to be parallel when they have common input and their outputs are combined by a summer. Fig. 7.58 shows the blocks that are in parallel, the overall transfer function can be obtained by adding individual transfer functions.

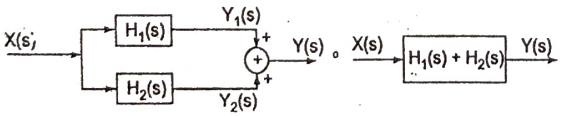


Fig. 7.58 Parallel connection of blocks.

From Fig. 7.58

$$\begin{aligned} Y(s) &= Y_1(s) + Y_2(s) \\ Y_1(s) &= H_1(s)X(s) \\ Y_2(s) &= H_2(s)X(s) \\ Y(s) &= H_1(s)X(s) + H_2(s)X(s) \\ &= X(s)[H_1(s) + H_2(s)] \end{aligned} \tag{7.151}$$

$$\frac{Y(s)}{X(s)} = H_1(s) + H_2(s) \tag{7.152}$$

7.12 Signal-flowgraph

Fig. 7.59a shows the block diagram representation of a system with transfer function $H(s)$. In signal - flow graph representation each block is represented by a directed line shown in Fig. 7.59b.

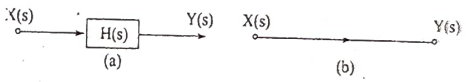


Fig. 7.59

In signal flow graph the systems are represented by a line with arrow and the system variables are represented by nodes. In addtion, all the summing points in block diagram are replaced by nodes in signal flow graph.

> ***Definition:*** A signal flow graph is a graphical representation of the relationship between the variables of a set of linear algebraic equations.

The different signal flow graph terms are explained below.

1. **Node:-** It is a system variable which is equal to the sum of all incoming signals at the node.
2. **Transmittance:-** Transmittance is gain between two nodes. Such gains are expressed in terms of transfer function.
3. **Branch:-** A branch is a directed line between nodes. The branch gain is known as transmittance.
4. **Input node or source:-** An input node or source is a node that has only out going branches. This corresponds to an independent variable.
5. **Output node or sink:-** It is a node that has only incoming branches. This corresponds to a dependent variable.

Mixed node:- A mixed node has both incoming and outgoing signals.

7. **Path:-** A path is a transversal of connected branches in the direction of the branch arrows such that no node traverse more than once.
8. **Loop:-** It is a closed path.

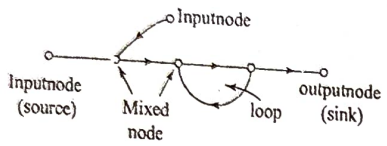


Fig. 7.60 A signal flow graph

7.13 System Realization

In this section we study different types of system realizations. They are 1. Direct form-I realization 2. Direct-form II realization 3. Cascade form realization 4. Parallel form realization. A transfer function can be realized using integrators and differentiators. However, differentiators are not used in realizing practical systems. The reason is that differentiators amplifies high frequency noise. The integrators suppress high frequency noise. Hence only integrators are used for realization of systems.

The transfer function of ideal integrator is given by

$$H(s) = \frac{1}{s} \tag{7.153}$$

Fig. 7.61 represents the transfer function of an ideal integrator

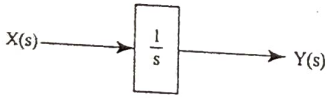


Fig. 7.61

Consider a system defined by differential equation

$$\frac{d^N y(t)}{dt^N} + a_{N-1}\frac{d^{N-1}y(t)}{dt^{N-1}} + \cdots\cdots + a_1\frac{dy(t)}{dt} + a_0\, y(t)$$
$$= b_M\frac{d^M x(t)}{dt^M} + b_{M-1}\frac{d^{M-1}x(t)}{dt^{M-1}} + \cdots\cdots + b_1\frac{dx(t)}{dt} + b_0 x(t) \tag{7.154}$$

The transfer function of the above system is

$$H(s) = \frac{Y(s)}{X(s)} = \frac{b_M s^M + b_{M-1}s^{M-1} + \cdots + b_1 s + b_0}{s^N + a_{N-1}s^{N-1} + \cdots + a_1 s + a_0} \tag{7.155}$$

If $M > N$, the system is non-causal, and the system is not physically realizable. Therefore the value of M must be less than or equal to N for physical realizability.

7.13.1 Direct form-I

Consider a system with transfer function

$$H(s) = \frac{Y(s)}{X(s)} = \frac{s^2+3s+2}{s^2+2s+2} \tag{7.156}$$

To realize the above system, first divide the numerator and denominator with s^2 (highest power of s). Then we obtain

$$\frac{Y(s)}{X(s)} = \frac{1+\frac{3}{s}+\frac{2}{s^2}}{1+\frac{2}{s}+\frac{2}{s^2}} \tag{7.157}$$

$$\Rightarrow \quad \left(1+\frac{2}{s}+\frac{2}{s^2}\right)Y(s) = X(s)\left[1+\frac{3}{s}+\frac{2}{s^2}\right] \tag{7.158}$$

$$Y(s)+\frac{2Y(s)}{s}+\frac{2Y(s)}{s^2} = X(s)+\frac{3X(s)}{s}+\frac{2X(s)}{s^2} \tag{7.159}$$

Let

$$X(s)+\frac{3X(s)}{s}+\frac{2X(s)}{s^2} = W(s) \tag{7.160}$$

Then

$$Y(s)+\frac{2Y(s)}{s}+\frac{2Y(s)}{s^2} = W(s) \tag{7.161}$$

$$\Rightarrow \quad Y(s) = W(s) - \frac{2}{s}Y(s) - \frac{2}{s^2}Y(s) \tag{7.162}$$

First we realize Eq. (7.160) using integrators, as shown in Fig. 7.62.

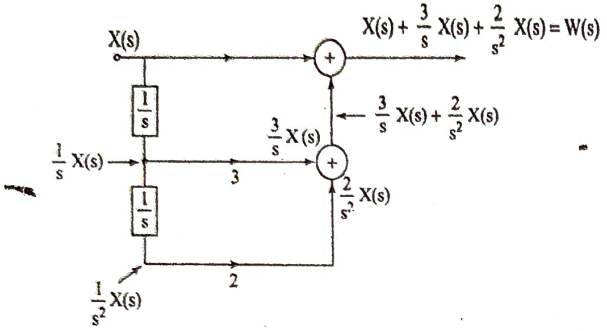


Fig. 7.62 Realization of Eq. (7.160)

The realization of Eq. (7.162) is

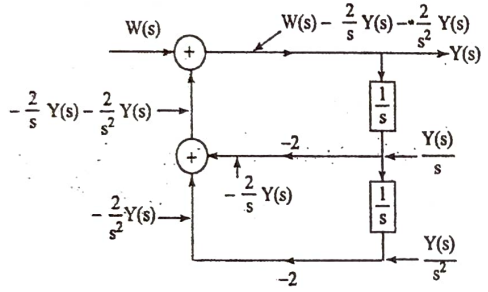


Fig. 7.63 Realization of Eq. (7.162)

Combine Fig. 7.62 and Fig. 7.63 to get the realization of the transfer function $H(s)$.

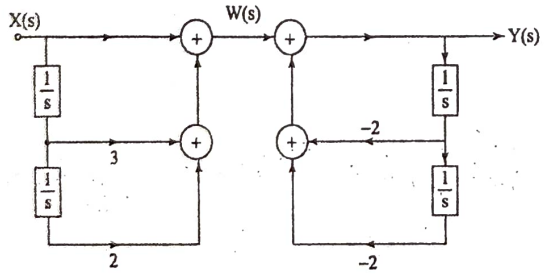


Fig. 7.64 Realization of $H(s)$

Solved Problem 7.58 Realize the transfer function of the system given in direct form I

$$H(s)=\frac{s+1}{s^2+3s+5} \tag{7.163}$$

Solution:

$$H(s)=\frac{Y(s)}{X(s)}=\frac{s+1}{s^2+3s+5}=\frac{\frac{1}{s}+\frac{1}{s^2}}{1+\frac{3}{s}+\frac{5}{s^2}} \tag{7.164}$$

$$\Rightarrow\quad Y(s)\left[1+\frac{3}{s}+\frac{5}{s^2}\right]=\left(\frac{1}{s}+\frac{1}{s^2}\right)X(s)$$

$$Y(s)+\frac{3Y(s)}{s}+\frac{5Y(s)}{s^2}=\frac{X(s)}{s}+\frac{X(s)}{s^2} \tag{7.165}$$

Let

$$\frac{X(s)}{s}+\frac{X(s)}{s^2}=W(s) \tag{7.166}$$

then

$$Y(s)+\frac{3Y(s)}{s}+\frac{5Y(s)}{s^2}=W(s) \tag{7.167}$$

The realization of Eq. (7.166) using integrators can be obtained as in Fig. 7.65.

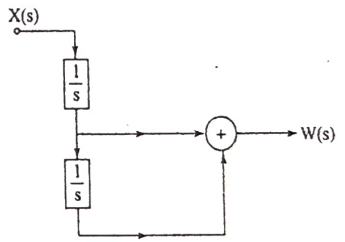


Fig. 7.65 Realization of Eq. 7.166.

Similarly the realization of Eq. (7.167) is obtained as in Fig. 7.66.

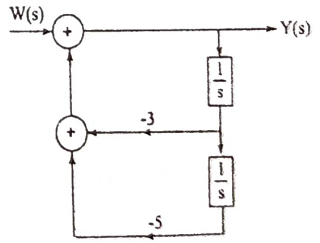


Fig. 7.66 Realization of Eq. 7.167.

Combining both Fig. 7.65 and Fig. 7.66, we get realization of $H(s)$

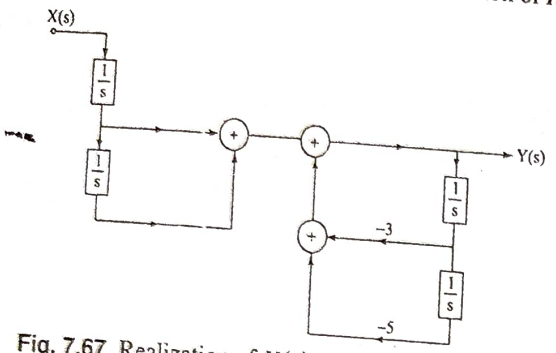


Fig. 7.67 Realization of $H(s)$ of solved problem 7.58

Solved Problem 7.59 Realize the system described by following differential equation in direct form I

$$\frac{d^3y(t)}{dt^3}+\frac{4d^2y(t)}{dt^2}+\frac{7dy(t)}{dt}+8y(t)=\frac{5d^2x(t)}{dt^2}+\frac{4dx(t)}{dt}+7x(t)$$

Solution:

Given

$$\frac{d^3y(t)}{dt^3}+\frac{4d^2y(t)}{dt^2}+\frac{7dy(t)}{dt}+8y(t)=\frac{5d^2x(t)}{dt^2}+\frac{4dx(t)}{dt}+7x(t) \quad (7.168)$$

Apply Laplace-transform with zero initial conditions yields

$$s^3Y(s)+4s^2Y(s)+7sY(s)+8Y(s)=5s^2X(s)+4sX(s)+7X(s) \quad (7.169)$$

Divide Eq. (7.169) throughout by s^3

$$Y(s)+\frac{4Y(s)}{s}+\frac{7Y(s)}{s^2}+\frac{8Y(s)}{s^3}=\frac{5X(s)}{s}+\frac{4X(s)}{s^2}+\frac{7X(s)}{s^3} \quad (7.170)$$

Let

$$\frac{5X(s)}{s}+\frac{4X(s)}{s^2}+\frac{7X(s)}{s^3}=W(s) \quad (7.171)$$

then $$Y(s)+\frac{4Y(s)}{s}+\frac{7Y(s)}{s^2}+\frac{8Y(s)}{s^3}=W(s) \quad (7.172)$$

$$\Rightarrow \quad Y(s)=W(s)-\frac{4Y(s)}{s}-\frac{7Y(s)}{s^2}-\frac{8Y(s)}{s^3} \quad (7.173)$$

Realizing Eq. (7.171) and Eq. (7.173) and combining we obtain realization of $H(s)$ as shown in Fig. 7.68.

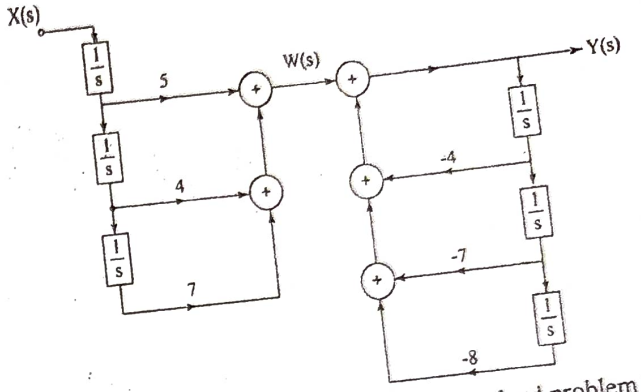


Fig. 7.68 Realization of the system given in solved problem 7.59.

Practice problem 7.16

Find direct form - I realization of the following systems with transfer function

(i) $H(s)=\frac{s^2+s+2}{s^3+4s^2+3s+2}$

(ii) $H(s)=\frac{s+1}{s^2+4s+5}$

Practice Problem 7.17

Find direct form - I realization of the following systems with differential equation

(i) $\frac{d^3y(t)}{dt^3}+5\frac{d^2y(t)}{dt^2}+6\frac{dy(t)}{dt}+9y(t)=\frac{dx(t)}{dt}+5x(t)$

(ii) $\frac{d^2y(t)}{dt^2}+6\frac{dy(t)}{dt}+4y(t)=3x(t)$

7.13.2 Direct form - II

Consider an LTI system with transfer function

$$H(s)=\frac{Y(s)}{X(s)}=\frac{s^2+4s+2}{s^2+5s+3} \quad (7.174$$

Divide the transfer function into two parts

$$\frac{Y(s)}{X(s)} = \frac{Y(s)}{W(s)} \cdot \frac{W(s)}{X(s)} \tag{7.175}$$

where $$\frac{W(s)}{X(s)} = \frac{1}{s^2+5s+3} \tag{7.176}$$

and $$\frac{Y(s)}{W(s)} = s^2+4s+2 \tag{7.177}$$

From Eq. (7.176) we get

$$X(s) = s^2W(s)+5sW(s)+3W(s)$$
$$\Rightarrow s^2W(s) = X(s)-5sW(s)-3W(s) \tag{7.178}$$

From Eq. (7.177) we can obtain

$$Y(s) = s^2W(s)+4sW(s)+2W(s) \tag{7.179}$$

To realize Eq. (7.178) assume $s^2W(s)$ is available. Pass this signal to successive integrators to get signals sW(s) and W(s) and then realize Eq. (7.178) as shown in Fig. 7.69.

The Eq. 7.179 can be realized as shown in Fig. 7.70.

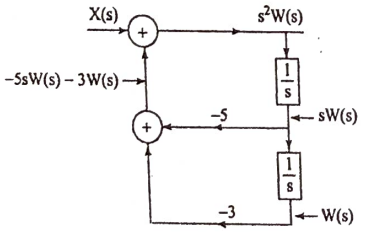


Fig. 7.69 Realization of Eq. (7.178).

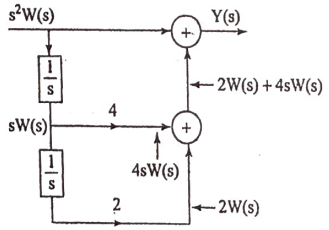
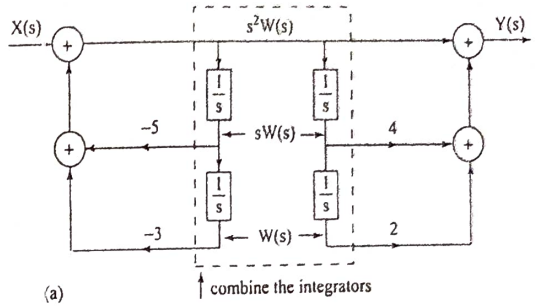


Fig. 7.70 Realization of Eq. (7.179).

Cascading Fig. (7.69) and Fig. (7.70), we obtain the realization of $H(s)$ as shown in Fig. 7.71b.



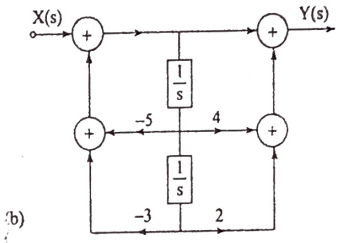


Fig. 7.71 Realization of $H(s)$

Solved Problem 7.60 Realize the systems given in direct form - II.

(i) Given

$$H(s) = \frac{s+1}{s^2+3s+5} \tag{7.180}$$

Let $$\frac{Y(s)}{W(s)} = s+1 \tag{7.181}$$

$$\Rightarrow \quad Y(s) = sW(s)+W(s) \tag{7.182}$$

and $$\frac{W(s)}{X(s)} = \frac{1}{s^2+3s+5} \tag{7.183}$$

$$\Rightarrow \quad s^2W(s) = X(s)-3sW(s)-5W(s) \tag{7.184}$$

Realizing Eq. (7.184) and Eq. (7.182) and combining them we get the realization of $H(s)$.

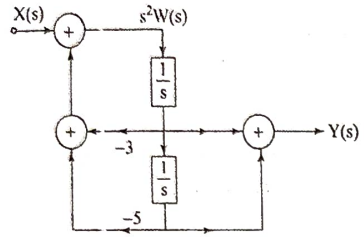


Fig. 7.72 Realization of the given system $H(s)$

(ii) Given $\frac{d^3y(t)}{dt^3}+4\frac{d^2y(t)}{dt^2}+7\frac{dy(t)}{dt}+8y(t)=5\frac{d^2x(t)}{dt^2}+4\frac{dx(t)}{dt}+7x(t)$

The transfer function of the system is

$$H(s)=\frac{Y(s)}{X(s)}=\frac{5s^2+4s+7}{s^3+4s^2+7s+8}$$

Let $\frac{Y(s)}{W(s)}=5s^2+4s+7$ and $\frac{W(s)}{X(s)}=\frac{1}{s^3+4s^2+7s+8}$

$$Y(s)=5s^2W(s)+4sW(s)+7W(s) \quad \text{and} \qquad (7.185)$$

$$s^3W(s)=X(s)-4s^2W(s)-7sW(s)-8W(s) \qquad (7.186)$$

Realizing Eq. (7.185) and Eq. (7.186) and combining we obtain realization of $H(s)$.

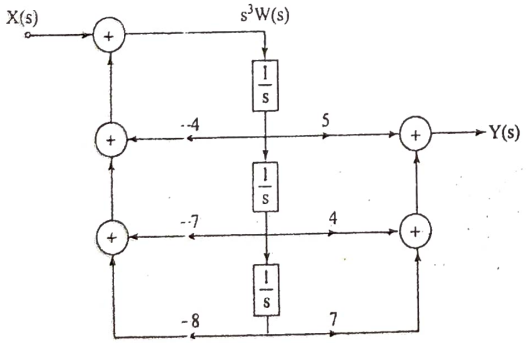


Fig. 7.73 Realization of solved problem 7.60(ii)

Practice problem 7.18

Realize the system described in practice problem 7.16 in direct form - II.

Practice Problem 7.19

Realize the system described in practice problem 7.17 in direct form - II.

7.13.3 Cascade form

In cascade form realization the given transfer function is expressed as a product of several transfer functions. Each of these transfer functions is realized in direct form-II and are cascaded.

Consider a system with transfer function

$$H(s)=\frac{5(s^2+3s+2)}{(s^2+8s+15)(s+0.5)} \qquad (7.187)$$

This can be expressed as a product of several transfer functions, that is

$$H(s)=\underbrace{\left(\frac{5}{s+0.5}\right)}_{H_1(s)}\underbrace{\left(\frac{s+1}{s+5}\right)}_{H_2(s)}\underbrace{\left(\frac{s+2}{s+3}\right)}_{H_3(s)} \qquad (7.188)$$

Now we can realize each of the transfer function as shown in Fig. 7.74.

The realization of $H_1(s)=\frac{5}{s+0.5}$ is shown in Fig. 7.74a

The realization of

$$H_2(s)=\frac{s+1}{s+5}$$

and

$$H_3(s)=\frac{s+2}{s+3}$$

...shown in Fig. 7.74b and 7.74c respectively.

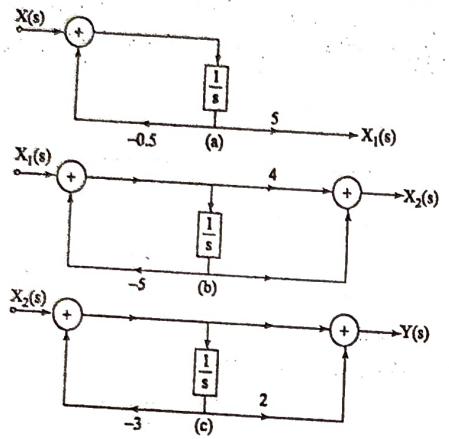


Fig. 7.74

These individual realizations can be connected in cascade to get the cascade realization of the transfer function $H(s)$, as shown in Fig. 7.75.

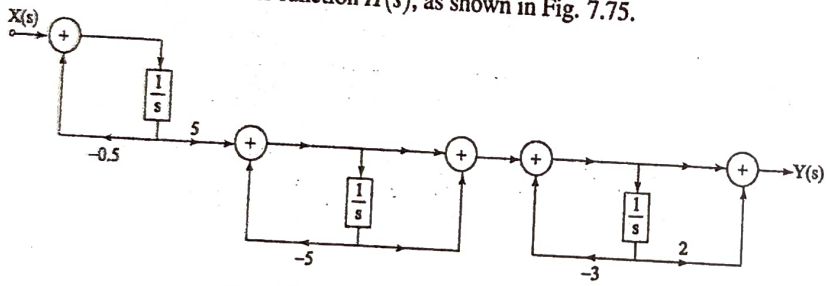


Fig. 7.75 Cascade realization of $H(s)$

Solved Problem 7.61 Realize

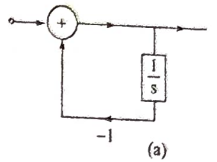
$$H(s) = \frac{s(s+2)}{(s+1)(s+3)(s+4)}$$ in cascade form

Solution:

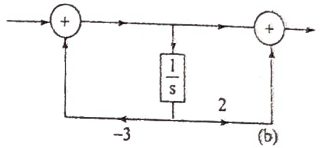
$$H(s) = \frac{s(s+2)}{(s+1)(s+3)(s+4)}$$

let $H_1(s) = \frac{s}{s+1}$; $H_2(s) = \frac{s+2}{s+3}$; $H_3(s) = \frac{1}{s+4}$

The realization of $H_1(s)$ is



The realization of $H_2(s)$ is



The realization of $H_3(s)$ is

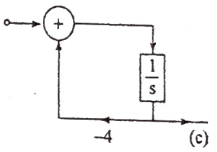


Fig. 7.76

Cascading all the above figures we get realization of $H(s)$.

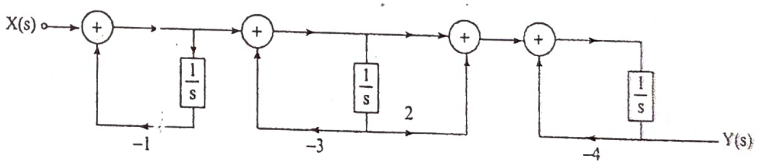


Fig. 7.77 Cascade Realization of $H(s)$.

Practice Problem 7.20

Realize the following systems in cascade form

(i) $H(s) = \frac{s+4}{(s+3)(s+2)}$

(ii) $H(s) = \frac{s^2+4s+3}{s^3+11s^2+38s+40}$

Practice Problem 7.21

Realize the following systems in cascade form

(i) $\frac{d^2y(t)}{dt^2}+5\frac{dy(t)}{dt}+6y(t)=\frac{d^2x(t)}{dt^2}+\frac{dx(t)}{dt}$

(ii) $\frac{d^3y(t)}{dt^3}+6\frac{d^2y(t)}{dt^2}+11\frac{dy(t)}{dt}+6y(t)=\frac{dx(t)}{dt}+4x(t)$

7.13.4 Parallel form realization

In parallel form realization, the given transfer function is expressed into its partial fractions and each factor is realized in direct form II and are connected in parallel.

Let us consider the transfer function

$$H(s)=\frac{s(s+2)}{(s+1)(s+3)(s+4)} \tag{7.189}$$

Now express the transfer function in partial fraction form as

$$H(s)=\frac{A}{s+1}+\frac{B}{s+3}+\frac{C}{s+4} \tag{7.190}$$

$$A=(s+1)\frac{s(s+2)}{(s+1)(s+3)(s+4)}\bigg|_{s=-1}=\frac{(-1)(-1+2)}{(-1+3)(-1+4)}=\frac{-1}{6}$$

$$B=(s+3)\frac{s(s+2)}{(s+1)(s+3)(s+4)}\bigg|_{s=-3}=\frac{(-3)(-3+2)}{(-3+1)(-3+4)}=\frac{-3}{2}$$

$$C=(s+4)\frac{s(s+2)}{(s+1)(s+3)(s+4)}\bigg|_{s=-4}=\frac{(-4)(-4+2)}{(-4+1)(-4+3)}=\frac{8}{3}$$

$$\Rightarrow\quad H(s)=\underbrace{\frac{-1}{6(s+1)}}_{H_1(s)}+\underbrace{\frac{3}{2(s+3)}}_{H_2(s)}+\underbrace{\frac{8}{3(s+4)}}_{H_3(s)} \tag{7.191}$$

To realize $H(s)$, realize $H_1(s)$, $H_2(s)$ and $H_3(s)$ and connect in parallel.

where the coefficients c_{ij} and d_{ij} are constants Y is $p\times 1$ output vector. C is $p\times m$ output matrix and D is $p\times m$ transmission matrix.

The state model of a linear-time invariant system is thus represented by the following equations

$$\dot{Q}=AQ+BX \quad \text{state equation} \tag{7.197}$$

$$Y=CQ+DX \quad \text{output equation} \tag{7.198}$$

To develop state variable equation for linear time invariant systems, first we must choose state variables. One way is to choose physical variables that associated with energy. The selection of state variable for different systems are given below.

1. For mechanical systems the potential and kinetic energy of a mass can be represented by position and velocity. Therefore, position and velocity can be chosen as state variables.
2. For an electric circuit like RLC network, capacitors and inductors are energy storage elements. Therefore the rate of change of current in an inductor and rate of change of voltage across a capacitor can be chosen as state variables.

Solved Problem 7.62 Write state variable equation for a mechanical system shown in Fig. 7.81.

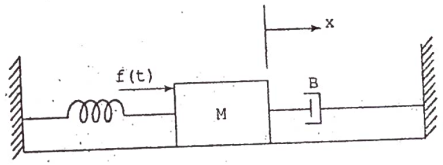


Fig. 7.81

Solution:

The differential equation of the system is

$$\frac{Md^2x(t)}{dt^2}+\frac{Bdx(t)}{dt}+k\,x(t)=f(t) \tag{7.199}$$

The position and velocity can be chosen as state variables of the system. Therefore let us define

$$q_1(t)=x(t) \quad \text{(position)} \tag{7.200}$$

$$q_2(t)=\dot{x}(t) \quad \text{(velocity)} \tag{7.201}$$

The loop equations can be written as

$$R_1 i_1 + q_1 = x(t) \tag{7.208}$$

$$R_2 i_2 + q_2 - q_1 = 0 \tag{7.209}$$

we also have

$$i_1 - i_2 = C_1 \dot{q}_1 \quad \text{and} \tag{7.210}$$

$$i_2 = C_2 \dot{q}_2 \tag{7.211}$$

Substituting Eq. (7.211) in Eq. (7.209)

$$R_2 C_2 \dot{q}_2 + q_2 - q_1 = 0$$

$$\Rightarrow \quad \dot{q}_2 = \frac{q_1}{R_2 C_2} - \frac{q_2}{R_2 C_2}. \tag{7.212}$$

From Eq. (7.208), we get

$$\begin{aligned} x(t) &= R_1 i_1 + q_1 \\ &= R_1 (C_1 \dot{q}_1 + i_2) + q_1 \qquad (\because i_1 = C_1 \dot{q}_1 + i_2) \\ &= R_1 C_1 \dot{q}_1 + R_1 C_2 \dot{q}_2 + q_1 \qquad (\because i_2 = C_2 \dot{q}_2) \\ &= R_1 C_1 \dot{q}_1 + R_1 C_2 \left[\frac{q_1}{R_2 C_2} - \frac{q_2}{R_2 C_2} \right] + q_1 \\ &= R_1 C_1 \dot{q}_1 + \left(1 + \frac{R_1}{R_2} \right) q_1 - \frac{R_1}{R_2} q_2 \end{aligned}$$

$$\Rightarrow \quad \dot{q}_1 = -\frac{1}{R_1 C_1} \left[\frac{R_1 + R_2}{R_2} \right] q_1 + \frac{1}{R_2 C_1} q_2 + \frac{x(t)}{R_1 C_1} \tag{7.213}$$

The state equations are

$$\begin{aligned} \dot{q}_1 &= -\frac{1}{R_1 C_1} \left[\frac{R_1 + R_2}{R_2} \right] q_1 + \frac{1}{R_2 C_1} q_2 + \frac{x(t)}{R_1 C_1} \\ \dot{q}_2 &= \frac{q_1}{R_2 C_2} - \frac{q_2}{R_2 C_2} \end{aligned} \tag{7.214}$$

and the output equation is

$$y = q_2 \tag{7.215}$$

The state equations in the matrix form is

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{R_1 C_1} \left(\frac{R_1 + R_2}{R_2} \right) & \frac{1}{R_2 C_1} \\ \frac{1}{R_2 C_2} & -\frac{1}{R_2 C_2} \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} + \begin{bmatrix} \frac{1}{R_1 C_1} \\ 0 \end{bmatrix} x(t) \tag{7.216}$$

$$y = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} \tag{7.217}$$

Solved Problem 7.64 For the circuit shown in Fig. 7.83 obtain state equation.

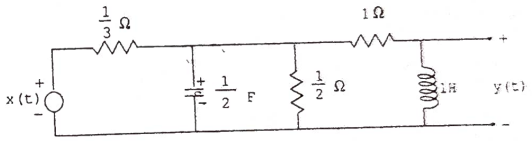


Fig. 7.83

Solution:

There are one capacitor and one inductor in the circuit diagram. Therefore voltage across capacitor and current through the inductor can be chosen as state variables.

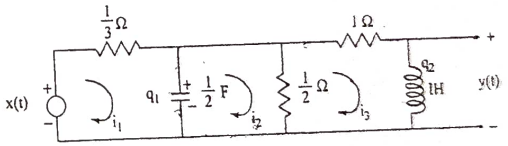


Fig. 7.84

The relationship between state variables and loop currents can be written as

$$\frac{1}{2} \dot{q}_1 = i_1 - i_2 \tag{7.218}$$

and

$$q_2 = i_3 \tag{7.219}$$

The loop equations are

$$\frac{1}{2}i_1 + q_1 = x(t) \tag{7.220}$$

$$\frac{1}{2}(i_2 - i_3) - q_1 = 0 \tag{7.221}$$

$$i_3 + q_2 + \frac{1}{2}(i_3 - i_2) = 0 \tag{7.222}$$

We have

$$\frac{1}{2}(i_2 - i_3) - q_1 = 0$$

$$\frac{1}{2}(i_2 - q_2) - q_1 = 0 \qquad (\because\ q_2 = i_3)$$

$$\Rightarrow \quad \frac{1}{2}(i_2 - q_2) = q_1$$

$$i_2 = 2q_1 + q_2 \tag{7.223}$$

and from Eq. (7.220) $\qquad i_1 = 3x(t) - 3q_1. \tag{7.224}$

Substituting Eq. (7.223) and Eq. (7.224) in Eq. (7.218), we get

$$\frac{1}{2}\dot{q}_1 = 3x(t) - 3q_1 - 2q_1 - q_2$$

$$= 3x(t) - 5q_1 - q_2$$

$$\Rightarrow \quad \dot{q}_1 = 6x(t) - 10q_1 - 2q_2$$

From Eq. (7.222), we have

$$\dot{q}_2 + \frac{1}{2}(i_3 - i_2) + i_3 = 0$$

$$\dot{q}_2 + \frac{3}{2}i_3 - \frac{1}{2}i_2 = 0$$

$$\dot{q}_2 + \frac{3}{2}q_2 - \frac{1}{2}(2q_1 + q_2) = 0 \qquad (\because\ i_2 = 2q_1 + q_2)$$

$$\dot{q}_2 + \frac{3}{2}q_2 - \frac{1}{2}q_2 - q_1 = 0$$

$$\dot{q}_2 + q_2 - q_1 = 0$$

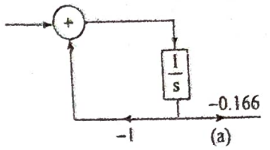
$$\dot{q}_2 = q_1 - q_2$$

The state equations are

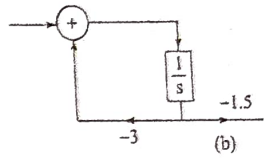
$$\dot{q}_1 = -10q_1 - 2q_2 + 6x(t)$$

$$\dot{q}_2 = q_1 - q_2 \tag{7.225}$$

Realization of $H_1(s)$



Realization of $H_2(s)$



Realization of $H_3(s)$

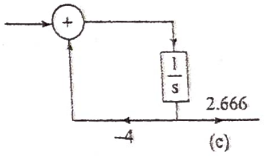


Fig. 7.78

Now the realization of $H(s)$ is given by

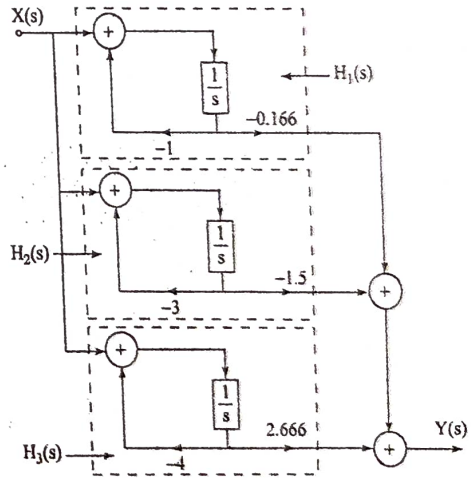


Fig. 7.79 Parallel form realization of $H(s)$.

Review questions

1. Define block diagram. What are the elements used in block diagram?
2. Define transfer function.
3. Derive the transfer function of a single loop feedback system
4. Derive the transfer function of an integrator.
5. Find the transfer function of system with
   (a) Cascade connection blocks
   (b) Parallel connection of blocks
6. Define signal flowgraph. What are the signal flow graph terms used?
7. What are the different types of system realizations? Explain in detail.
8. Why differentiators are not used in system realization?

Practice Problem 7.22

Realize the system described in practice problem 7.20 in parallel form.

Practice Problem 7.23

Realize the system described in practice problem 7.21 in parallel form.

7.14 State Space Analysis

In chapter 4 we studied the transfer function method of representing a system. It is simple and attractive. But it has certain drawbacks. They are

1. The transfer function describes only the zero-state response of a system.
2. It describes only the relationship between the input and output of a system, but does not provide any information regarding the internal state of a system.
3. It is limited to single-input and single-output systems.
4. It is applicable only for linear time-invariant systems.

These drawbacks of transfer function approach can be overcome by state variable approach.

7.14.1 State and state variable of a system

The state of a system is a minimal set of variables known as state variables, such that the knowledge of these variables for $t = t_0$, together with the knowledge of the input for $t \geq t_0$, provides the state and output for the system for $t > t_0$.

Let us consider the system shown in Fig. 7.80. The system has m inputs, p outputs and n state variables. The m inputs are $x_1(t), x_2(t) \cdots x_m(t)$; p outputs are $y_1(t), y_2(t), \cdots y_p(t)$ and n state variables are $q_1(t), q_2(t) \cdots q_n(t)$. Usually we are concerned with how the system behaviour change with time. Therefore we define rate of change of state variables as $\dot{q}_1$, $\dot{q}_2$, $\dot{q}_3 \cdots \dot{q}_n$. The time behaviour of state variables and the output of the system depends on the present state variables and the input applied. Hence we get two equations. One is relating the state variables to the inputs known as state equation and the other is relating the output variables interms of state variables and inputs known as output equation.

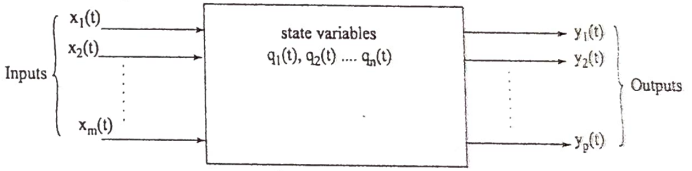


Fig. 7.80

Let us consider a continuous-time LTI system. The state variable equation for the system is of the form

$$\begin{aligned}
\dot{q}_1 &= a_{11}q_1(t) + a_{12}q_2(t) + \cdots a_{1n}q_n(t) + b_{11}x_1(t) + b_{12}x_2(t) + \cdots + b_{1m}x_m(t) \\
\dot{q}_2 &= a_{21}q_1(t) + a_{22}q_2(t) + \cdots a_{2n}q_n(t) + b_{21}x_1(t) + b_{22}x_2(t) + \cdots + b_{2m}x_m(t) \\
&\vdots \qquad\qquad \vdots \qquad\qquad \vdots \qquad\qquad \vdots \\
\dot{q}_n &= a_{n1}q_1(t) + a_{n2}q_2(t) + \cdots a_{nn}q_n(t) + b_{n1}x_1(t) + b_{n2}x_2(t) + \cdots + b_{nm}x_m(t)
\end{aligned} \tag{7.192}$$

The Eq. (7.192) can be represented in matrix form as

$$\underbrace{\begin{bmatrix}\dot{q}_1\\ \dot{q}_2\\ \vdots\\ \dot{q}_n\end{bmatrix}}_{\dot{Q}} = \underbrace{\begin{bmatrix}a_{11} & a_{12} & a_{13} & \cdots & a_{1n}\\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n}\\ \vdots & \vdots & & & \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn}\end{bmatrix}}_{A} \underbrace{\begin{bmatrix}q_1(t)\\ q_2(t)\\ \vdots\\ q_n(t)\end{bmatrix}}_{Q} + \underbrace{\begin{bmatrix}b_{11} & b_{12} & \cdots & b_{1m}\\ b_{21} & b_{22} & \cdots & b_{2m}\\ \vdots & \vdots & \vdots & \vdots\\ b_{n1} & b_{n2} & \cdots & b_{nm}\end{bmatrix}}_{B} \underbrace{\begin{bmatrix}x_1(t)\\ x_2(t)\\ \vdots\\ x_m(t)\end{bmatrix}}_{X} \tag{7.193}$$

where the coefficients a_{ij} and b_{ij} are constants. In vector matrix form Eq. (7.193) can be written as

$$\dot{Q} = AQ + BX \qquad \text{(state equation)} \tag{7.194}$$

The vector Q is called $n \times 1$ state vector and X is $m \times 1$ input vector, A is an $n \times n$ square matrix, B is an n $\times$ 1 column vector.

Similarly the output of the system can be expressed as linear combination of state variables and inputs given by

$$\begin{aligned}
y_1 &= c_{11}q_1(t) + c_{12}q_2(t) + \cdots + c_{1n}q_n(t) + d_{11}x_1(t) + \cdots + d_{1m}x_m(t)\\
y_2 &= c_{21}q_1(t) + c_{22}q_2(t) + \cdots + c_{2n}q_n(t) + d_{21}x_1(t) + \cdots + d_{2m}x_m(t)\\
&\vdots \qquad\qquad \vdots \qquad\qquad \vdots \qquad\qquad \vdots\\
y_p &= c_{p1}q_1(t) + c_{p2}q_2(t) + \cdots + c_{pn}q_n(t) + d_{p1}x_1(t) + \cdots + d_{pm}x_m(t)
\end{aligned} \tag{7.195}$$

The Eq. (7.195) can be represented in matrix form as

$$\underbrace{\begin{bmatrix}y_1\\ y_2\\ \vdots\\ y_p\end{bmatrix}}_{Y} = \underbrace{\begin{bmatrix}c_{11} & c_{12} & \cdots & c_{1n}\\ c_{21} & c_{22} & \cdots & c_{2n}\\ \vdots & \vdots & & \\ c_{p1} & c_{p2} & \cdots & c_{pn}\end{bmatrix}}_{C} \underbrace{\begin{bmatrix}q_1(t)\\ q_2(t)\\ \vdots\\ q_n(t)\end{bmatrix}}_{Q} + \underbrace{\begin{bmatrix}d_{11} & d_{12} & \cdots & d_{1m}\\ d_{21} & d_{22} & \cdots & d_{2m}\\ \vdots & \vdots & & \\ d_{p1} & d_{p2} & \cdots & d_{pm}\end{bmatrix}}_{D} \underbrace{\begin{bmatrix}x_1(t)\\ x_2(t)\\ \vdots\\ x_m(t)\end{bmatrix}}_{X} \tag{7.196}$$

The state equations in the matrix form is

$$\begin{bmatrix}\dot{q}_1\\ \dot{q}_2\end{bmatrix} = \begin{bmatrix}-10 & -2\\ 1 & -1\end{bmatrix}\begin{bmatrix}q_1\\ q_2\end{bmatrix} + \begin{bmatrix}6\\ 0\end{bmatrix}x(t) \tag{7.226}$$

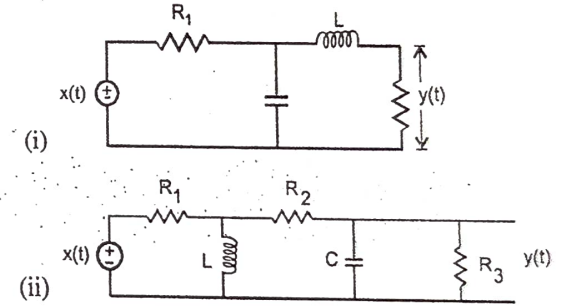
The output equation is

$$\begin{aligned}
y(t) &= \dot{q}_2\\
&= q_1 - q_2
\end{aligned} \tag{7.227}$$

$$\Rightarrow \quad y = [1, \quad -1]\begin{bmatrix}q_1\\ q_2\end{bmatrix} \tag{7.228}$$

Practice Problem 7.25

Obtain the state variable representation for the circuits shown below



7.14.3 State-space representation from system's transfer function

In section 7.13 we discussed the method for developing state variable equation using physical quantities of the system (like voltage across capacitor, current through inductor in electrical networks) as state variables. In this section we discuss an alternate approach of obtaining state variable equation from the system's transfer function.

Let us consider a system with transfer function

$$H(s) = \frac{b_3s^3 + b_2s^2 + b_1s + b_0}{s^3 + a_2s^2 + a_1s + a_0} \tag{7.229}$$

The direct form-II realization of the system is shown in Fig. 7.85.

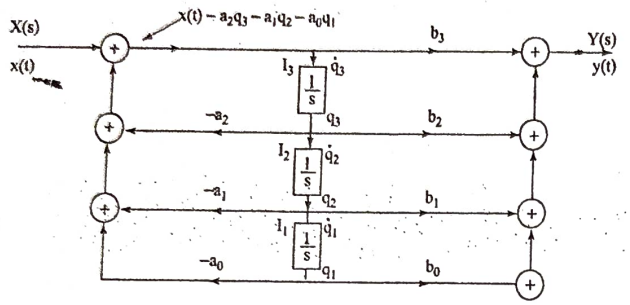


Fig. 7.85

To find state variable equation, assume that the output of each integrator as a state variable. Let the three integrator outputs are q_1, q_2, q_3 as shown in Fig. 7.85.

Then we can write

$$q_2 = \dot{q}_1 \quad (\because \text{ The output of integrator } I_1 \text{ is } q_1. \text{ Therefore the input is } \dot{q}_1 \text{ or } q_2) \tag{7.230}$$

$$\dot{q}_2 = q_3 \quad (\because \text{ The output of integrator } I_2 \text{ is } q_2. \text{ Therefore the input is } \dot{q}_2 \text{ or } q_3) \tag{7.231}$$

$$\dot{q}_3 = x - a_2 q_3 - a_1 q_2 - a_0 q_1 \tag{7.232}$$

and

$$\begin{aligned} y &= b_3\dot{q}_3 + b_2 q_3 + b_1 q_2 + b_0 q_1 \\ &= b_3[x - a_2 q_3 - a_1 q_2 - a_0 q_1) + b_2 q_3 + b_1 q_2 + b_0 q_1 \\ &= (b_0 - b_3 a_0) q_1 + (b_1 - b_3 a_1) q_2 + (b_2 - b_3 a_2) q_3 + b_3 x \end{aligned} \tag{7.233}$$

The state equations are

$$\begin{aligned} \dot{q}_1 &= q_2 \\ \dot{q}_2 &= q_3 \\ \dot{q}_3 &= -a_0 q_1 - a_1 q_2 - a_2 q_3 + x(t) \end{aligned} \tag{7.234}$$

The output equation is

$$y = (b_0 - b_3 a_0) q_1 + (b_1 - b_3 a_1) q_2 + (b_2 - b_3 a_2) q_3 + b_3 x \tag{7.235}$$

Representing the equations in matrix form we get

$$\underbrace{\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix}}_{\dot{Q}} = \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix}}_{Q} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}_{B} x(t) \tag{7.236}$$

$$y = [(b_0 - b_3 a_0), \quad (b_1 - b_3 a_1), \quad (b_2 - b_3 a_2)] \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} + b_3 x \tag{7.237}$$

Solved Problem 7.65 Find the state space representation of the following system whose differential equation representation is

$$\frac{d^3y(t)}{dt^3} + \frac{3d^2y(t)}{dt^2} + \frac{5dy(t)}{dt} + 6y(t) = \frac{d^2x(t)}{dt^2} + \frac{6dx(t)}{dt} + 5x(t)$$

Solution:

Given

$$\frac{d^3y(t)}{dt^3} + \frac{3d^2y(t)}{dt^2} + \frac{5dy(t)}{dt} + 6y(t) = \frac{d^2x(t)}{dt^2} + \frac{6dx(t)}{dt} + 5x(t) \tag{7.238}$$

Applying Laplace transform on both sides with zero initial conditions yields

$$s^3Y(s) + 3s^2Y(s) + 5sY(s) + 6Y(s) = s^2X(s) + 6sX(s) + 5X(s)$$

$$\Rightarrow \quad H(s) = \frac{Y(s)}{X(s)} = \frac{s^2 + 6s + 5}{s^3 + 3s^2 + 5s + 6} \tag{7.239}$$

The direct form-II realization of the system is given by

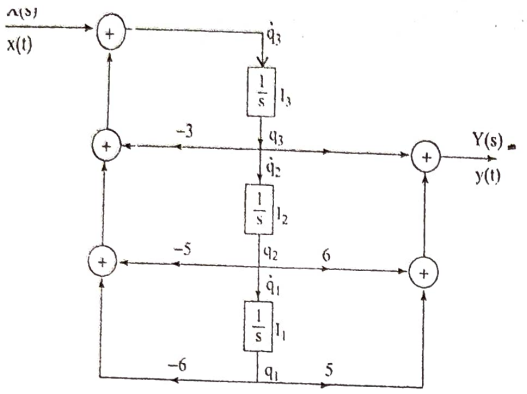


Fig. 7.86

Let us take the output of integrators as q_1, q_2, q_3 as shown in Fig. 7.86.
Then we can write

$$\dot{q}_1 = q_2$$
$$\dot{q}_2 = q_3$$
$$\dot{q}_3 = -6q_1 - 5q_2 - 3q_3 + x(t) \tag{7.240}$$

The output

$$y(t) = q_3 + 6q_2 + 5q_1 \tag{7.241}$$

The matrix representation is

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -5 & -3 \end{bmatrix}}_{A} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}_{B} x(t) \tag{7.242}$$

$$y = [1 \quad 6 \quad 5] \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} \tag{7.243}$$

$$\Rightarrow A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -5 & -3 \end{bmatrix}; B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad C = [1 \quad 6 \quad 5]$$

Solved Problem 7.66 For the given transfer function of system obtain state space representation

$$H(s) = \frac{s^2 + 3s + 4}{s^2 + 7s + 13} \tag{7.244}$$

Solution:

Given

$$H(s) = \frac{s^2 + 3s + 4}{s^2 + 7s + 13}$$
$$H(s) = \frac{Y(s)}{X(s)} = \frac{Y(s)}{W(s)} \cdot \frac{W(s)}{X(s)}$$

Let $\frac{Y(s)}{W(s)} = s^2 + 3s + 4$ and $\frac{W(s)}{X(s)} = \frac{1}{s^2 + 7s + 13}$

Now

$$Y(s) = s^2 W(s) + 3sW(s) + 4W(s) \quad \text{and} \tag{7.245}$$
$$s^2 W(s) = X(s) - 7sW(s) - 13W(s) \tag{7.246}$$

Realizing Eq. (7.245) and Eq. (7.246) and combining we get direct form- II realization of the system as shown in Fig. 7.87

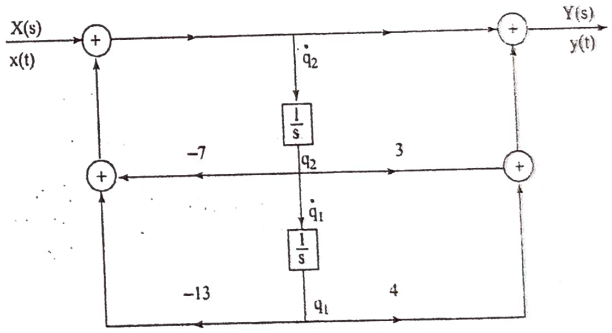


Fig. 7.87

Let us take the output the integrators as state variables q_1 and q_2 as shown in Fig. 7.87. Then we can write the state equation as

$$\dot{q}_1 = q_2 \tag{7.247}$$
$$\dot{q}_2 = -7q_2 - 13q_1 + x(t) \tag{7.248}$$

and the output equation is

$$\begin{aligned} y(t) &= \dot{q}_2 + 3q_2 + 4q_1 \\ &= [-7q_2 - 13q_1 + x(t)] + 3q_2 + 4q_1 \\ &= -4q_2 - 9q_1 + x(t) \end{aligned} \tag{7.249}$$

Now the state matrix can be written as

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ -13 & -7 \end{bmatrix}}_{A} \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{B} x(t) \tag{7.250}$$

and the output matrix is

$$y(t) = \underbrace{[-9, -4]} \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} + x(t) \tag{7.251}$$

$$A = \begin{bmatrix} 0 & 1 \\ -13 & -7 \end{bmatrix}; \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}; \quad C = [-9 \quad -4]; \quad D = [1]$$

Practice Problem 7.26

Obtain state space representation of the following system with transfer function

$$H(s) = \frac{s+2}{s^2+4s+8}$$

7.14.4 Derivation of transfer function from state model

Consider the state model

$$\dot{Q} = AQ + BX \tag{7.252a}$$

$$y = CQ + DX \tag{7.252b}$$

Taking Laplace transform on both sides for Eq. (7.252) yields

$$sQ(s) - q(0) = AQ(s) + BX(s) \tag{7.253a}$$

and

$$Y(s) = CQ(s) + DX(s) \tag{7.253b}$$

From Eq. (7.253a) we find

$$sQ(s) - AQ(s) = q(0) + BX(s) \tag{7.254}$$

Since A and B are matrices, Eq. (7.254) can be written as

$$(sI - A)Q(s) = q(0) + BX(s) \tag{7.255}$$

where I is a unit matrix.

$$\text{Now} \quad Q(s) = (sI - A)^{-1}q(0) + (sI - A)^{-1}BX(s) \tag{7.256}$$

Assuming zero initial conditions, we get

$$Q(s) = (sI - A)^{-1}BX(s). \tag{7.257}$$

Substituting Eq. (7.257) in Eq. (7.253b) yields

$$Y(s) = C(sI - A)^{-1}BX(s) + DX(s) \tag{7.258}$$

$$\Rightarrow \quad H(s) = \frac{Y(s)}{X(s)} = C(sI - A)^{-1}B + D \tag{7.259}$$

$$\boxed{H(s) = \frac{adj(sI - A)}{det\,(sI - A)}B + D} \tag{7.260}$$

Solved Problem 7.67 Find the transfer function for the system which is represented in state space representation

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} x(t)$$

$$y = [0 \quad 1] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

Solution:

We have

$$H(s) = C(sI - A)^{-1}B + D \tag{7.261}$$

Given

$$A=\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix};\quad B=\begin{bmatrix}0\\ 1\end{bmatrix};\quad C=[0\quad 1];\quad D=[0]$$

$$sI-A=s\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}-\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}$$

$$=\begin{bmatrix}s & 0\\ 0 & s\end{bmatrix}-\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}$$

$$=\begin{bmatrix}s-2 & 1\\ 1 & s-2\end{bmatrix} \tag{7.262}$$

$$(sI-A)^{-1}=\frac{1}{(s-2)^2-1}\begin{bmatrix}s-2 & -1\\ -1 & s-2\end{bmatrix}$$

$$=\frac{1}{s^2-4s+3}\begin{bmatrix}s-2 & -1\\ -1 & s-2\end{bmatrix}$$

$$=\begin{bmatrix}\frac{s-2}{s^2-4s+3} & \frac{-1}{s^2-4s+3}\\ \frac{-1}{s^2-4s+3} & \frac{s-2}{s^2-4s+3}\end{bmatrix} \tag{7.263}$$

$$C(sI-A^{-1})B=[0\quad 1]\begin{bmatrix}\frac{s-2}{s^2-4s+3} & \frac{-1}{s^2-4s+3}\\ \frac{-1}{s^2-4s+3} & \frac{s-2}{s^2-4s+3}\end{bmatrix}\begin{bmatrix}0\\ 1\end{bmatrix}$$

$$=\begin{bmatrix}\frac{-1}{s^2-4s+3} & \frac{s-2}{s^2-4s+3}\end{bmatrix}\begin{bmatrix}0\\ 1\end{bmatrix}$$

$$=\frac{s-2}{s^2-4s+3} \tag{7.264}$$

$$\Rightarrow\quad \boxed{H(s)=\frac{s-2}{s^2-4s+3}}$$

7.14.5 Time domain solution of state equation

The state equation of a system is given by

$$\dot{Q}=AQ+BX \tag{7.265}$$

Let us assume a homogeneous state equation of the form

$$\dot{Q}=AQ \tag{7.266}$$

Assume the solution for Eq. (7.266) is of the form

$$q(t)=b_0t+b_2t^2+\ldots+b_kt^k+b_{k+1}t^{k+1}+\cdots \tag{7.267}$$

$$\text{then } \dot{q}(t)=b_1+2b_2t+\ldots+kb_k^{k-1}+(k+1)b_{k+1}t^k+\cdots \tag{7.268}$$

Substituting Eq. (7.267) and Eq. (7.268) into Eq. (7.266) yields

$$b_1+2b_2t+3b_3t^2+\ldots+kb_kt^{k-1}+(k+1)b_{k+1}t^{k+1}\cdots$$
$$=Ab_0+Ab_1t+Ab_2t^2+Ab_3t^3+\ldots+Ab_kt^k+Ab_{k+1}t^{k+1}\cdots \tag{7.269}$$

Comparing the coefficients of Eq. (7.269) we obtain

$$b_1=Ab_0$$
$$b_2=\frac{1}{2}Ab_1=\frac{1}{2}A^2b_0$$
$$b_3=\frac{Ab_2}{3}=\frac{A}{3}\left(\frac{1}{2}\right)A^2b_0=\frac{A^3}{3!}b_0$$
$$\vdots$$
$$b_k=\frac{A^k}{k!}b_0$$
$$b_{k+1}=\frac{A^{k+1}}{(k+1)!}b_0 \tag{7.270}$$

Substituting the above values in Eq. (7.267), we get

$$q(t)=b_0+Ab_0t+\frac{1}{2!}A^2b_0t^2+\frac{1}{3!}A^3b_t^3+\ldots\frac{1}{k!}b_kA^kt^k+\frac{1}{(k+1)!}b_{k+1}A^{k+1}t^{k+1}$$
$$=\left(1+At+\frac{A^2t^2}{2!}+\frac{A^3t^3}{3!}+\ldots+\frac{A^kt^k}{k!}+\frac{A^{k+1}t^{k+1}}{(k+1)!}\cdots\right)b_0$$
$$=e^{At}b_0 \tag{7.271}$$

Substituting $t=0$ in Eq. (7.271), we get

$$q(0)=b_0 \tag{7.272}$$

$$\Rightarrow\ q(t)=e^{At}x(0) \tag{7.273}$$

where e^{At} is a matrix known as state transition matrix and is represented by $\phi(t)$

$$\text{That is } \phi(t)=e^{At} \tag{7.274}$$

Now consider the Eq. (7.265)

$$\dot{Q} = AQ + BX$$
$$\dot{Q} - AQ = BX \tag{7.275}$$

Multiplying both sides with e^{-At}

$$e^{-At}[\dot{Q} - AQ] = e^{-At}BX$$
$$e^{-At}\dot{Q} - Ae^{-At}Q = e^{-At}BX$$
$$\Rightarrow e^{At}\frac{d}{dt}(Q) - Q\frac{d}{dt}(e^{-At}) = e^{-At}BX$$
$$\frac{d}{dt}[e^{-At}Q] = e^{-At}BX \tag{7.276}$$

Integrating on both sides from 0 to t

$$e^{-At}Q\Big|_0^t = \int_0^t e^{-A\tau}BX(\tau)d\tau \qquad \boxed{X \text{ and } Q \text{ are functions of time}}$$

$$e^{-At}[Q(t) - Q(0)] = \int_0^t e^{-A\tau}BX(\tau)d\tau$$

$$Q(t) = e^{At}Q(0) + \int_0^t e^{-A(t-\tau)}X(\tau)d\tau$$

$$= \phi(t)Q(0) + \int_0^t \phi(t-\tau)BX(\tau)d\tau \tag{7.277}$$

This is the desired solution. The integral in Eq. (7.277) is called as the convolution integral. The first term on the right hand side of equation is the response due to the input $u(t) = 0$. Hence it is the zero input response. The second term depends only on the input, not on the initial state vector. Hence it is known as zero state response.

Properties of the transition matrix

1. From Eq. (7.273) we have

$$Q(t) = e^{At}Q(0)$$
$$= \phi(t)Q(0) \tag{7.278}$$

at $t = 0$

$$Q(0) = \phi(0)Q(0) \tag{7.279}$$

from which we have

$$\boxed{Q(0) = I}$$

2.

$$\phi(t) = e^{At} = (e^{-At})^{-1} = [\phi(-t)]^{-1} \tag{7.280}$$
$$\text{that is } \phi^{-1}(t) = \phi(-t) \tag{7.281}$$

3.

$$\phi(t_1 + t_2)e^{A(t_1+t_2)} = e^{At_1}e^{At_2} = \phi(t_1)\phi(t_2) \tag{7.282}$$

7.14.6 Laplace transform solution of state equation

Consider the state equation

$$\dot{Q} = AQ + BX \tag{7.283}$$

and the output equation

$$Y = CQ + DX \tag{7.284}$$

Taking Laplace transform on both sides yields

$$sQ(s) - Q(0) = AQ(s) + BX(s)$$
$$(sI - A)q(s) = Q(0) + BX(s)$$

where I is the identity matrix

$$Q(s) = (sI - a)^{-1}[Q(0) + BX(s)] \tag{7.285}$$
$$= \phi(s)[Q(0) + BX(s)] \tag{7.286}$$
$$\text{where } \phi(s) = (sI - A)^{-1} \tag{7.287}$$
$$\Rightarrow Q(s) = \phi(s)Q(0) + \phi(s)BX(s) \tag{7.288}$$

and

$$Q(t) = L^{-1}[\phi(s)Q(0)] + L^{-1}[\phi(s)BX(s)] \tag{7.289}$$

Solved Problem 7.68 A linear time-invariant system is characterised by the state equation

$$\begin{bmatrix}\dot{q}_1 \\ \dot{q}_2\end{bmatrix} = \begin{bmatrix}1 & 0 \\ 1 & 1\end{bmatrix}\begin{bmatrix}q_1 \\ q_2\end{bmatrix} + \begin{bmatrix}0 \\ 1\end{bmatrix}u$$

where u is a unit step function.

Compute the solution of the equation assuming initial conditions

$$Q(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Solution:

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix};\ B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$Q(s) = L^{-1}[\phi(s)Q(0)] + L^{-1}[\phi(s)BX(s)]$$

we know $\phi(s) = (sI - A)^{-1}$

$$sI - A = s\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} s-1 & 0 \\ -1 & s-1 \end{bmatrix}$$

$$(sI - A)^{-1} = \frac{1}{(s-1)^2}\begin{bmatrix} s-1 & 0 \\ 1 & s-1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{s} & 0 \\ (\frac{1}{s-1})^2 & \frac{1}{s-1} \end{bmatrix}$$

$$\phi(s)Q(0) = \begin{bmatrix} \frac{1}{s-1} & 0 \\ (\frac{1}{s-1})^2 & \frac{1}{s-1} \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{s-1} \\ (\frac{1}{s-1})^2 \end{bmatrix}$$

$$L^{-1}[\phi(s)Q(0)] = L^{-1}\begin{bmatrix} \frac{1}{s-1} \\ (\frac{1}{s-1})^2 \end{bmatrix} = \begin{bmatrix} e^t \\ te^t \end{bmatrix}$$

$$\phi(s)B = \begin{bmatrix} \frac{1}{s-1} & 0 \\ (\frac{1}{s-1})^2 & \frac{1}{s-1} \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{1}{s-1} \end{bmatrix}$$

$$\phi(s)BX(s) = \begin{bmatrix} 0 \\ \frac{1}{s-1} \end{bmatrix}\begin{bmatrix} \frac{1}{s} \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{1}{s(s-1)} \end{bmatrix}$$

$$L^{-1}[\phi(s)BX(s)] = \begin{bmatrix} 0 \\ -1 + e^t \end{bmatrix}$$

$$\Rightarrow Q(t) = \begin{bmatrix} e^t \\ te^t \end{bmatrix} + \begin{bmatrix} 0 \\ -1 + e^t \end{bmatrix}$$

$$Q(t) = \begin{bmatrix} e^t \\ te^t + e^t - 1 \end{bmatrix}$$

Practice problem 7.27

A Linear time invariant system is characterized by the state equation

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} q_1 \\ q_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$

where u is a unit step function.

Compute the solution of the equation assuming initial conditions $Q(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

Solved Problem 7.69 Consider an LTI system with input $x(t) = e^{-t}u(t)$ and impulse response $h(t) = e^{-2t}u(t)$.

(a) Determine Laplace transform of $x(t)$ and $h(t)$.

(b) Using convolution property, determine the Laplace transform $Y(s)$ of the output $y(t)$.

(c) From the Laplace transform of $Y(s)$ as obtained in part (b), determine $y(t)$.

(d) Verify your result in part (c) by explicitly convolving using $x(t)$ and $h(t)$

AU 2006

Solution:

(a) Given $x(t) = e^{-t}u(t)$ and $h(t) = e^{-2t}u(t)$

$$L[x(t)] = L[e^{-t}u(t)] = \frac{1}{s+1}$$

$$L[h(t)] = L[e^{-2t}u(t)] = \frac{1}{s+2}$$

(b) Using convolution property we have

$$y(t) = x(t) * h(t)$$

$$Y(s) = L[x(t) * h(t)] = X(s)H(s)$$

$$= \frac{1}{(s+1)(s+2)}$$

(c) Finding partial fraction expansion we get

$$Y(s) = \frac{A}{s+1} + \frac{B}{s+2} = \frac{1}{s+1} - \frac{1}{s+2}$$

Taking inverse Laplace transform

(d)
$$y(t) = e^{-t}u(t) - e^{-2t}u(t)$$

$$x(t) * h(t) = e^{-t}u(t) * e^{-2}u(t)$$

$$= \int_0^t e^{-\tau} e^{-2(t-\tau)}\, d\tau$$

$$= e^{-2t} \int_0^t e^{\tau}\, dt = e^{-2t}\, e^{\tau}|_0^t = e^{-2t}[e^t - 1]$$

$$= (e^{-t} - e^{-2t})u(t)$$

Solved Problem 7.70 Consider a continuous time LTI system

$$\frac{d^2y(t)}{dt^2} - \frac{dy(t)}{dt} - 2y(t) = x(t)$$

(i) Find the system function $H(s)$.

(ii) Determine of the impulse response $h(t)$ for (i) the system is causal (ii) the system is stable and (iii) the system is neither causal nor stable.

AU Dec 2008

Solution:

(i) Taking Laplace transform on both sides of the differential equation

$$s^2Y(s) - sY(s) - 2Y(s) = X(s)$$

The system function

$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s^2 - s - 2}$$

(ii)

$$H(s) = \frac{1}{s^2 - s - 2} = \frac{1}{(s-2)(s+1)} = \frac{A}{s-2} + \frac{B}{s+1}$$

$$= \frac{1/3}{s-2} - \frac{1/3}{s+1}$$

The pole-zero diagram of $H(s)$ is shown in Fig.7.88

From the system to be causal ROC is $Re(s) > 2$.

Then

$$h(t) = \frac{1}{3}e^{2t}u(t) - \frac{1}{3}e^{-t}u(t)$$

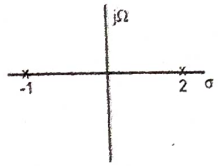


Fig.7.88

For the system to be stable,

ROC is $-1 < Re(s) < 2$. Therefore

$$h(t) = -\frac{1}{3}e^{2t}u(-t) - \frac{1}{3}e^{-t}u(t)$$

For the system to be neither stable or neither causal the ROC is $Re(s) < -1$

$$h(t) = -\frac{1}{3}e^{2t}u(-t) + \frac{1}{3}e^{-t}u(-t)$$

Solved Problem 7.71 Show that the Laplace transform of $x(t) = e^{-at}\left[A\cos bt + \frac{B-Aa}{b}\sin bt\right]u(t)$ is $X(s) = \frac{As+B}{s^2+2as+c}$ where $c = b^2 + a^2$ and $a > 0$

JNTU March 2006, Nov 2003.

Solution:

$$x(t) = e^{-at}\left[A\cos bt + \frac{B-Aa}{b}\sin bt\right]u(t)$$

$$= Ae^{-at}\cos btu(t) + \frac{B-Aa}{b}e^{-at}\sin btu(t)$$

We know

$$L[e^{-at}\cos btu(t)] = \frac{s+a}{(s+a)^2+b^2}$$

and

$$L[e^{-at}\sin btu(t)] = \frac{b}{(s+a)^2+b^2}$$

$$X(s) = A\frac{s+a}{(s+a)^2+b^2} + \frac{B-Aa}{b}\frac{b}{(s+a)^2+b^2}$$

$$= \frac{As+aA+B-Aa}{(s+a)^2+b^2} = \frac{As+B}{s^2+2as+a^2+b^2}$$

Comparing with the given $X(s)$ we find that $c = a^2 + b^2$.

Short Questions and Answers

1. Define the Laplace transform.

The Laplace transform of a signal $x(t)$ is defined as

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt$$

where s is complex frequency denoted by $s = \sigma + j\Omega$.

2. What is the condition for convergence of the Laplace transform?

The necessary condition for convergence of the Laplace transform is absolutely integrable of $x(t)e^{-\sigma t}$. That is, $X(s)$ exists if

$$\int_{-\infty}^{\infty} |x(t)e^{-\sigma t}| < \infty$$

3. What is Region of convergence (ROC)?

The Region of convergence of $X(s)$ is the set of all values of σ for which the Laplace transform converges.

4. What is Laplace transform of the following signals?

(i) $\delta(t)$ (ii) $u(t)$ (iii) $e^{-at}u(t)$ (iv) $\cos\Omega_0 t u(t)$ (v) $\sin\Omega_0 t u(t)$

$$L[\delta(t)] = 1$$

$$L[u(t)] = \frac{1}{s}; \quad L[e^{-at}u(t)] = \frac{1}{s+a}; \quad L[\cos\Omega_0 t u(t)] = \frac{s}{s^2+\Omega_0^2}$$

$$L[\sin\Omega_0 t u(t)] = \frac{\Omega_0}{s^2+\Omega_0^2}$$

5. State the time shifting property of Laplace transform.

$$\text{If} \quad L[x(t)] = X(s)$$

$$\text{then}$$

$$L[x(t-t_0)] = e^{-st_0}X(s)$$

6. State the frequency shifting property of Laplace transform.

$$\text{If} \quad L[x(t)] = X(s)$$

$$\text{then}$$

$$L[e^{-at}x(t)] = X(s+a)$$

7. State time convolution property of Laplace transform.

$$\text{If} \quad L[x_1(t)] = X_1(s) \quad \text{and} \quad L[x_2(t)] = X_2(s)$$

$$\text{then}$$

$$L[x_1(t) * x_2(t)] = X_1(s)X_2(s)$$

8. State the initial value theorem and final value theorem of Laplace transform.

Initial value theorem:

The initial value theorem allows calculate $x(0)$ directly from the transform $X(s)$

$$\lim_{t\to 0} x(t) = \lim_{s\to\infty} sX(s)$$

Final value theorem

$$\lim_{t\to\infty} x(t) = \lim_{s\to 0} sX(s)$$

9. Define transfer function. AU Nov 2003

The transfer function of the system is the ratio of the Laplace transform of the output signal to the Laplace transform of the input signal with all initial conditions are zero.

10. Define (i) Natural response (ii) Forced response.

(i) The natural response of the system is the part of the total response which is due to initial conditions of the system alone.

(ii) The forced response of the system is the part of the total response which is due to input alone

11. Define poles and zeros of a transfer function.

The transfer function of a system is the ratio of two polynomials. The roots of the numerator polynomial are called the zeros of the transfer function and the roots of the denominator polynomial are called poles of the transfer function.

12. Define (i) Transient response (ii) Steady state response.

The forced response consists of two parts.

The part of forced response which is due to poles of the system is known as transient response. The transient response vanishes after a sufficiently long time.

The part of the forced response which is due to poles of input signals $X(s)$ is known as steady state response. This response exists after the transients lies out.

13. What are the transfer functions of the following?

(i) An ideal integrator (ii) An ideal delay of T seconds.

(i) The transfer function of an ideal integrator is $\frac{1}{s}$

(ii) The transfer function of an ideal delay of T seconds is e^{-sT}

14. What is the condition for stability of a system?

For a system to be stable, the poles of the transfer function must be in the left half of s-plane.

15. What is the relationship between Fourier transform and Laplace transform? (Au April 2003)

Or

Define Laplace transform. In what way it is different from Fourier Transform. AU Nov 2008

The Laplace transform of a signal $x(t)$ as given by

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt \qquad \text{(a)}$$

The Fourier transform of a signal $x(t)$ is given by

$$X(j\Omega) = \int_{-\infty}^{\infty} x(t)e^{j\Omega t}dt \qquad \text{(b)}$$

Eq. (a) and Eq. (b) are identical when $s = j\Omega$. That is $X(j\Omega)$ is equal to $X(s)$ when $s = j\Omega$. In the s-plane this corresponds the points on imaginary axis.

$$X(j\Omega) = X(s)\Big|_{s=j\Omega}$$

16. Define (i) block diagram, (ii) signal flowgraph.

(i) A block diagram is interconnection of subsystems representing certain basic mathematical operation in such a way that the overall diagram represents the system's mathematical model.

(ii) A signal flow graph is a graphical representation of the relationships between the variables of a set of linear algebraic equations.

17. What are the different types of realizations?

1. Direct form I realization
2. Direct form II realization
3. Cascade form realization
4. Parallel-form realization

18. Why differentiators are not used in realizing practicals systems?

The differentiators are not used in realizing practical systems because differentiators amplifies high frequency noise.

19. What are the drawbacks of transfer function method of representing a system?

The drawbacks of transfer function method of representing a system are

1. The transfer function describes only the zero state response of a system.
2. It describes only the relationship between the input and output of a system, but does not provide any information regarding the internal state of a system.
3. It is limited to single-input and single output systems.
4. It is applicable only for linear time invariant systems.

20. Define state of a system.

The state of a system is a minimal set of variables known as state variables, such that the knowledge of these variables for $t = t_0$, together with the knowledge of the input for $t \geq t_0$, provides the state and output for the system for $t > t_0$.

21. Determine Laplace transform of $x(t) = e^{-at}\sin\omega t u(t)$. AU 2004, 2008

We know the Laplace transform of

$$\sin\omega t\, u(t) = \frac{\omega}{s^2+\omega^2}$$

Using frequency shifting property,

$$L\left[e^{-at}\sin\omega t\, u(t)\right] = \frac{\omega}{(s+a)^2+\omega^2}$$

22. Find Laplace transform of $\delta(t)+u(t)$.

$$L[\delta(t)] = 1 \quad \text{and} \quad L[u(t)] = \frac{1}{s}$$

$$\Rightarrow L[\delta(t)+u(t)] = 1+\frac{1}{s}$$

23. What is the Laplace transform of the function? AU Nov 2009

$$x(t) = u(t) - u(t-2)$$

$$L[u(t)] = \frac{1}{s}$$

Using time shifting property

$$L[u(t-2)] = \frac{e^{-2s}}{s}$$

$$L[x(t)] = \frac{1-e^{-2s}}{s}$$

24. Find the Laplace transform of unit ramp function. AU Nov 2008

$$r(t) = tu(t)$$

$$L[r(t)] = \int_0^\infty te^{-st}dt$$

$$= \frac{1}{s}te^{-st}\Big|_0^\infty - \frac{1}{s^2}e^{-st}\Big|_0^\infty = \frac{1}{s^2}$$

Multiple Choice Questions

1. The output of a system $V(s) = Z(s)I(s)$ where $V(s), Z(s)$ and $I(s)$ are the Laplace transforms of the corresponding time function $v(t), z(t)$ and $i(t)$. The voltage $v(t)$ is

   (a) $v(t) = z(t)i(t)$ (b) $v(t) = \int i(\tau)z(t-\tau)d\tau$

   (c) $v(t) = \int i(t)z(t+\tau)d\tau$ (d) $v(t) = z(t)+i(t)$

2. The Laplace transform of a unit ramp function starting at $t = a$ is

   (a) $\frac{1}{(s+a)^2}$ (b) $\frac{e^{-as}}{(s+a)^2}$ (c) $\frac{e^{-as}}{s^2}$ (d) $\frac{a}{s^2}$

3. If $L[f(t)] = \frac{2(s+1)}{s^2+2s+5}$, then $f(0^+)$ and $f(\infty)$ are given by

   (a) 0, 2 respectively (b) 2, 0 respectively

   (c) 0,1 respectively (d) $\frac{2}{5}$, 0 respectively

4. The inverse Laplace transform of the function $\frac{s+5}{(s+1)(s+3)}$ is

   (a) $2e^{-t}-e^{-3t}$ (b) $2e^{-t}+e^{-3t}$

   (c) $e^{-t}-2e^{-3t}$ (d) $e^{-t}+e^{-3t}$

5. The Laplace transform of $e^{at}\cos(bt)$ is equal to

   (a) $\frac{s+a}{(s+a)^2+b^2}$ (b) $\frac{s-a}{(s-a)^2+b^2}$

   (c) $\frac{s-a}{(s+a)^2+b^2}$ (d) $\frac{(s+a)}{(s-b)^2}$

6. If $L[f(t)] = F(s)$ then $L[f(t-T)]$ is equal to

   (a) $e^{sT}F(s)$ (b) $e^{-sT}F(s)$ (c) $\frac{F(s)}{1+e^{sT}}$ (d) $\frac{F(s)}{1-e^{-sT}}$

7. The transfer function of a system is given by $H(s) = \frac{1}{s^2(s+2)}$. The impulse response of the system is

   (a) $(t^2 * e^{-2t})u(t)$ (b) $(t * e^{-2t})u(t)$

   (c) $t^2e^{-2t}u(t)$ (d) $te^{-2t}u(t)$

8. The Laplace transform of a continuous-time signal is $X(s) = \frac{5-s}{s^2-s-2}$. If the Fourier transform of this signal exists, then $x(t)$ is

   (a) $e^{2t}u(t)-2e^{-t}u(t)$ (b) $-e^{2t}u(-t)+2e^{-t}u(t)$

   (c) $-e^{2t}u(-t)-2e^{-t}u(t)$ (d) $e^{2t}u(-t)-2e^{-t}u(t)$

9. The Laplace transform of $i(t)$ is given by

$$I(s) = \frac{2}{s(1+s)}$$

As $t \to \infty$, the value of $i(t)$ tends to

(a) 0 (b) 1 (c) 2 (d) ∞

10. The following table gives some time functions and their Laplace transforms

	$f(t)$	$F(s)$
1.	$\delta(t)$	s
2.	$u(t)$	$\frac{1}{s}$
3.	$tu(t)$	$\frac{2}{s^2}$
4.	$t^2u(t)$	$\frac{2}{s^3}$

of these, the correctly matched pair is

(a) 2 and 4 (b) 1 and 4 (c) 3 and 4 (d) 1 and 2

11. The final value of $L^{-1}\left[\frac{2s+1}{s^4+8s^3+16s^2+s}\right]$ is

(a) ∞ (b) 2 (c) 1 (d) zero

12. If $x(t)$ and its first derivative are Laplace transformable and the Laplace transform of $x(t)$ is $X(s)$, then $\lim_{x\to 0} x(t)$ is given by

(a) $\lim_{s\to\infty} sX(s)$ (b) $\lim_{s\to 0} sX(s)$ (c) $\lim_{s\to\infty} \frac{X(s)}{s}$ (d) $\lim_{s\to 0} \frac{X(s)}{s}$

13. If $\delta(t)$ denotes a unit impulse, then the Laplace transform of $\frac{d^2\delta(t)}{dt^2}$ will be

(a) 1 (b) s^2 (c) s (d) s^{-2}

14. Given that $h(t) = 10e^{-10t}u(t)$ and $f(t) = \sin 10tu(t)$, the Laplace transform of the signal $y(t) = \int_0^t f(\tau)h(t-\tau)d\tau$ is given by

(a) $\frac{10}{(s+10)(s^2+100)}$ (b) $\frac{10(s+10)}{s^2+100}$ (c) $\frac{100}{(s+10)(s^2+100)}$

(d) $\frac{1}{(s+10)(s^2+100)}$

15. The impulse response of a single-pole system would approach a non-zero constant as $t \to \infty$ if and only if the pole is located in the s-plane

(a) on the negative real axis (b) at the origin
(c) on the positive real axis (d) on the imaginary axis

16. If $x(t)$ and $\frac{d}{dt}x(t)$ are Laplace transformable and $\lim_{s\to\infty} x(t)$ exists, then $\lim_{t\to\infty} x(t)$ is equal to

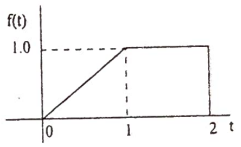
(a) $\lim_{s\to 0} sX(s)$ (b) $\lim_{s\to\infty} sX(s)$ (c) $\lim_{s\to\infty} \frac{X(s)}{s}$ (d) $\lim_{s\to 0} \frac{X(s)}{s}$

17. The Laplace transform of $\sin(\Omega t + \alpha)$ is

(a) $\frac{\alpha}{s^2+\alpha^2}\exp\left(\frac{s}{\alpha\Omega}\right)$ (b) $\frac{\Omega}{s^2+\Omega^2}\exp\left(\frac{\alpha s}{\Omega}\right)$

(c) $\frac{s}{s^2+\alpha^2}\exp\left(\frac{s}{\alpha\Omega}\right)$ (d) $\frac{\Omega}{s^2+\Omega^2}\exp\left(\frac{s}{\alpha\Omega}\right)$

18. The function $f(t)$ shown in the given figure will have Laplace transform as



(a) $\frac{1}{s^2} - \frac{1}{s}e^{-s} - \frac{1}{s^2}e^{-2s}$

(b) $\frac{1}{s^2}(1 - e^{-s} - e^{-2s})$

(c) $\frac{1}{s}[1 - e^{-s} - e^{-2s}]$

(d) $\frac{1}{s^2}[1 - e^{-s} - se^{-2s}]$

19. Inverse Laplace transform of the function $\frac{2s+5}{s^2+5s+6}$ is

(a) $2e^{-2.5}\cosh(0.5t)$ (b) $e^{-2t} - e^{-3t}$

(c) $2e^{-2.5}\sinh(0.5t)$ (d) $2e^{-2.5t}\cos 0.5t$

20. The output of a linear system to a unit step input $u(t)$ is t^2e^{-2t}. The system function $H(s)$ is

(a) $\frac{2}{s^2(s+2)}$ (b) $\frac{2}{(s+2)^2}$ (c) $\frac{2}{(s+2)^3}$ (d) $\frac{2s}{(s+2)^3}$

21. The transfer function of zero order hold is

(a) $\frac{1-e^{-sT}}{s}$ (b) $\frac{1}{s}$ (c) $\frac{e^{-sT}}{s}$ (d) $\frac{1-e^{sT}}{s}$

Answers to Multiple Choice Questions

1. (b)	7. (b)	13. (b)	19. (a)
2. (c)	8. (c)	14. (c)	20. (d)
3. (b)	9. (c)	15. (b)	21. (a)
4. (a)	10. (a)	16. (a)	
5. (a)	11. (c)	17. (b)	
6. (b)	12. (a)	18. (d)	

8

Fourier Analysis of Discrete-Time Signals

8.1 Introduction

In chapter 6, we have studied the representation of continuous-time periodic signals as a sum of sinusoids or complex exponentials. In this chapter we will study the representation of a discrete-time periodic signal as weighted sum of complex exponentials. This gives us an insight to frequency analysis of discrete-time signals.

Consider the exponential $e^{j\omega n} = \cos\omega n + j\sin\omega n$, which is periodic if and only if ω is a multiple of π. That is, $e^{j\omega n}$ is not necessarily be periodic for any ω as in the continuous-time case. We know that the continuous-time exponential $e^{j\Omega t}$ is periodic with period $T = \frac{2\pi}{\Omega}$. Similarly the signal $e^{jk\Omega t}$ is also periodic with period T (not necessarily be fundamental period). Thus there are infinite different complex exponentials with period T. This is not the case with discrete-time exponential. If $e^{j\omega n}$ is periodic, then $e^{jk\omega n}$ is also periodic for every integer k. However there are only a finite number of discrete complex exponentials. This can be proved as shown below. Let $\omega = \frac{2\pi}{N}$, then we can define, the discrete exponential as

$$\phi_k(n) = e^{jk\left(\frac{2\pi}{N}\right)n} \quad \text{for} \quad k, 0, \pm1, \pm2 \ldots \tag{8.1}$$

Note that n denotes time instants and k denotes different exponentials.

For any integer k, we have

$$\begin{aligned}\phi_{k+N}(n) = e^{j(k+N)\frac{2\pi}{N}n} &= e^{jk\frac{2\pi}{N}n}\, e^{j2\pi n} \\ &= e^{j2\pi kn/N}\end{aligned} \tag{8.2}$$

since $e^{j2\pi n} = 1$ for every integer n.

Thus $e^{jk\left(\frac{2\pi}{N}\right)n}$ is periodic in k with period N. Thus there are only N distinct complex exponentials. From Eq. (8.2) we can say that $\phi_k(n), \phi_{k\pm N}(n), \phi_{k\pm 2N}(n)$ all denote the same sequence.